Deployment of Online Food Ordering and Delivery Application

A DevOps Implementation for Scalable Food Delivery Platform

Domain: Food Delivery and E-Commerce
Project Overview:
Zomato is a leading online food delivery and restaurant discovery platform. To handle increasing user demand and improve deployment efficiency, the company is transitioning from
a monolithic architecture to a microservices-based architecture using DevOps practices.

The goal is to implement a CI/CD pipeline, containerized deployments, and real-time
monitoring to ensure high availability, faster releases, and automated scalability. AWS will be
used as the primary cloud provider for hosting and deployment.

Business Goals:

- Faster and reliable deployments to production.
- Reduced manual intervention in infrastructure management.
- Improved scalability and performance monitoring.
- Automated rollback in case of failures.

Current Challenges:

- Manual Deployments: Time-consuming and error-prone.
- Monitoring Gaps: No real-time visibility into application health.

CI/CD Pipeline Implementation:

The following tools will be used to automate the deployment process:

Tools and Technologies

- Git – Version control for tracking code changes.
- Jenkins – CI/CD pipeline automation (build, test, deploy).
- Docker – Containerization of the application.
- Kubernetes (K8s) – Orchestration for scalable deployments.
- AWS (EC2, EKS, ECR) – Cloud infrastructure for hosting.
- Prometheus & Grafana – Monitoring and alerting.

Pipeline Workflow:

- GitHub Webhook Trigger
- Code push to the main/master branch triggers Jenkins automatically.
- Jenkins fetches the code
- Docker Image Creation
- The application is packaged into a Docker image and pushed to AWS

ECR/DockerHub.

- Kubernetes Deployment (Staging Environment)
- The Docker image is deployed to a Kubernetes cluster (AWS EKS).
- Monitoring & Validation
- Prometheus collects metrics.
- Grafana visualizes performance dashboards.
- If tests pass, the build is promoted to production.

Production Deployment:

- The same Docker image is deployed to the production Kubernetes cluster.
- Auto-scaling ensures high availability during peak loads.

DevOps Project: Zomato Application Deployment

Project Overview

This project demonstrates a complete **end-to-end DevOps implementation** for deploying a Zomato clone application using modern DevOps tools and practices.

The goal of the project is to automate the entire software delivery lifecycle from code integration to deployment and monitoring while ensuring:

- High code quality
- Strong security checks
- Automated CI/CD pipeline
- Containerized deployment
- Kubernetes orchestration
- Real-time monitoring
- GitOps-based continuous delivery

This project simulates a real-world production environment where automation, scalability, and observability are critical.

Project Objectives

- Build a CI/CD pipeline using Jenkins
- Implement code quality checks using SonarQube
- Perform vulnerability scanning using OWASP, Trivy, and Docker Scout
- Containerize the application using Docker
- Deploy the application on AWS infrastructure
- Orchestrate workloads using Kubernetes (EKS)
- Implement GitOps using ArgoCD

Monitor infrastructure and application health using Prometheus & Grafana

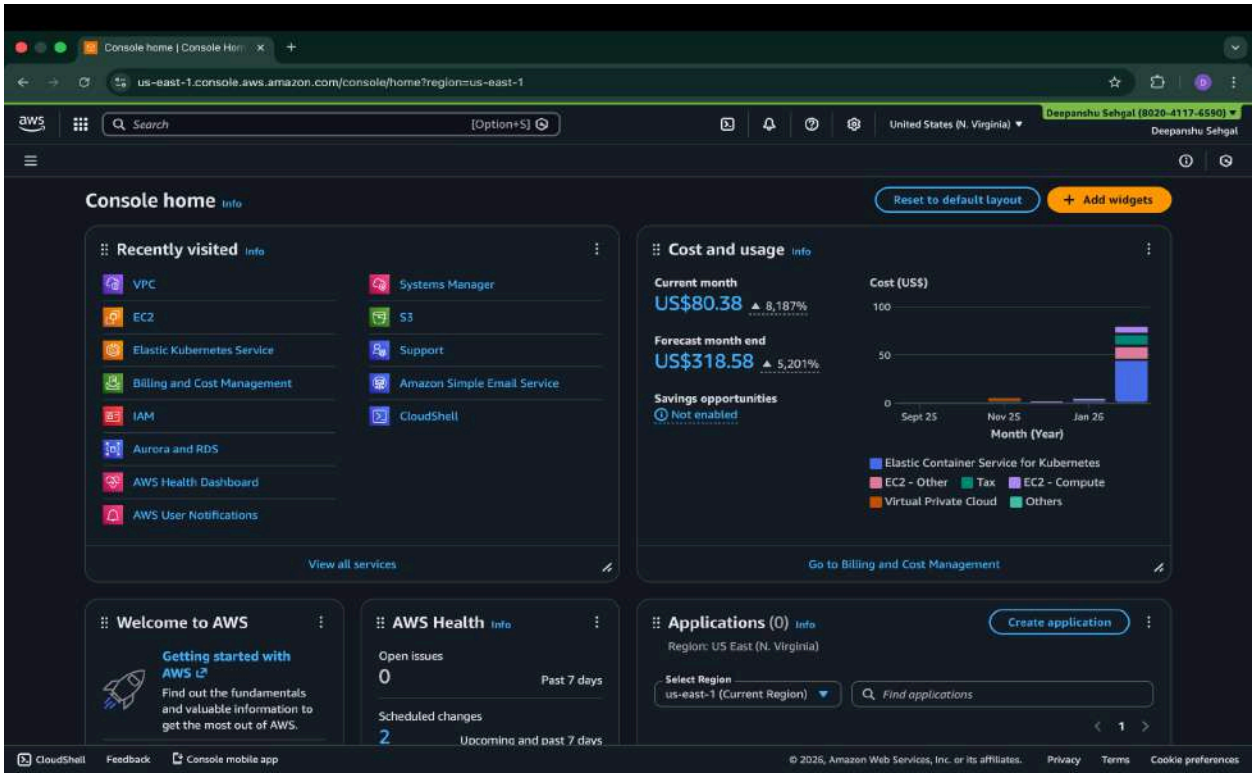
Phase 1: Infrastructure Setup

An Ubuntu 24.04 EC2 instance was launched as the primary DevOps server.

Key Setup Tasks

- Configured security groups
- Opened required ports:
    - 8080 (Jenkins)
    - 9000 (SonarQube)
    - 3000 (App/Grafana)
    - 9090 (Prometheus)
    - 9100 (Node Exporter)
- Installed AWS CLI
- Updated system packages

This prepared the base infrastructure for DevOps tooling.



Home | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Overview:

Search [Option+S]

United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2

- Dashboard
- EC2 Global View
- Events
- Instances
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager New
- Images
- Elastic Block Store
- Network & Security
  - Security Groups
  - Elastic IPs
  - Placement Groups
  - Key Pairs
  - Network Interfaces

Compute

Amazon Elastic Compute Cloud (EC2)

Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

Launch a virtual server

Launch instance

View dashboard

Get started walkthroughs

Get started tutorial

Benefits and features

EC2 offers ultimate scalability and control

Fully resizable compute capacity to support virtually any workload. This service is best if you want:

- Highest level of control of the entire technology stack, allowing full integration with all AWS services
- Widest variety of server size options
- Widest availability of operating systems to choose from including Linux, Windows, and macOS
- Global scalability

Find out more about EC2

Additional actions

View running instances

Migrate a server

Pricing (US)

EC2 pricing options

Use the AWS pricing calculator

Manage budgets

CloudShell | Feedback | Console Mobile App

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Launch an instance | EC2 | us

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search [Option+S]

United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Zomato Server

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type — Free tier eligible

ami-0b6c6ebed2801a5cb (64-bit (x86)) / ami-0071c8c431eea0edb (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...read more

ami-0b6c6ebed2801a5cb

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer

Cancel | Launch instance | Preview code

CloudShell | Feedback | Console Mobile App

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Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws | Search [Option+S] | United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2 > Instances > Launch an instance

Instance type Info | Get advice

Instance type

t2.large
Family: t2 2 vCPU 8 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.1208 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0963 USD per Hour
On-Demand RHEL base pricing: 0.1216 USD per Hour
On-Demand SUSE base pricing: 0.1928 USD per Hour On-Demand Linux base pricing: 0.0928 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

jenkins machine

Create new key pair

Network settings Info

Edit

Network | Info

vpc-0907d014797195056

Subnet | Info

No preference (Default subnet in any availability zone)

Auto-assign public IP | Info

Enable

Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb

Virtual server type (instance type)
t2.large

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer

Cancel | Launch instance | Preview code

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Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws | Search [Option+S] | United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2 > Instances > Launch an instance

Network settings Info

Edit

Network | Info

vpc-0907d014797195056

Subnet | Info

No preference (Default subnet in any availability zone)

Auto-assign public IP | Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Common security groups | Info

Select security groups

launch-wizard-1 sg-0b3a59aa6a111d9b6
VPC: vpc-0907d014797195056

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Configure storage Info

Advanced

1x 30 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb

Virtual server type (instance type)
t2.large

Firewall (security group)
launch-wizard-1

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer

Cancel | Launch instance | Preview code

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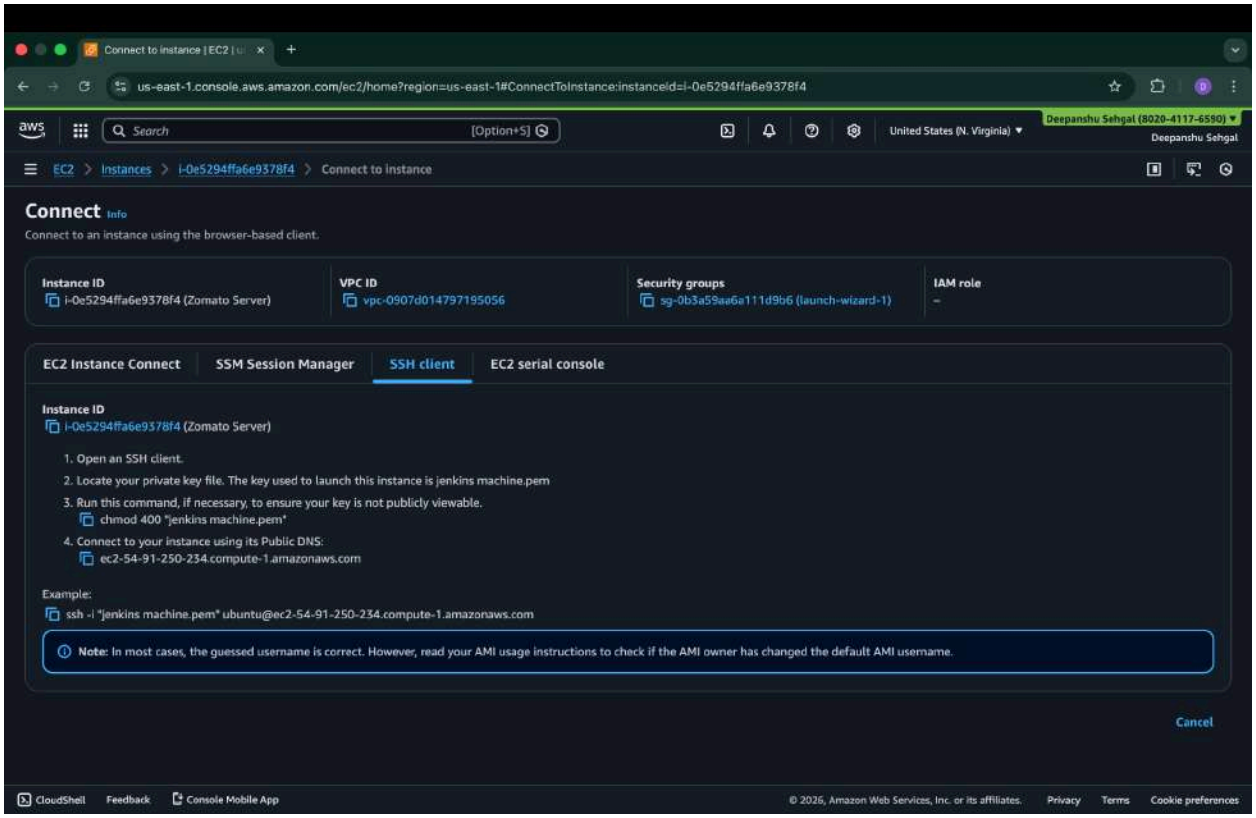
Connect to instance | EC2
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-0e5294ffa6e9378f4
Search
[Option+S]
United States (N. Virginia)
Deepanshu Sehgal (8020-4117-6590)
Deepanshu Sehgal
EC2 > Instances > i-0e5294ffa6e9378f4 > Connect to instance
Connect Info
Connect to an instance using the browser-based client.
Instance ID
i-0e5294ffa6e9378f4 (Zomato Server)
VPC ID
vpc-0907d014797195056
Security groups
sg-0b3a59aa6a111d9b6 (launch-wizard-1)
IAM role
-
EC2 Instance Connect
SSM Session Manager
SSH client
EC2 serial console
Instance ID
i-0e5294ffa6e9378f4 (Zomato Server)
1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is jenkins machine.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "jenkins machine.pem"
4. Connect to your instance using its Public DNS:
ec2-54-91-250-234.compute-1.amazonaws.com
Example:
ssh -i "jenkins machine.pem" ubuntu@ec2-54-91-250-234.compute-1.amazonaws.com
Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.
Cancel
CloudShell
Feedback
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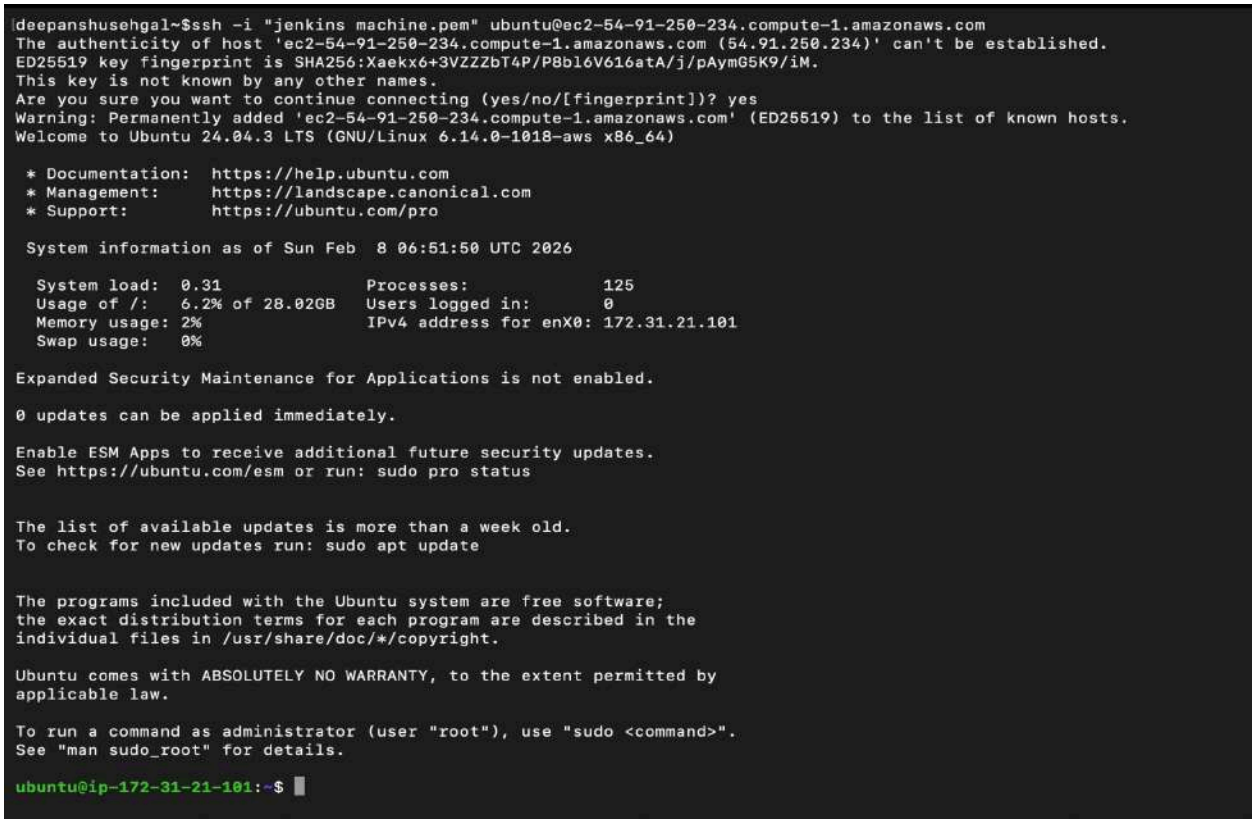

deepanshusehgal~$ssh -i "jenkins machine.pem" ubuntu@ec2-54-91-250-234.compute-1.amazonaws.com
The authenticity of host 'ec2-54-91-250-234.compute-1.amazonaws.com (54.91.250.234)' can't be established.
ED25519 key fingerprint is SHA256:Xaekx6+3VZZZbT4P/P8bl6V616atA/j/pAymG5K9/iM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-91-250-234.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Sun Feb 8 06:51:50 UTC 2026
System load: 0.31
Usage of /: 6.2% of 28.02GB
Memory usage: 2%
Swap usage: 0%
Processes: 125
Users logged in: 0
IPv4 address for enX0: 172.31.21.101
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-172-31-21-101:~$

Phase 2: Jenkins CI/CD Setup

Jenkins Installation

Jenkins was installed on Ubuntu and configured as the CI/CD engine.

Plugins Installed

- Git Plugin
- SonarQube Scanner
- OWASP Dependency Check
- Docker Pipeline
- Email Extension Plugin

Tool Configuration

- JDK 17
- NodeJS
- Sonar Scanner
- Docker integration

This allowed Jenkins to perform builds, scans, and deployments.

Update the packages
Install AWS CLI

```
ubuntu@ip-172-31-21-101:~$ sudo su
root@ip-172-31-21-101:/home/ubuntu# sudo apt update -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
78 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-21-101:/home/ubuntu# sudo apt install unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 78 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 0s (8287 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 71752 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-21-101:/home/ubuntu# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 63.6M  100 63.6M    0     0   108M      0 --:--:-- --:--:-- --:--:--  108M
root@ip-172-31-21-101:/home/ubuntu#
```

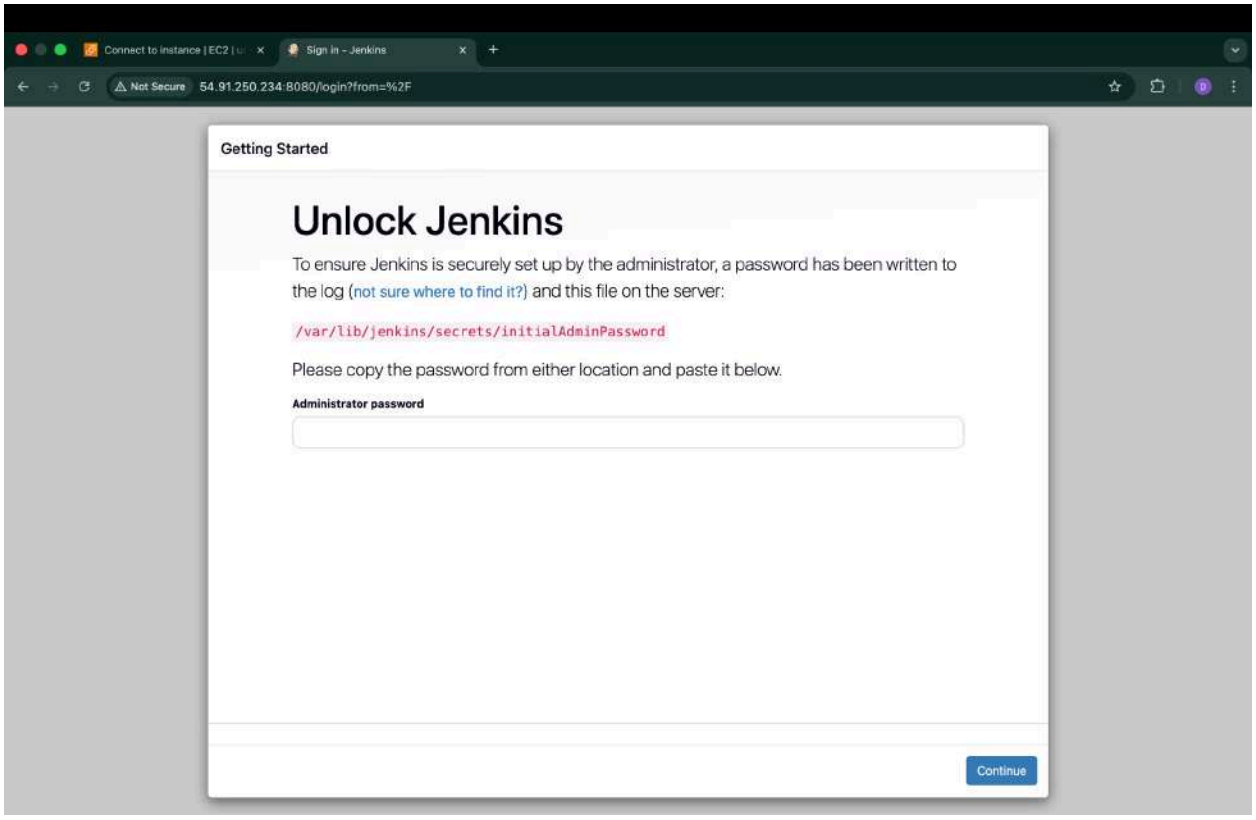
```
root@ip-172-31-21-101:/home/ubuntu# sudo ./aws/install
You can now run: /usr/local/bin/aws --version
root@ip-172-31-21-101:/home/ubuntu#
```

- Install Jenkins on Ubuntu

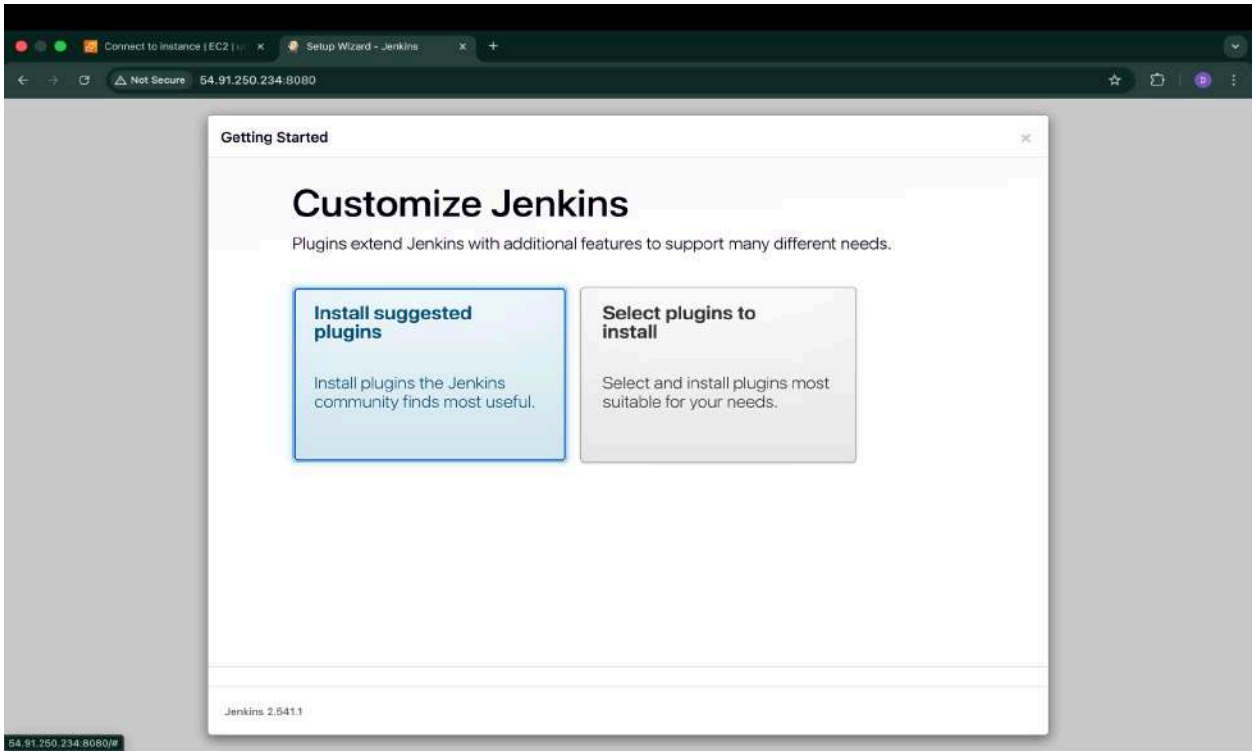
```
root@ip-172-31-21-101:/home/ubuntu# vi jenkins.sh
```

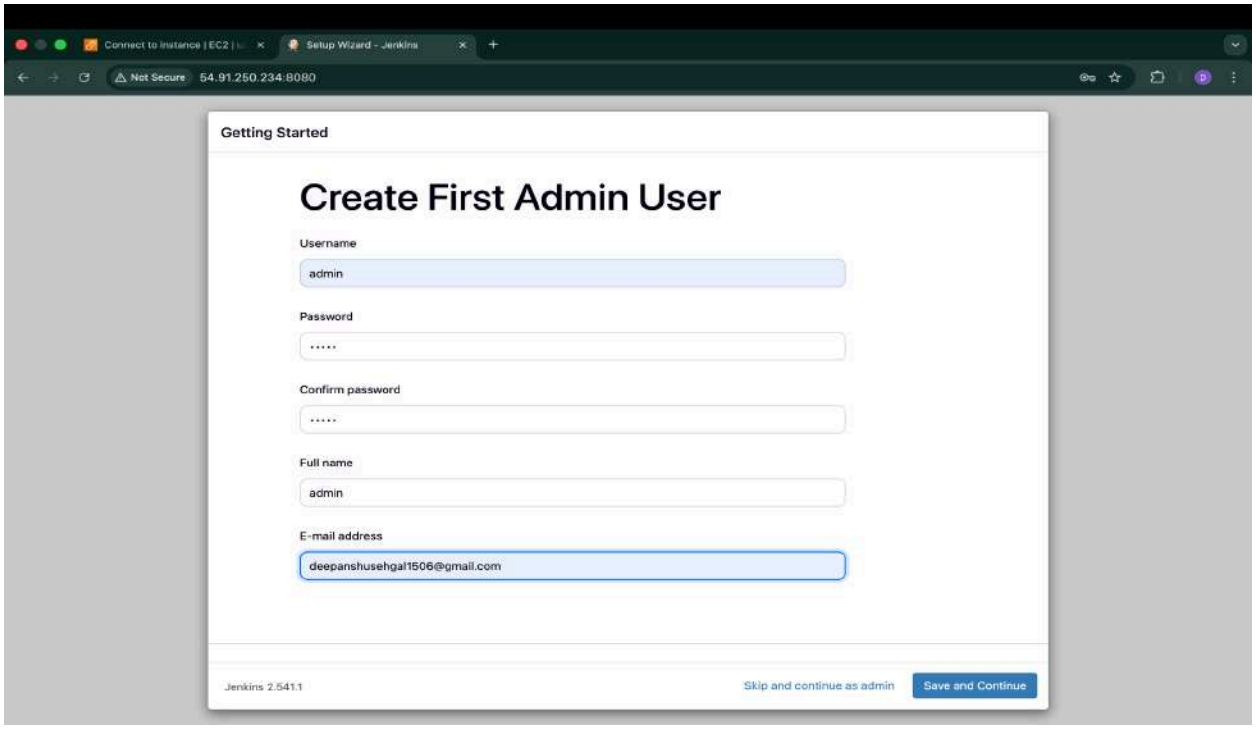
```
#!/bin/bash
sudo apt update -y
wget -O - https://packages.adoptium.net/artifactory/api/gpg/key/public | sudo tee /etc/apt/keyrings/adoptium.asc
echo "deb [signed-by=/etc/apt/keyrings/adoptium.asc] https://packages.adoptium.net/artifactory/deb $(awk -F= '/^VERSION_CODENAME/{print$2}' /etc/os-release) main
" | sudo tee /etc/apt/sources.list.d/adoptium.list
sudo apt update -y
sudo apt install temurin-17-jdk -y
/usr/bin/java --version
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/
null
sudo apt-get update -y
sudo apt-get install jenkins -y
sudo systemctl start jenkins
sudo systemctl status jenkins
~
~
~
~
~
~
~
```

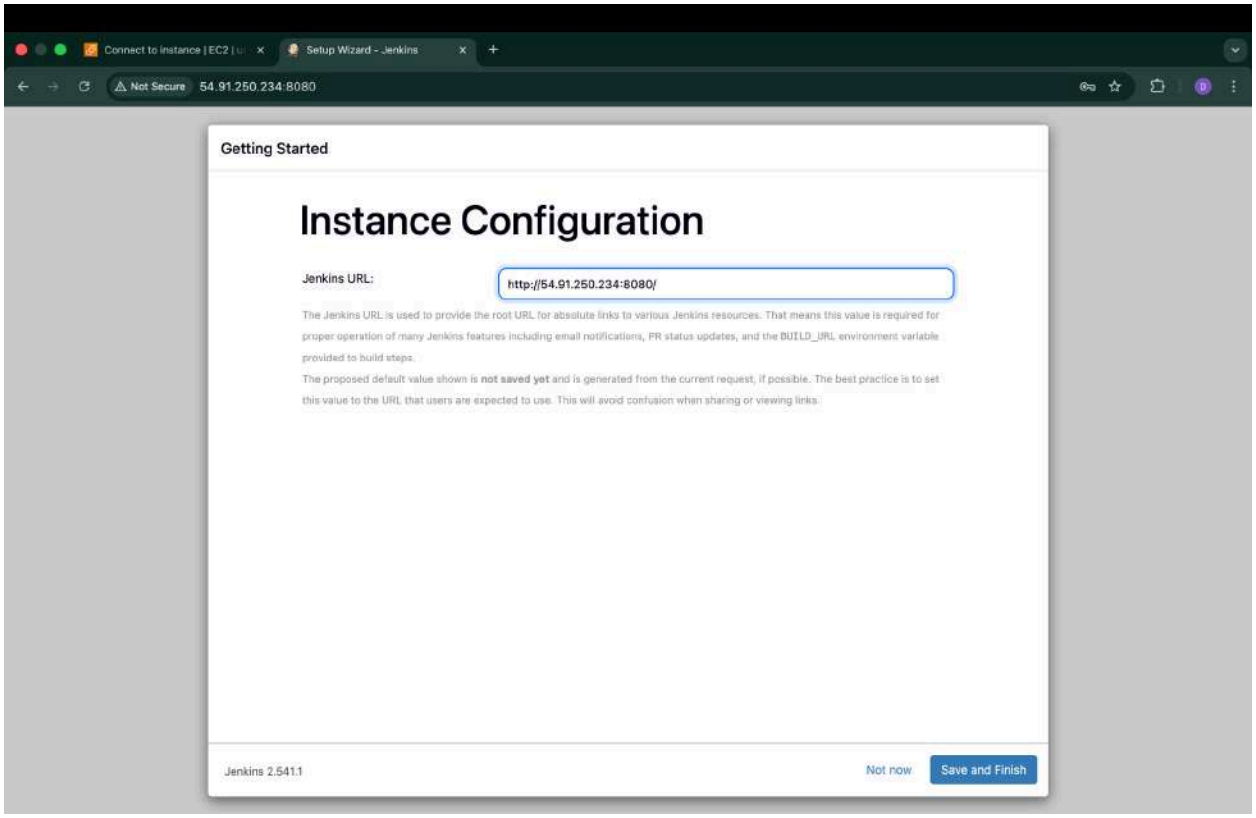
```
root@ip-172-31-21-101:/home/ubuntu# vi jenkins.sh
root@ip-172-31-21-101:/home/ubuntu# vi jenkins.sh
root@ip-172-31-21-101:/home/ubuntu# chmod +x jenkins.sh
root@ip-172-31-21-101:/home/ubuntu# ls
aws  awscliv2.zip  jenkins.sh
root@ip-172-31-21-101:/home/ubuntu# ./jenkins.sh
```

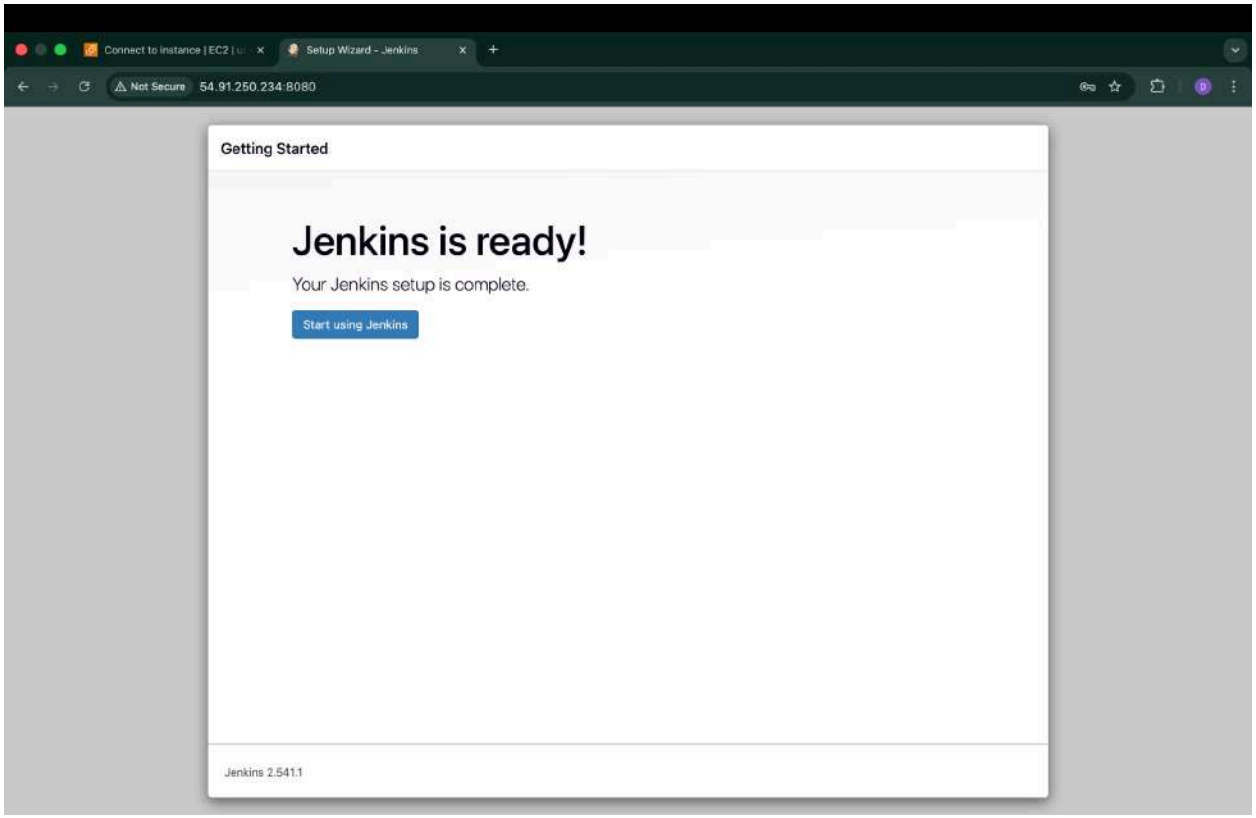


```
root@ip-172-31-21-101:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword
64f1560fc3024b46b68710acf479f3bc
root@ip-172-31-21-101:/home/ubuntu#
```






Connect to instance | EC2 |
Setup Wizard - Jenkins
Not Secure
54.91.250.234:8080
Getting Started
Instance Configuration
Jenkins URL:
http://54.91.250.234:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.541.1
Not now
Save and Finish


Connect to instance | EC2 |
Setup Wizard - Jenkins
Not Secure
54.91.250.234:8080
Getting Started
Jenkins is ready!
Your Jenkins setup is complete.
Start using Jenkins
Jenkins 2.541.1

Connect to instance | EC2 | u

Dashboard - Jenkins

Not Secure 54.91.250.234:8080

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status 0/2

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job +

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.541.1

Phase 3: Code Quality & Security

Security and quality were integrated into the pipeline.

SonarQube

- Static code analysis
- Quality Gate enforcement
- Code smell & bug detection
- Token-based authentication for Jenkins integration

OWASP Dependency Check

- Identified vulnerable dependencies
- Generated security reports

Trivy Scan

- File system vulnerability scanning
- Detected CVEs in dependencies

Docker Scout

- Container image analysis
- Security recommendations

This ensured a **shift-left security approach**.

Now Install Docker

```
root@ip-172-31-21-101:/home/ubuntu# vi docker.sh
```

```
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
sudo usermod -aG docker ubuntu
sudo chmod 777 /var/run/docker.sock
newgrp docker
sudo systemctl status docker
~
~
~
~
~
~
```

```
root@ip-172-31-21-101:/home/ubuntu# vi docker.sh
root@ip-172-31-21-101:/home/ubuntu# ls
aws  awscliv2.zip  docker.sh  jenkins.sh
root@ip-172-31-21-101:/home/ubuntu# chmod +x docker.sh
root@ip-172-31-21-101:/home/ubuntu# ./docker.sh
```

```
root@ip-172-31-21-101:/home/ubuntu# docker --version
Docker version 29.2.1, build a5c7197
root@ip-172-31-21-101:/home/ubuntu#
```

Install Trivy on Ubuntu

```
root@ip-172-31-21-101:/home/ubuntu# vi trivy.sh
```

```
sudo apt-get install wget apt-transport-https gnupg
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor | sudo tee /usr/share/keyrings/trivy.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb generic main" | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
sudo apt-get install trivy
~
~
```

```
root@ip-172-31-21-101:/home/ubuntu# vi trivy.sh
root@ip-172-31-21-101:/home/ubuntu# ls
aws  awscliv2.zip  docker.sh  jenkins.sh  trivy.sh
root@ip-172-31-21-101:/home/ubuntu# cat trivy.sh
sudo apt-get install wget apt-transport-https gnupg
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor | sudo tee /usr/share/keyrings/trivy.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb generic main" | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
sudo apt-get install trivy
root@ip-172-31-21-101:/home/ubuntu# chmod +x trivy.sh
root@ip-172-31-21-101:/home/ubuntu# ls
aws  awscliv2.zip  docker.sh  jenkins.sh  trivy.sh
root@ip-172-31-21-101:/home/ubuntu# ./trivy.sh
```

```
root@ip-172-31-21-101:/home/ubuntu# trivy --version
Version: 0.69.1
root@ip-172-31-21-101:/home/ubuntu#
```

Install Docker Scout

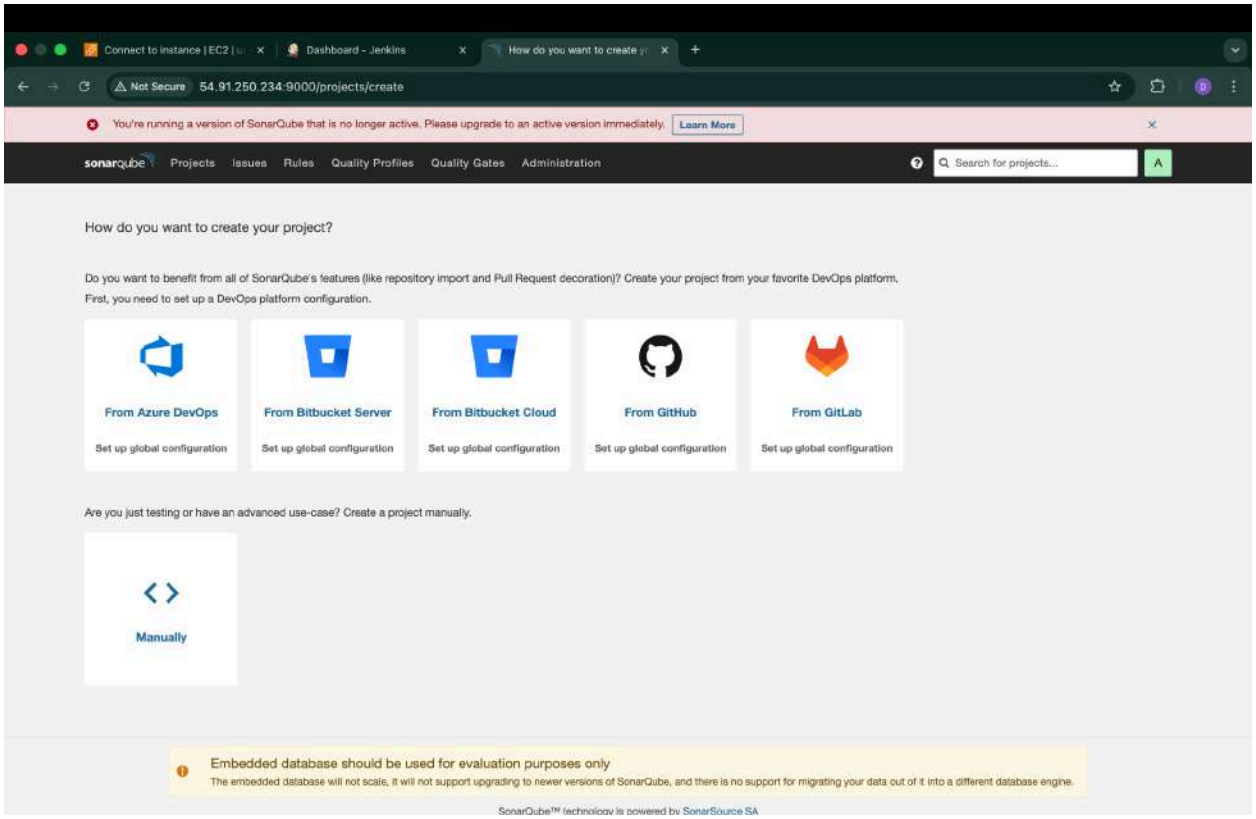
```
root@ip-172-31-21-101:/home/ubuntu# docker login -u deepanshu112

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-21-101:/home/ubuntu#
```

```
root@ip-172-31-21-101:/home/ubuntu# curl -sSfL https://raw.githubusercontent.com/docker/scout-cli/main/install.sh|sh -s -- -b /usr/local/bin
[info] fetching release script for tag='v1.19.0'
[info] using release tag='v1.19.0' version='1.19.0' os='linux' arch='amd64'
[info] installed /usr/local/bin/docker-scout
root@ip-172-31-21-101:/home/ubuntu#
```

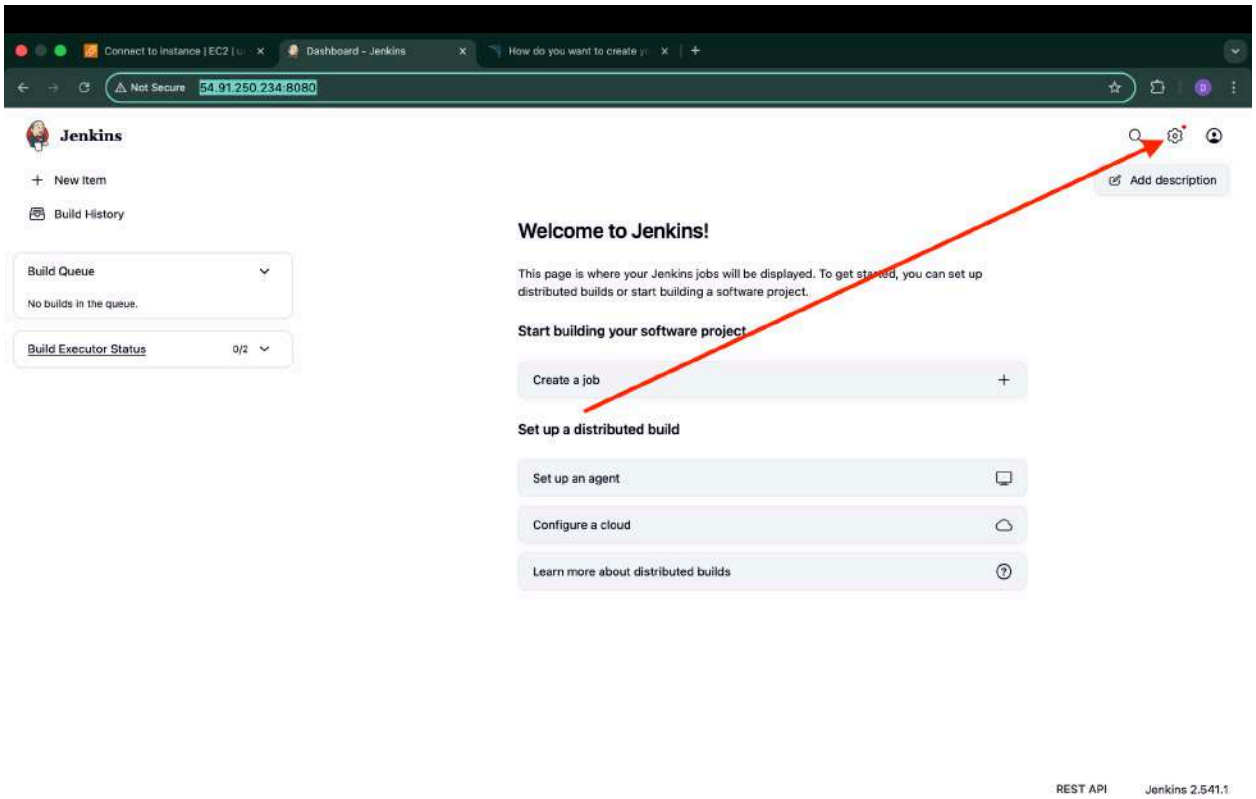
```
root@ip-172-31-21-101:/home/ubuntu# sudo chmod 777 /var/run/docker.sock
root@ip-172-31-21-101:/home/ubuntu#
```

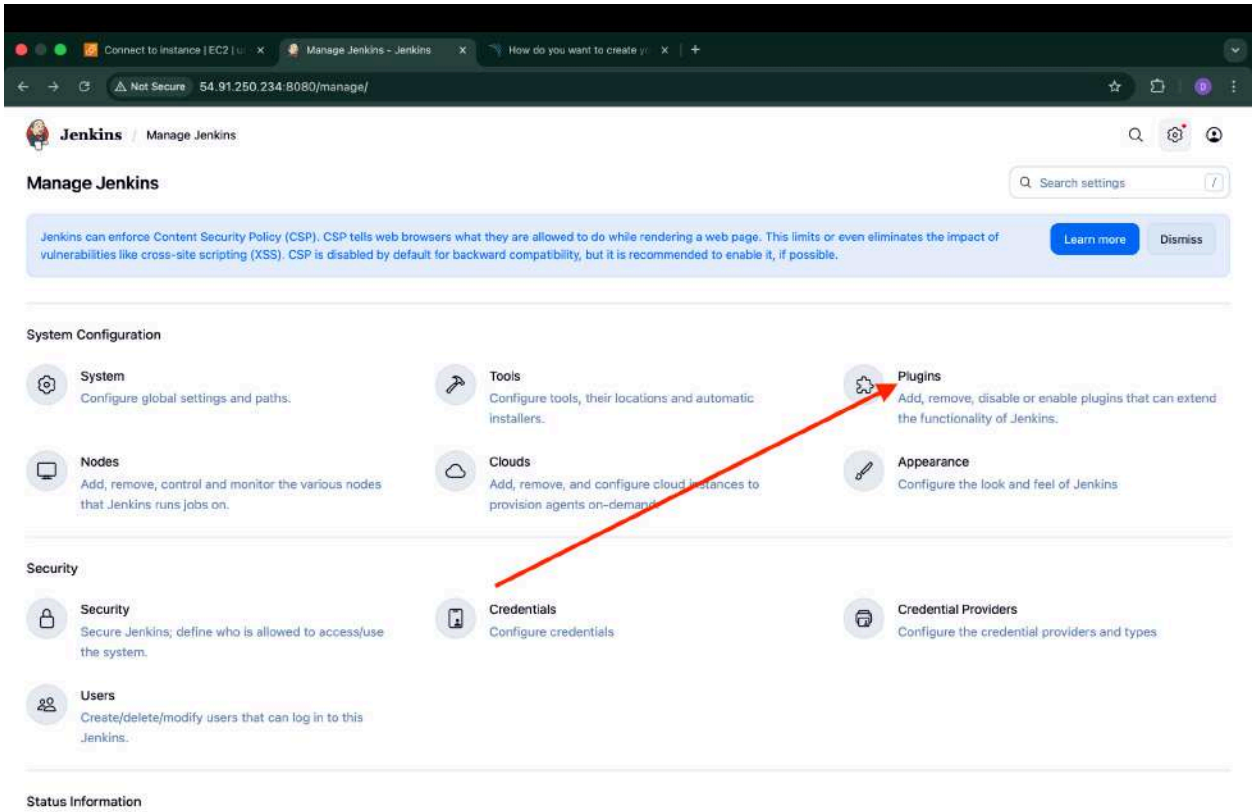
Install SonarQube using Docker

```
ubuntu@ip-172-31-21-101:~$ sudo su
root@ip-172-31-21-101:/home/ubuntu# docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
Unable to find image 'sonarqube:lts-community' locally
lts-community: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
e24a8b9e652f: Pull complete
1df735f481ad: Pull complete
5d5a1fad7028: Pull complete
eb27e3a98da1: Pull complete
c7ad1fe61e07: Pull complete
f3929ce9ef98: Pull complete
60d98d907669: Pull complete
ce2bd68be4fc: Download complete
600f318704dc: Download complete
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:lts-community
40b28e9fc5d140265a1247569b5543ff7afcbc4db5913097874 2e76256043bac
root@ip-172-31-21-101:/home/ubuntu# docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS          PORTS                                          NAMES
40b28e9fc5d1   sonarqube:lts-community   "/opt/sonarqube/dock…"   13 seconds ago   Up 8 seconds    0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp    sonar
root@ip-172-31-21-101:/home/ubuntu#
```



Installation of Plugins in Jenkins





Connect to instance | EC2 | u... | Available plugins - Plugins - | How do you want to create y...

Not Secure 54.91.250.234:8080/manage/pluginManager/available

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Search available plugins

Install

Install	Name ↓	Released	Health
☑	Pipeline: Stage View 2.39 User Interface Pipeline Stage View Plugin.	27 days ago	100
☑	Eclipse Temurin installer 146.v1898676a_f04e Provides an installer for the JDK tool that downloads the Eclipse Temurin™ build based upon OpenJDK from the Adoptium Working Group.	7 mo 8 days ago	96
☑	SonarQube Scanner 2.18.2 External Site/Tool Integrations · Build Reports This plugin allows an easy integration of SonarQube, the open source platform for Continuous Inspection of code quality.	1 mo 29 days ago	84
☑	Docker 1308.vff6e33248305 Cloud Providers · Cluster Management · docker This plugin integrates Jenkins with Docker	2 mo 19 days ago	100
☑	Docker Commons 457.v0f62a_94f11a_3 Library plugins (for use by other plugins) · docker Provides the common shared functionality for various Docker-related plugins.	8 mo 13 days ago	93
☑	Docker Pipeline 634.vedc7242b_eda_7 pipeline · DevOps · Deployment · docker Build and use Docker containers from pipelines. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	3 mo 27 days ago	76

Connect to instance | EC2 | u... | Available plugins - Plugins - | How do you want to create y...

Not Secure 54.91.250.234:8080/manage/pluginManager/available

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

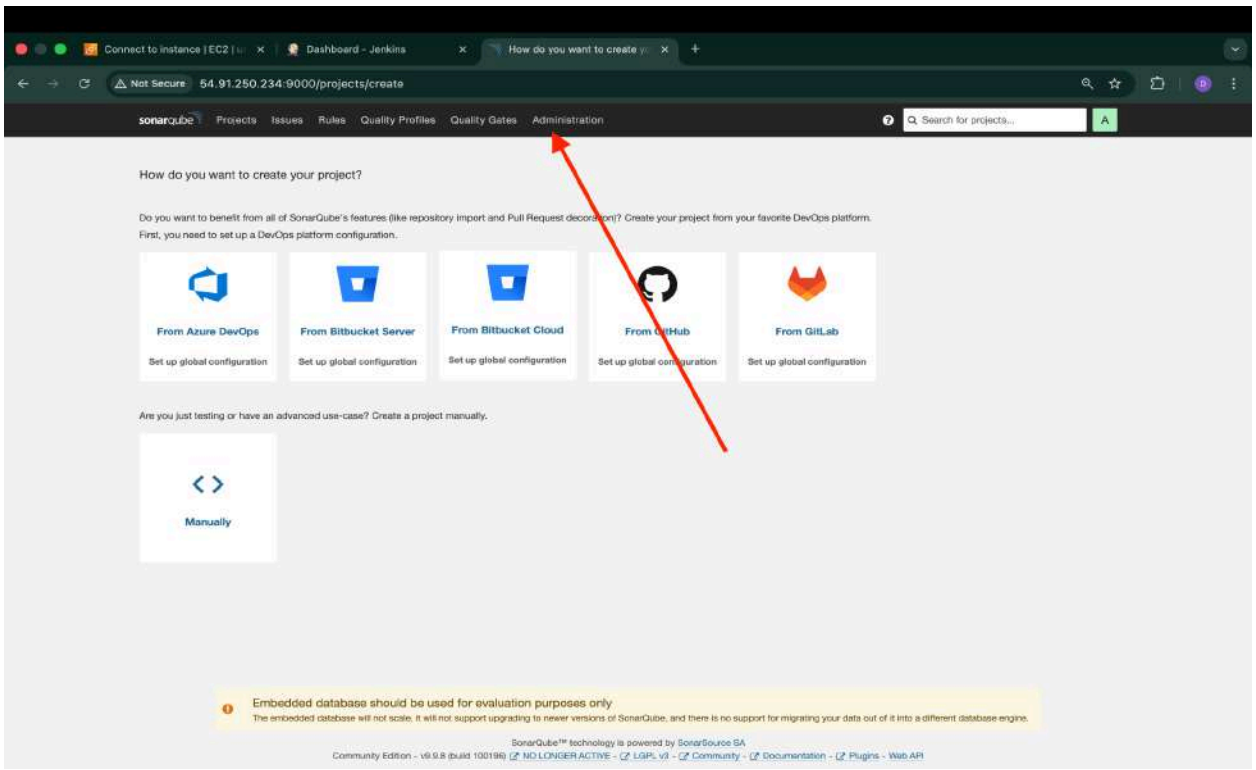
Search available plugins

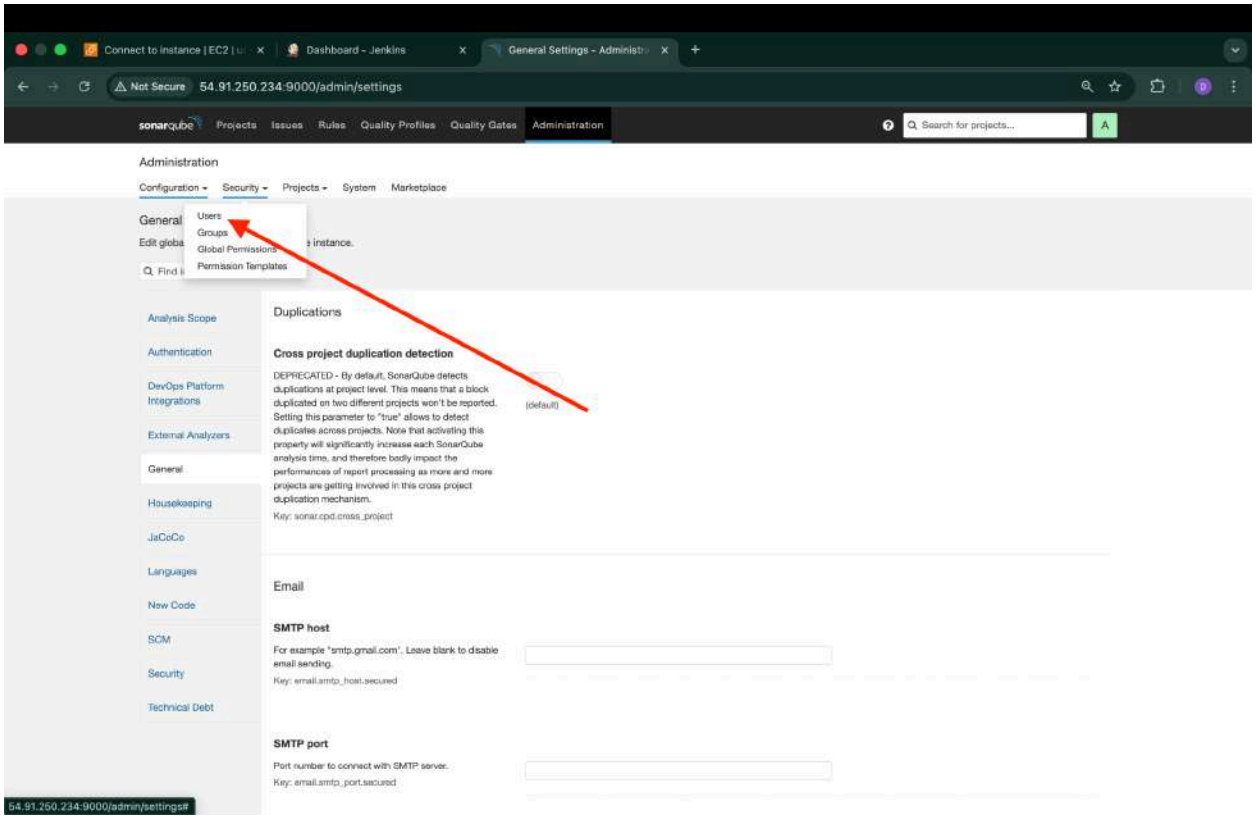
Install

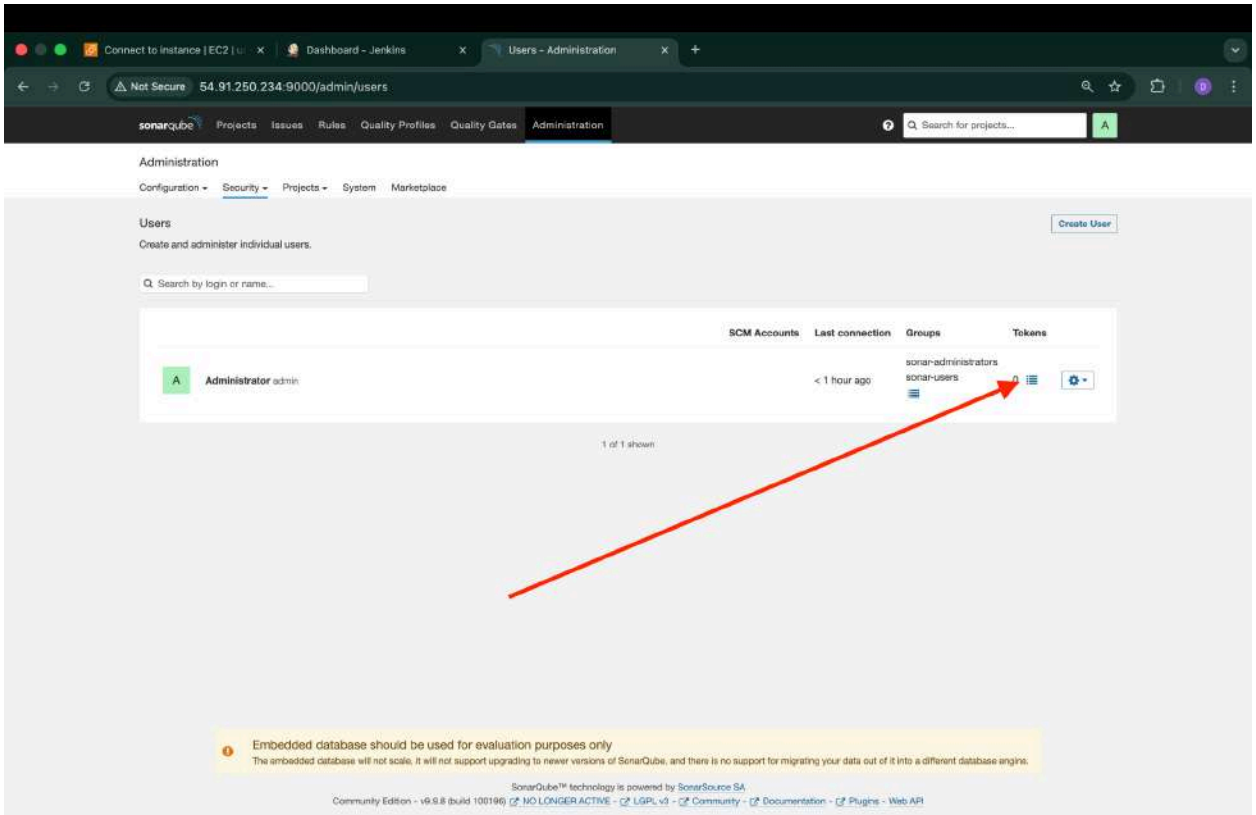
☑	Docker API 3.7.0-133.v93b_8fb_c17a_77 Library plugins (for use by other plugins) · docker This plugin provides docker-java API for other plugins.	2 mo 18 days ago	100
☑	docker-build-step 2.12 Build Tools · docker This plugin allows to add various docker commands to your job as build steps. Warning: This plugin version may not be safe to use. Please review the following security notices: • CSRF vulnerability and missing permission check	1 yr 8 mo ago	62
☑	Email Extension Template 249.v81c6ea_f33f2e Build Notifiers · emailext This plugin allows administrators to create global templates for the Extended Email Publisher. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	19 days ago	80
☑	OWASP Dependency-Check 5.6.2 Security · DevOps · Build Tools · Build Reports This plug-in can independently execute a Dependency-Check analysis and visualize results. Dependency-Check is a utility that identifies project dependencies and checks if there are any known, publicly disclosed, vulnerabilities.	2 mo 25 days ago	100
☑	NodeJS 1.6.6 npm NodeJS Plugin executes NodeJS script as a build step.	2 mo 25 days ago	100
☑	Prometheus metrics 852.v317db_5d17a_b_0		

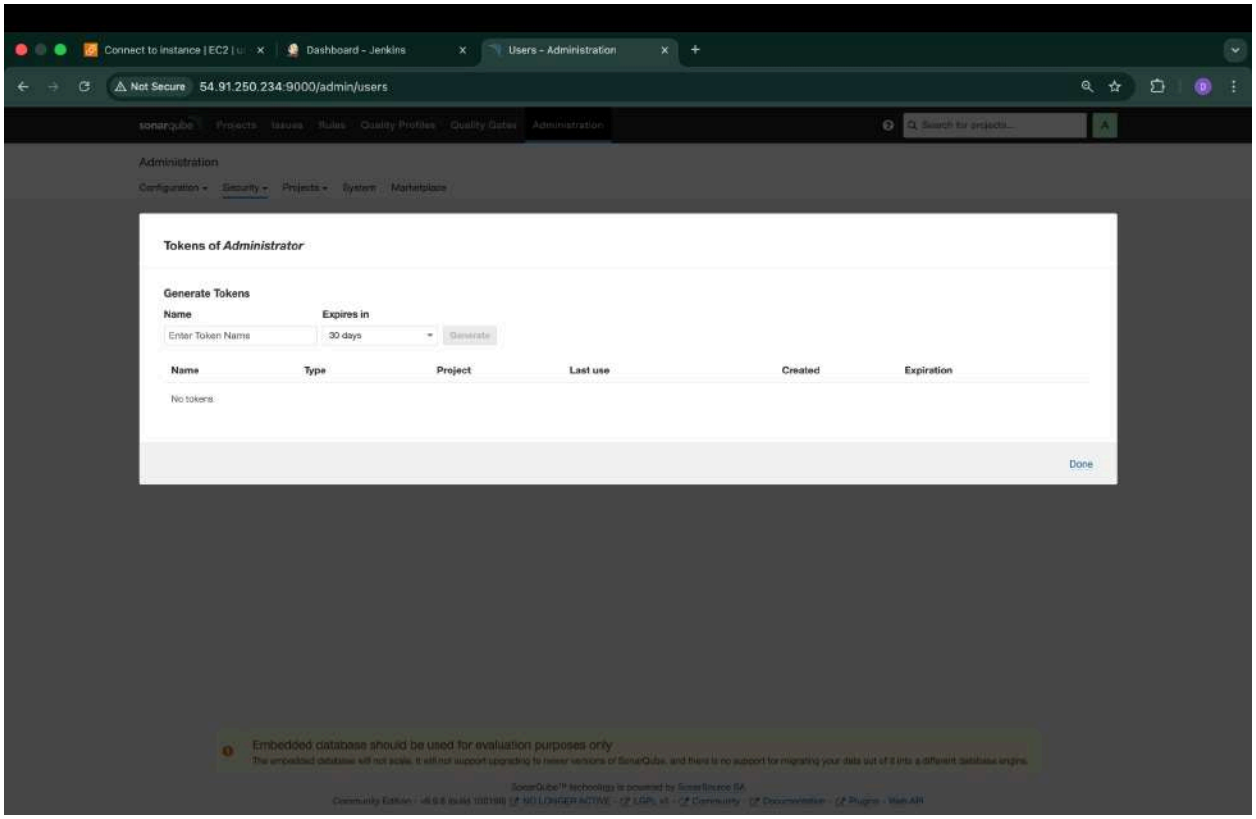
https://plugins.jenkins.io/dependency-check-jenkins-plugin

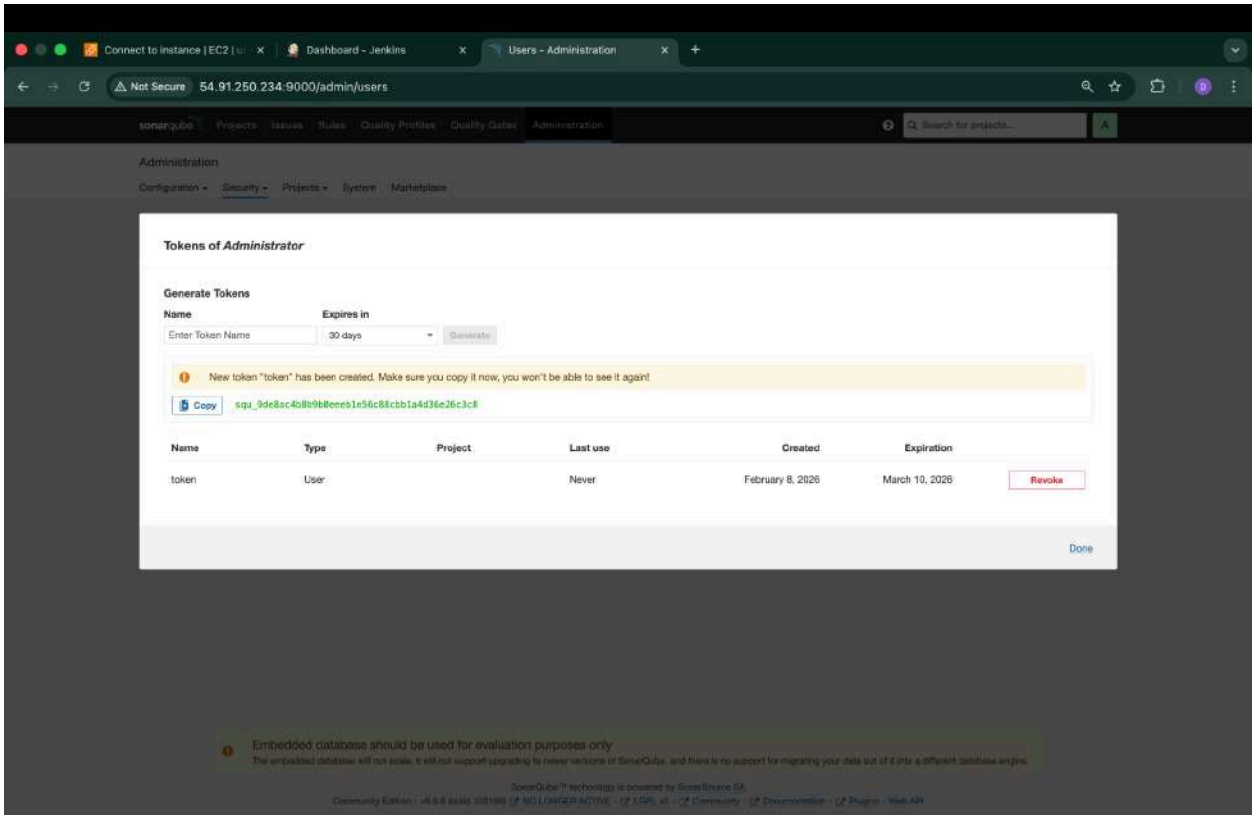
SonarQube configuration in Jenkins



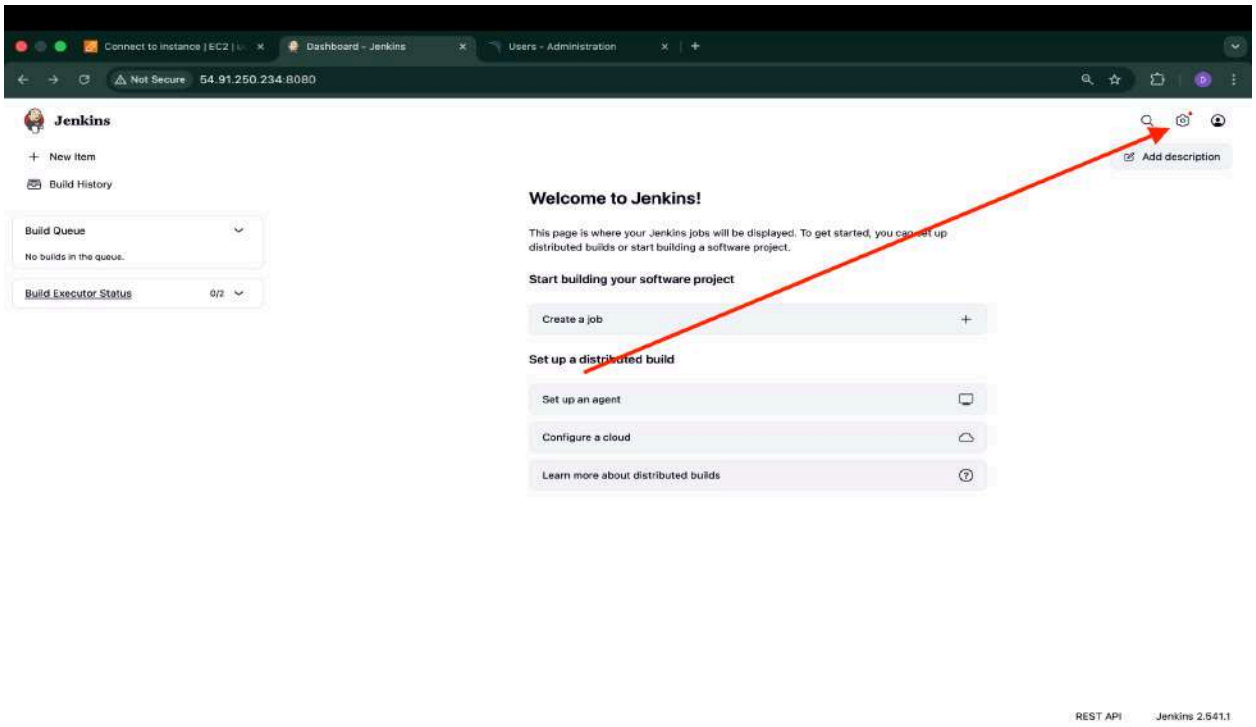


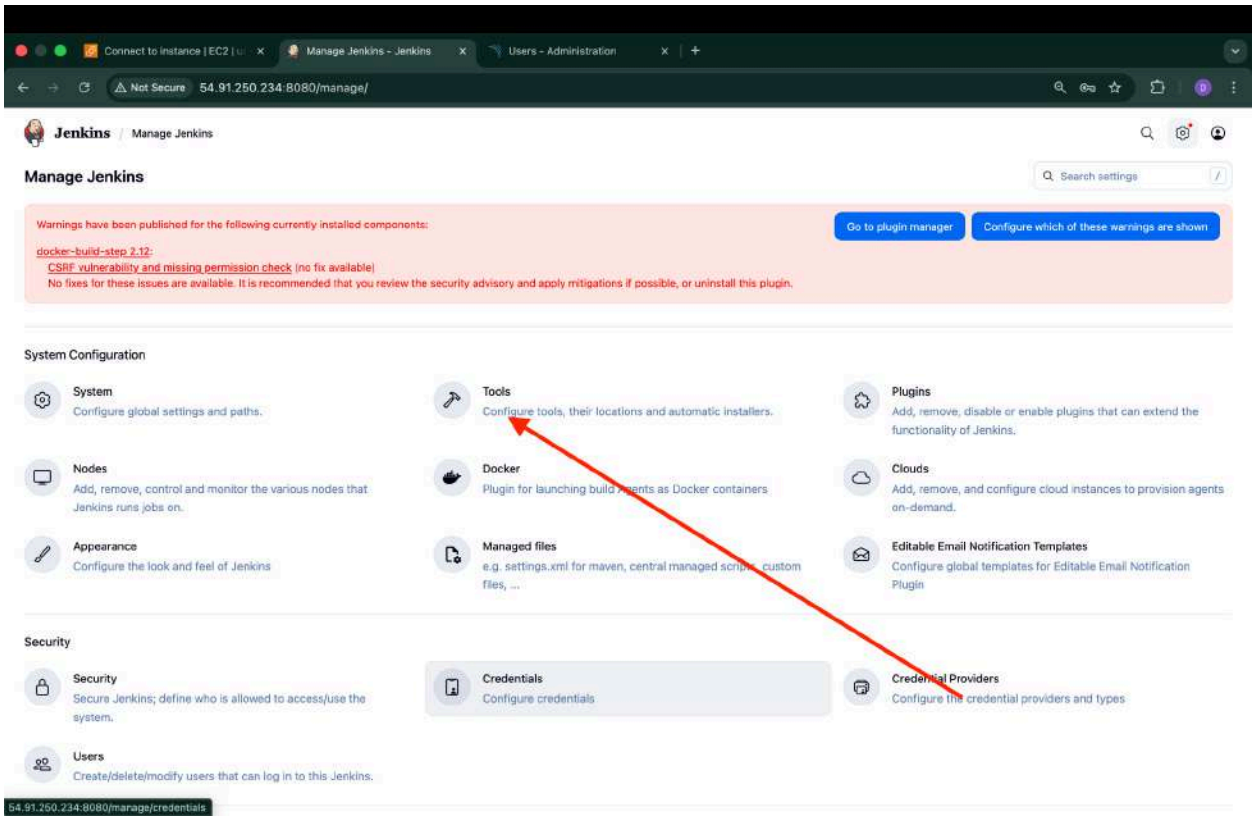

Connect to instance | EC2 |
Dashboard - Jenkins
Users - Administration
Not Secure
54.91.250.234:9000/admin/users
sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
Administration
Configuration
Security
Projects
System
Marketplace
Users
Create and administer individual users.
Create User
Search by login or name...
SCM Accounts
Last connection
Groups
Tokens
Administrator admin
< 1 hour ago
sonar-administrators
sonar-users
1 of 1 shown
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA
Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API


Connect to instance | EC2 |
Dashboard - Jenkins
Users - Administration
Not Secure
54.91.250.234:9000/admin/users
Administration
Tokens of Administrator
Generate Tokens
Name
Expires in
Enter Token Name
30 days
Generate
Name
Type
Project
Last use
Created
Expiration
No tokens
Done
Embedded database should be used for evaluation purposes only


Connect to instance | EC2 | u
Dashboard - Jenkins
Users - Administration
Not Secure
54.91.250.234:9000/admin/users
Administration
Tokens of Administrator
Generate Tokens
Name
Expires in
Enter Token Name
30 days
Generate
New token "token" has been created. Make sure you copy it now, you won't be able to see it again!
Copy
squ_9de8ac4b8b9b0eeeb1e56c88cbb1a4d36e26c3c8
Name
Type
Project
Last use
Created
Expiration
token
User
Never
February 8, 2026
March 10, 2026
Revoke
Done
Embedded database should be used for evaluation purposes only

Tools Configuration in Jenkins





JDK installations

+ Add JDK

JDK

Name

jdk17

Install automatically ?

Install from adoptium.net ?

Version ?

jdk-17.0.8.1+1

+ Add Installer

+ Add JDK

SonarQube Scanner installations

+ Add SonarQube Scanner

SonarQube Scanner

Name

sonar-scanner

Install automatically ?

Install from Maven Central

Version

SonarQube Scanner 8.0.1.6346

+ Add Installer

+ Add SonarQube Scanner

NodeJS installations

+ Add NodeJS

NodeJS

Name

node23

Install automatically ?

Install from nodejs.org

Version

NodeJS 23.1.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

Force 32bit architecture

Global npm packages to install

Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax `packageName@version`

Global npm packages refresh hours

Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache

72

+ Add Installer

Dependency-Check installations

+ Add Dependency-Check

Dependency-Check

Name

DP-Check

Install automatically ?

Install from github.com

Version

dependency-check 12.2.0

+ Add Installer

+ Add Dependency-Check

Docker installations

+ Add Docker

Docker

Name

docker

Install automatically ?

Download from docker.com

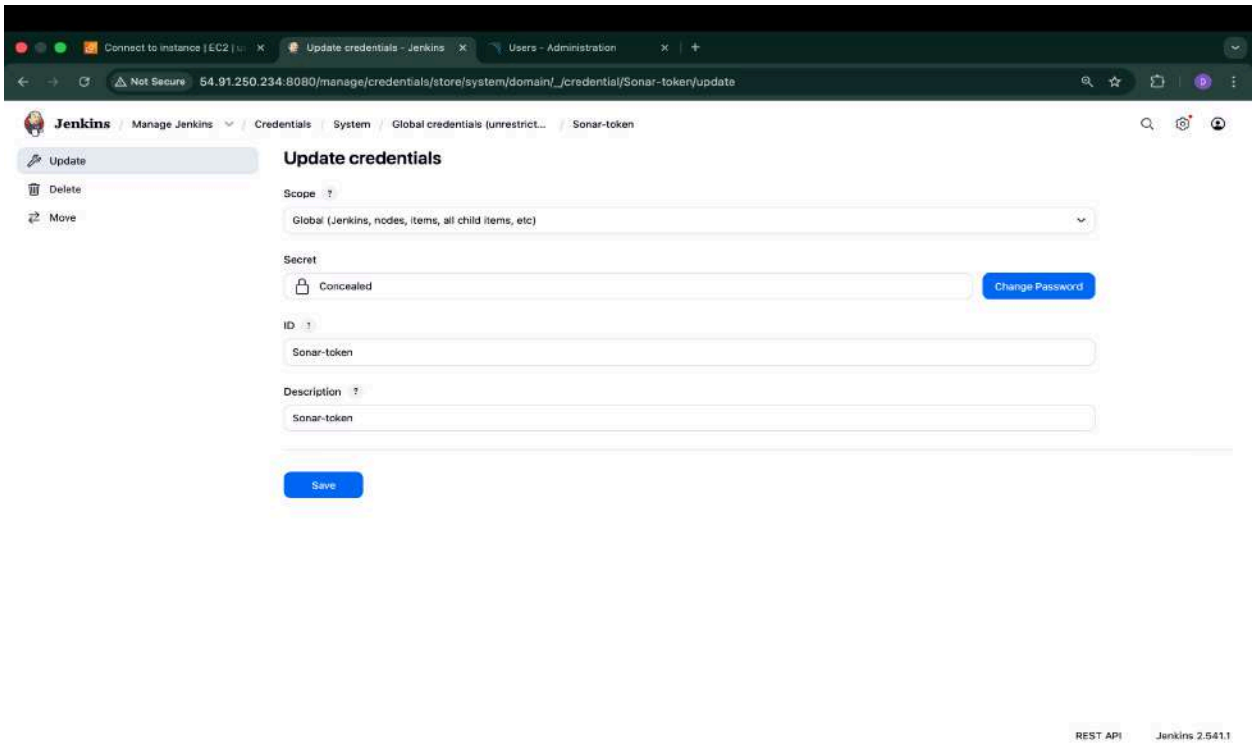
Docker version ?

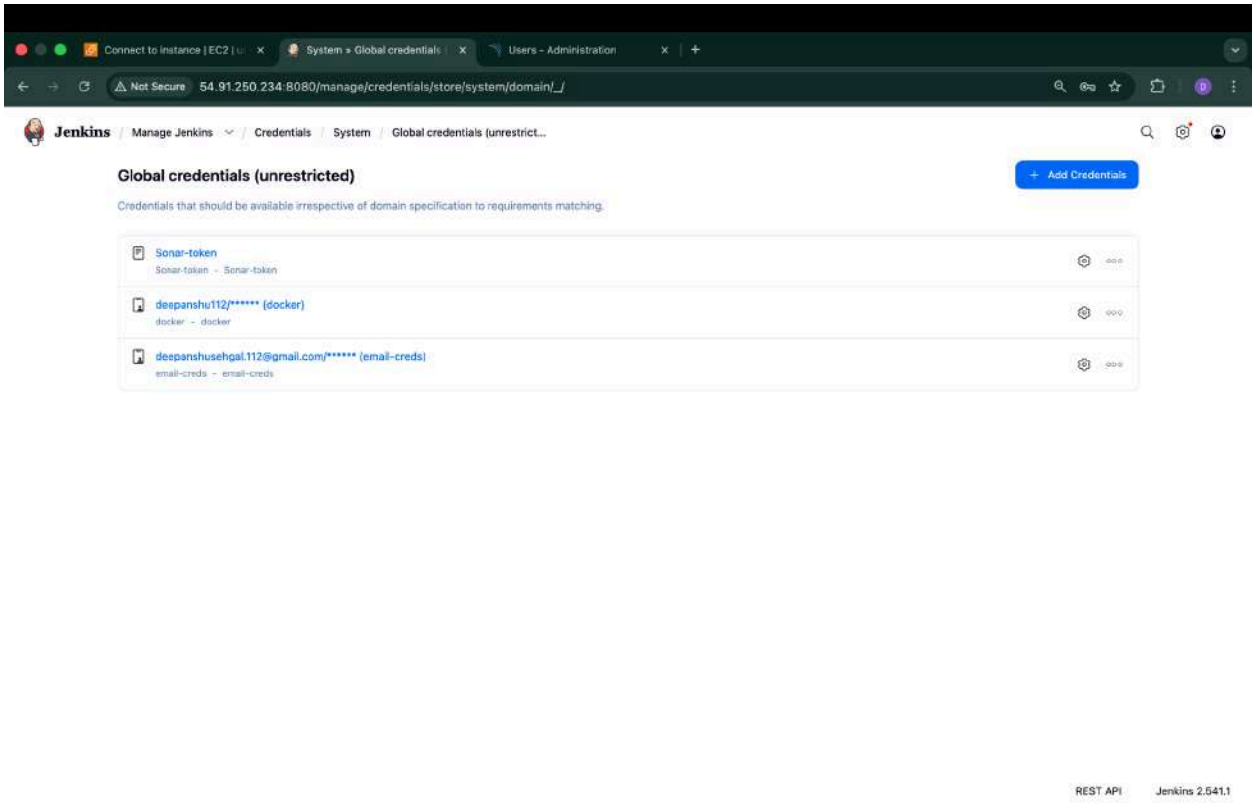
latest

+ Add Installer

+ Add Docker

Configuration of SonarQube Token in Jenkins





Connect to instance | EC2 | u | Manage Jenkins - Jenkins | Users - Administration

Not Secure 54.91.250.234:8080/manage/

Jenkins / Manage Jenkins

Manage Jenkins

Search settings

Warnings have been published for the following currently installed components:

docker-build-step 2.12:
CSRF vulnerability and missing permission check (no fix available)
No fixes for these issues are available. It is recommended that you review the security advisory and apply mitigations if possible, or uninstall this plugin.

Go to plugin manager | Configure which of these warnings are shown

System Configuration

System
Configure global settings and paths.

Tools
Configure tools, their locations and automatic installers.

Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Docker
Plugin for launching build Agents as Docker containers

Clouds
Add, remove, and configure cloud instances to provision agents on-demand.

Appearance
Configure the look and feel of Jenkins

Managed files
e.g. settings.xml for maven, central managed scripts, custom files, ...

Editable Email Notification Templates
Configure global templates for Editable Email Notification Plugin

Security

Security
Secure Jenkins; define who is allowed to access/use the system.

Credentials
Configure credentials

Credential Providers
Configure the credential providers and types

Users
Create/delete/modify users that can log in to this Jenkins.

SonarQube servers

If checked, job administrators will be able to inject a SonarQube server configuration as environment variables in the build.

☐ Environment variables

SonarQube installations
List of SonarQube installations

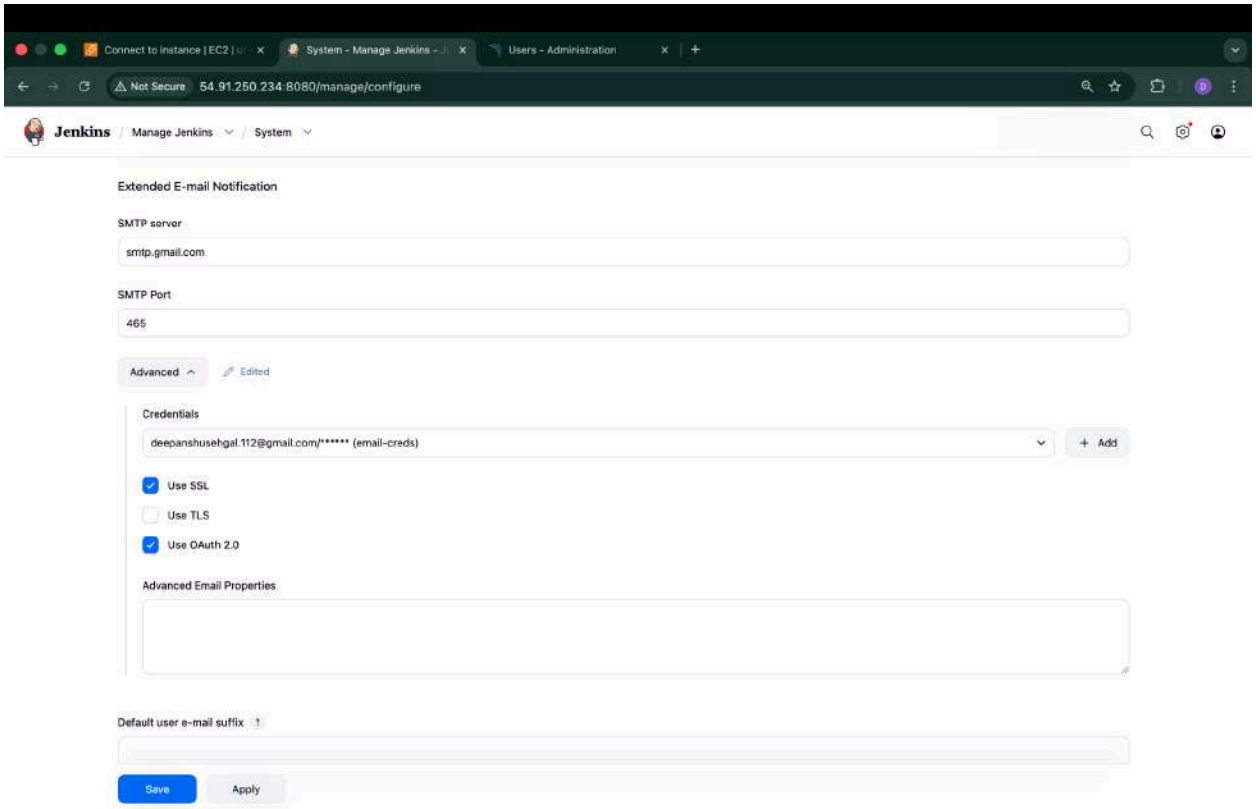
Name
sonar-server

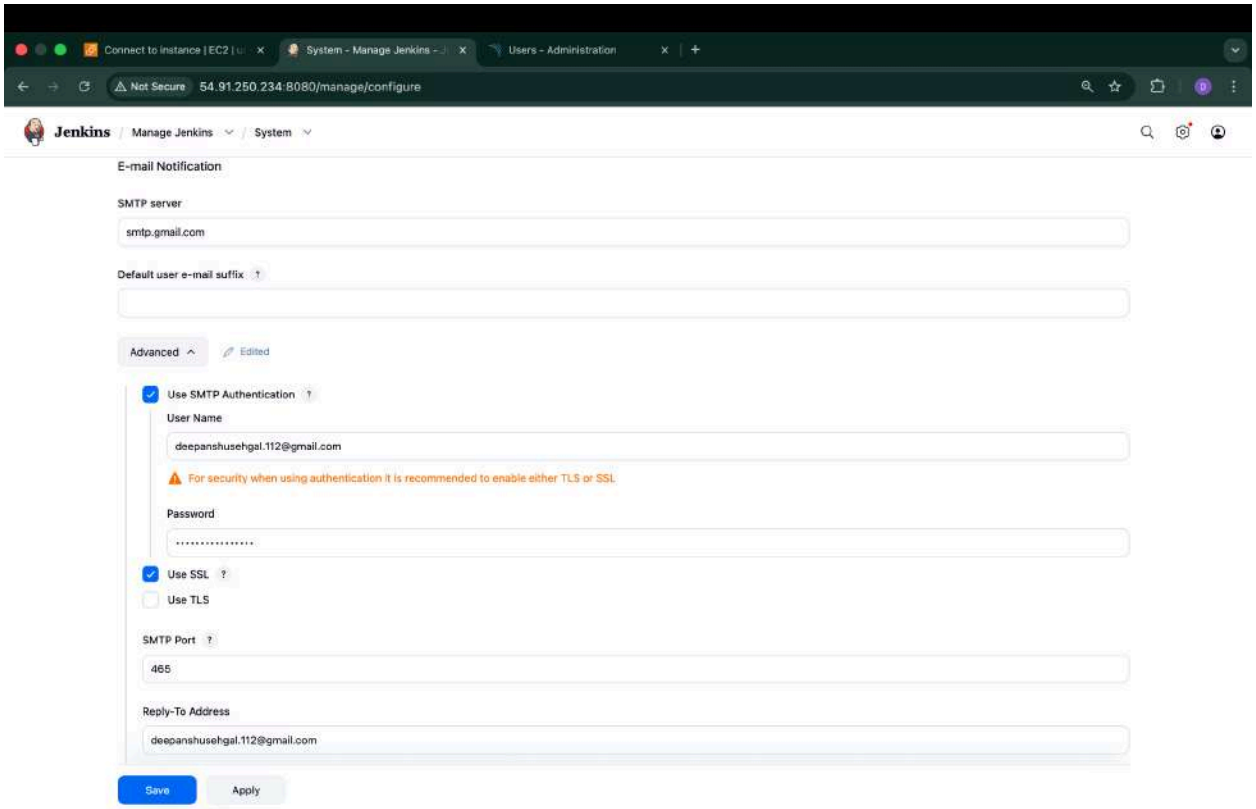
Server URL
Default is http://localhost:9000
http://54.91.250.234:9000

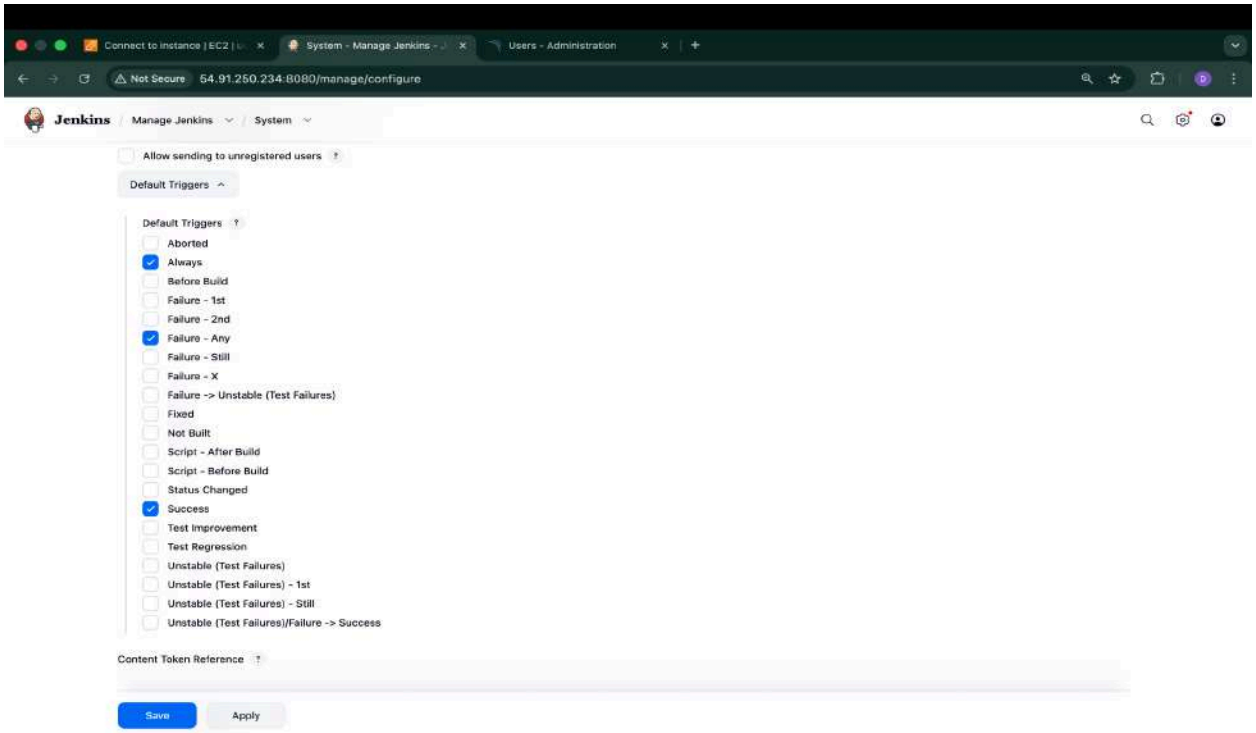
Server authentication token
SonarQube authentication token. Mandatory when anonymous access is disabled.
Sonar-token | + Add

Advanced

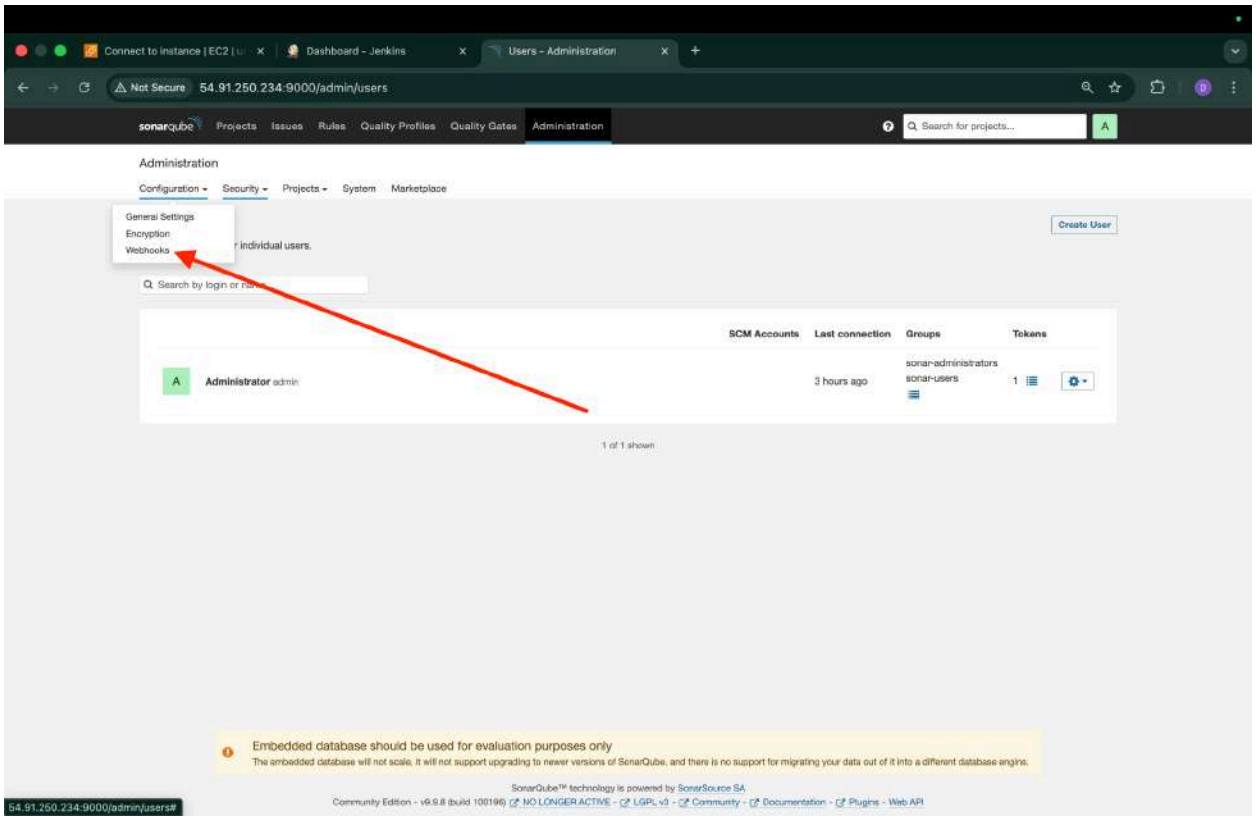
+ Add SonarQube

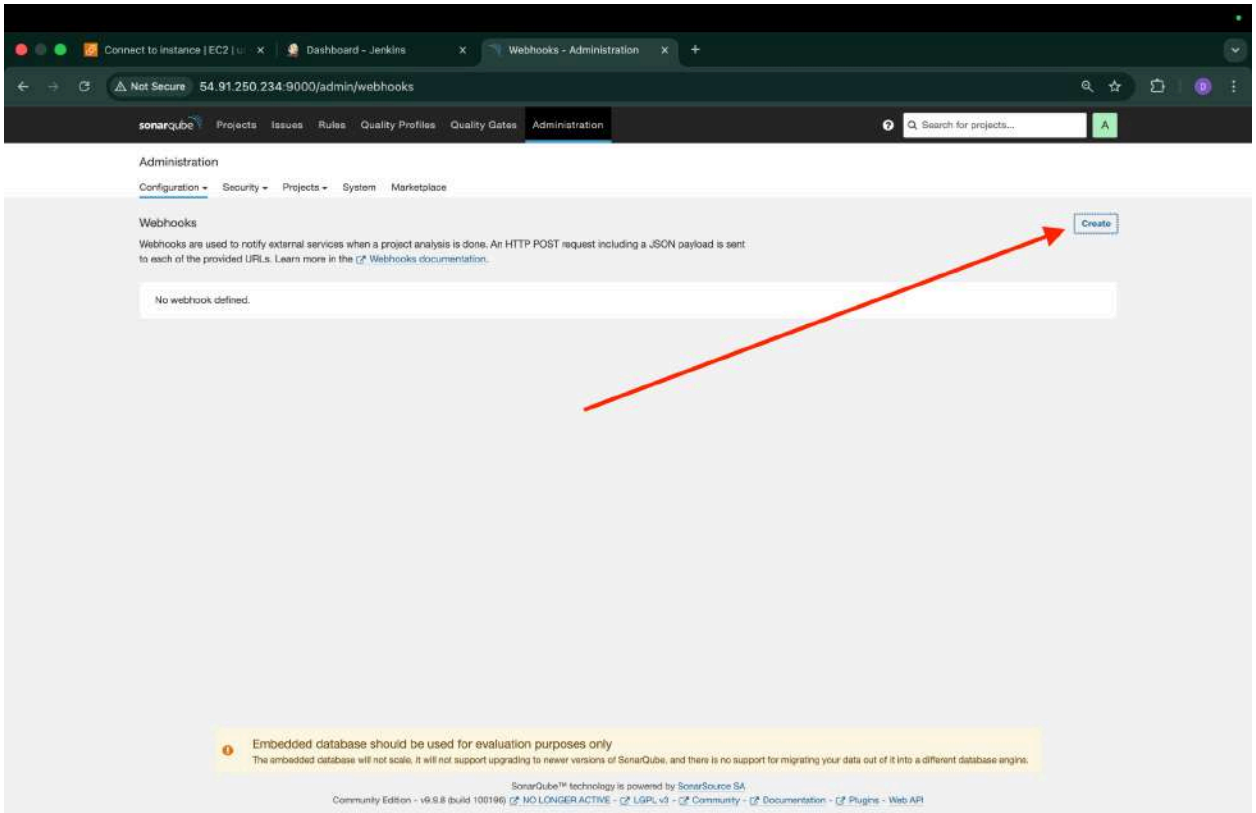

Connect to instance | EC2 |
System - Manage Jenkins -
Users - Administration
Not Secure
54.91.250.234:8080/manage/configure
Jenkins
Manage Jenkins
System
Extended E-mail Notification
SMTP server
smtp.gmail.com
SMTP Port
465
Advanced
Edited
Credentials
deepanshusehgal.112@gmail.com/****** (email-creds)
Add
Use SSL
Use TLS
Use OAuth 2.0
Advanced Email Properties
Default user e-mail suffix
Save
Apply

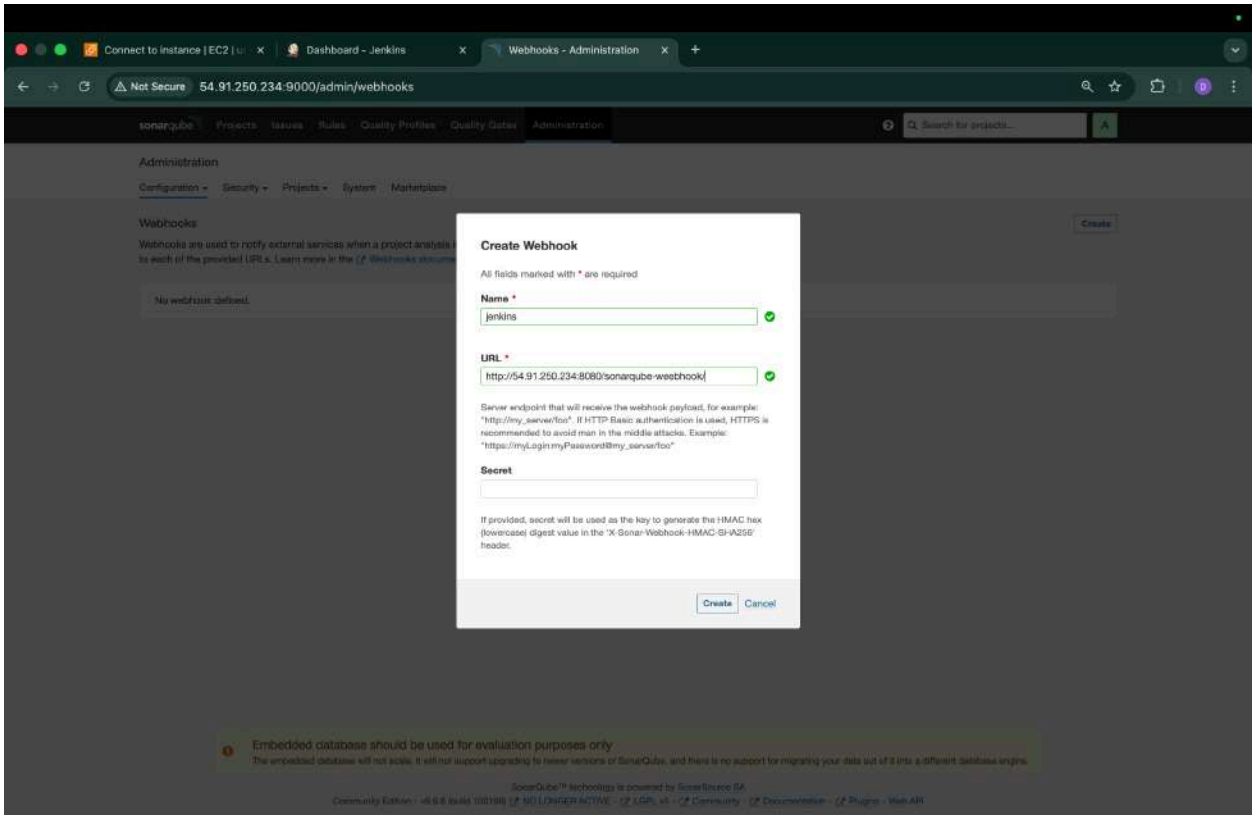

Connect to instance | EC2 |
System - Manage Jenkins -
Users - Administration
Not Secure
54.91.250.234:8080/manage/configure
Jenkins
Manage Jenkins
System
E-mail Notification
SMTP server
smtp.gmail.com
Default user e-mail suffix
Advanced
Edited
Use SMTP Authentication
User Name
deepanshusehgal.112@gmail.com
For security when using authentication it is recommended to enable either TLS or SSL
Password
Use SSL
Use TLS
SMTP Port
465
Reply-To Address
deepanshusehgal.112@gmail.com
Save
Apply

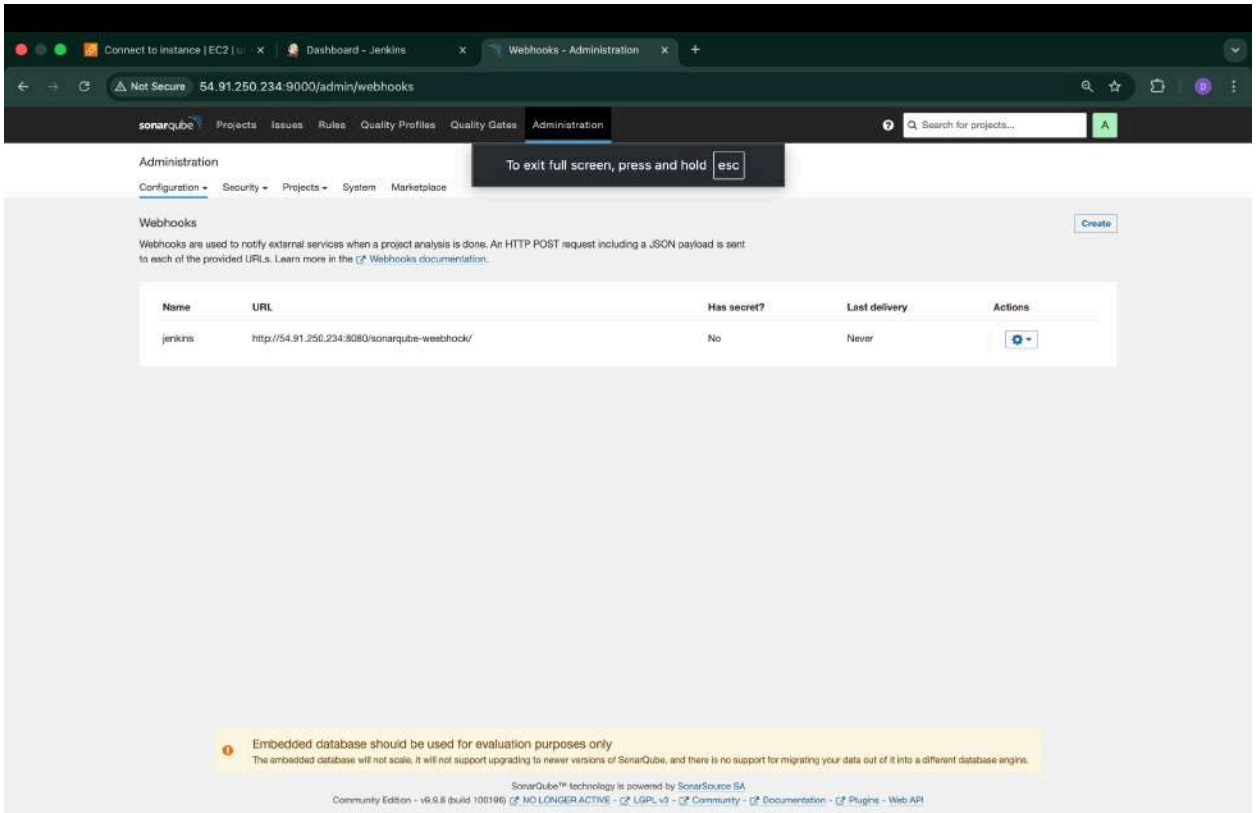


Create webhook in SonarQube




Connect to instance | EC2 |
Dashboard - Jenkins
Webhooks - Administration
Not Secure
54.91.250.234:9000/admin/webhooks
sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
Administration
Configuration
Security
Projects
System
Marketplace
Webhooks
Webhooks are used to notify external services when a project analysis is done. An HTTP POST request including a JSON payload is sent to each of the provided URLs. Learn more in the Webhooks documentation.
Create
No webhook defined.
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA.
Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API


Connect to instance | EC2 |
Dashboard - Jenkins
Webhooks - Administration
Not Secure
54.91.250.234:9000/admin/webhooks
Administration
Webhooks
Create Webhook
All fields marked with * are required
Name *
jenkins
URL *
http://54.91.250.234:8080/sonarqube-webhook/
Server endpoint that will receive the webhook payload, for example: "http://my_server/foo". If HTTP Basic authentication is used, HTTPS is recommended to avoid man in the middle attacks. Example: "https://myLogin:myPassword@my_server/foo"
Secret
If provided, secret will be used as the key to generate the HMAC hex (lowercase) digest value in the 'X-Sonar-Webhook-HMAC-SH-A256' header.
Create
Cancel
Embedded database should be used for evaluation purposes only



Phase 4: Containerization

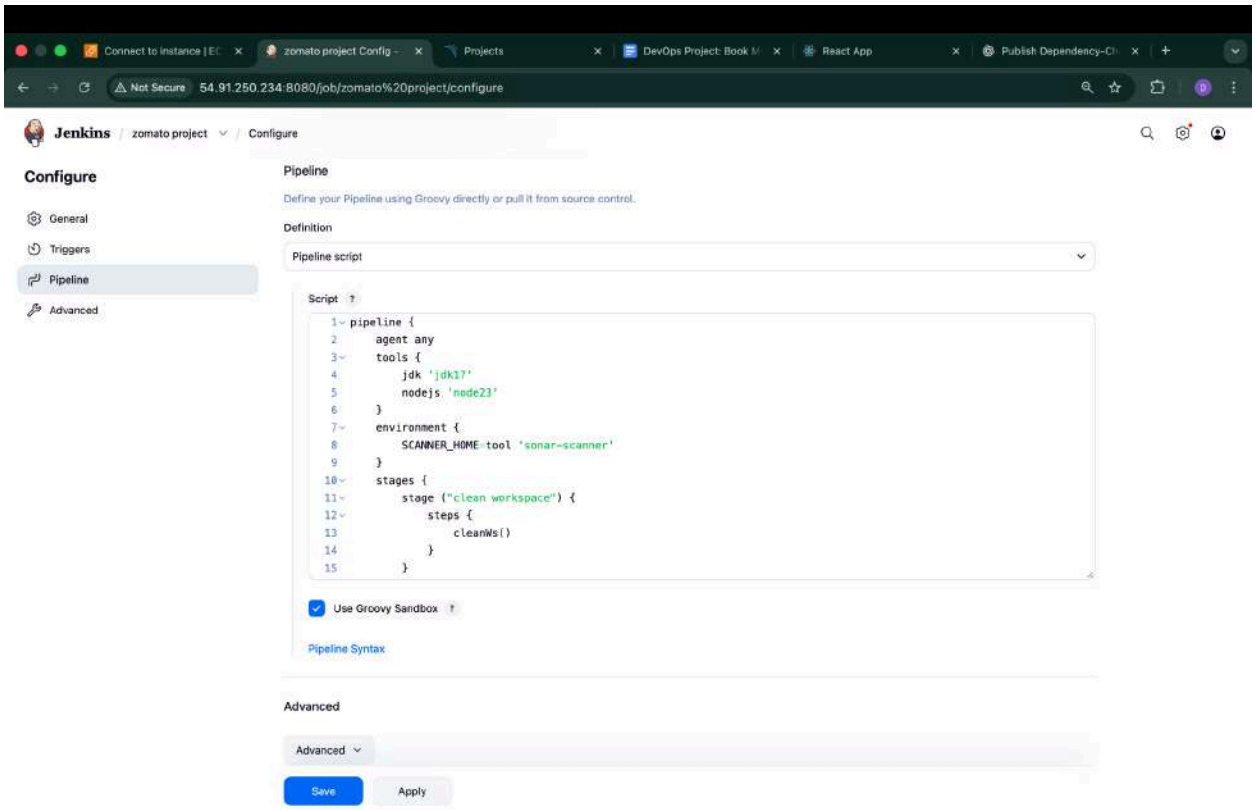
The application was containerized using Docker.

Steps

- Dockerfile created
- Image built locally
- Tagged appropriately
- Pushed to DockerHub
- Deployed as a running container

This enabled portability and consistent deployments.

Now Triger the pipeline

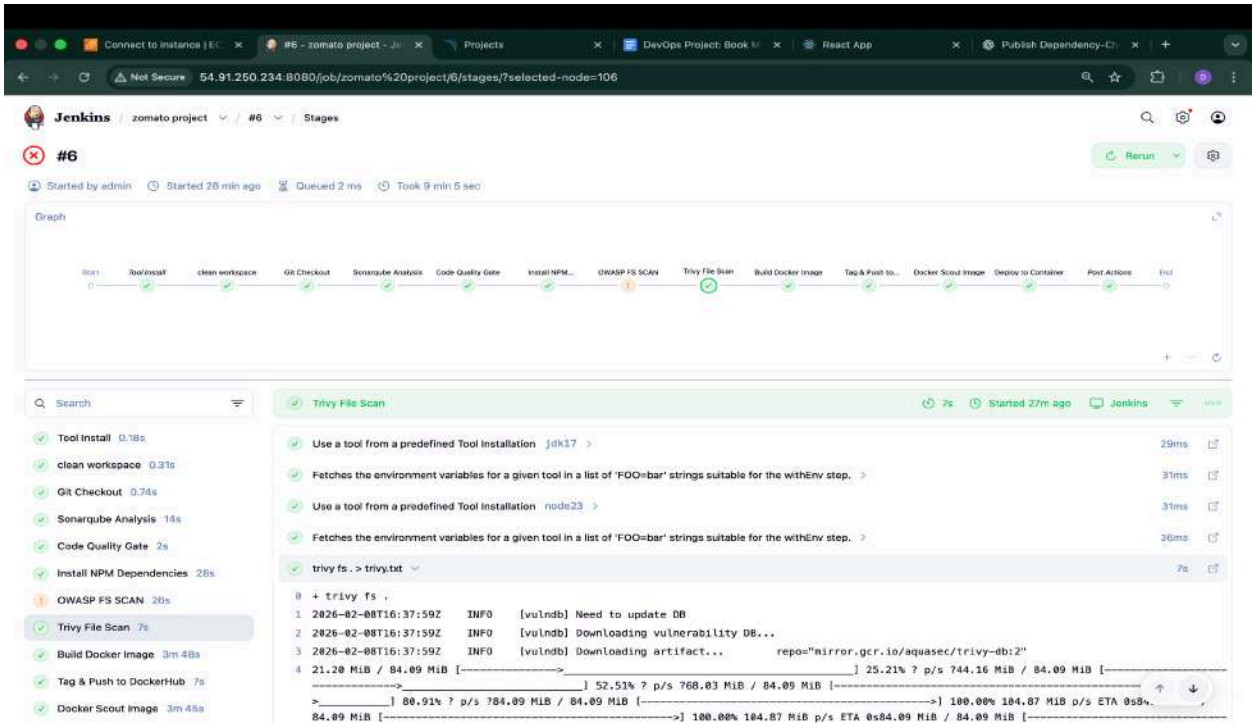


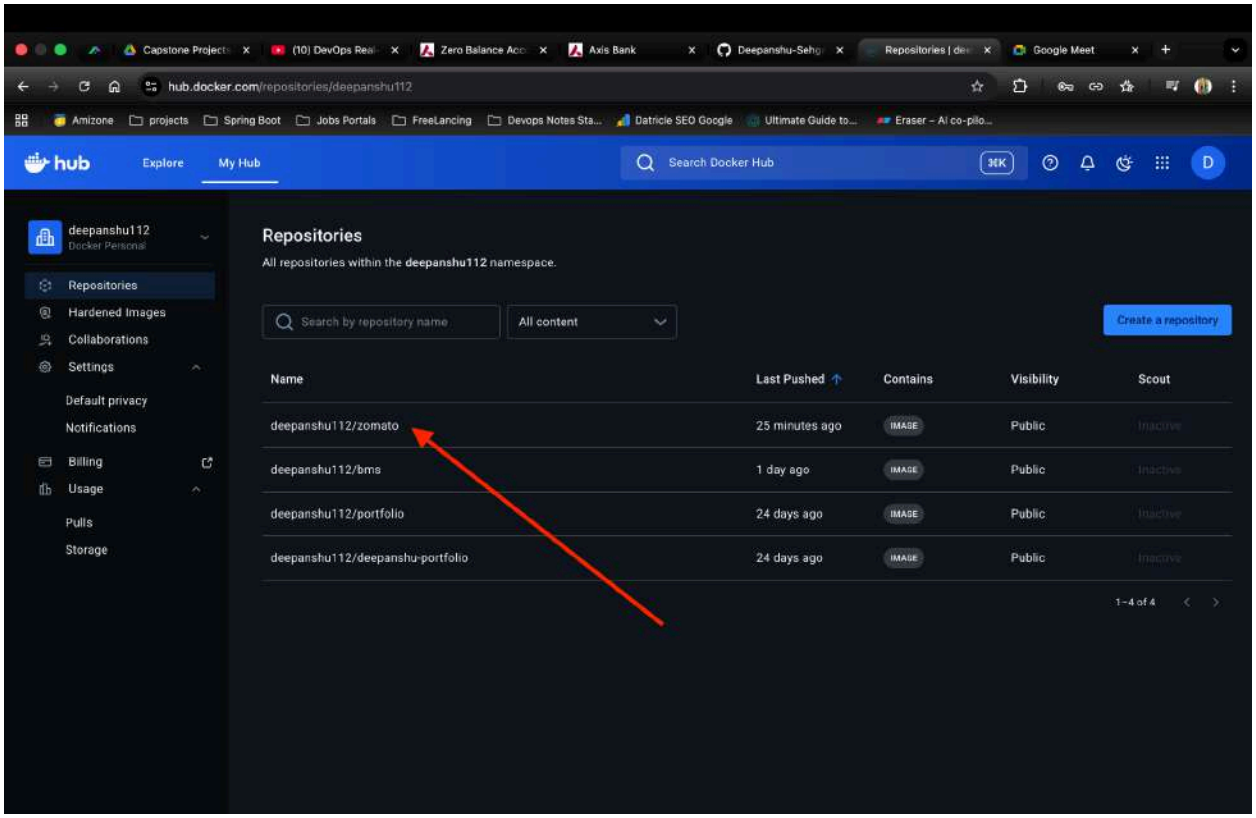
CI/CD Pipeline Workflow

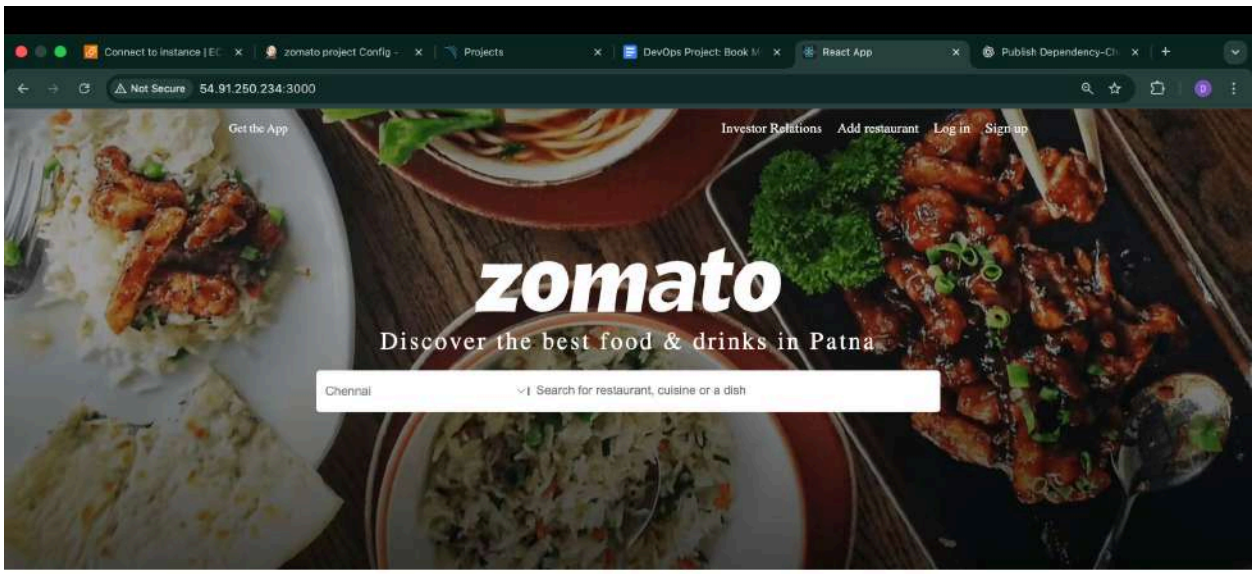
The Jenkins pipeline executed:

1. Clean workspace
2. Git checkout
3. SonarQube analysis
4. Quality Gate validation
5. NPM dependency installation
6. OWASP scan
7. Trivy scan
8. Docker image build
9. Push to DockerHub
10. Docker Scout scan
11. Container deployment
12. Email notification

Fully automated pipeline with zero manual intervention.




hub.docker.com/repositories/deepanshu112
hub
Explore
My Hub
Search Docker Hub
deepanshu112
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories
All repositories within the deepanshu112 namespace.
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
deepanshu112/zomato
25 minutes ago
IMAGE
Public
deepanshu112/bms
1 day ago
IMAGE
Public
deepanshu112/portfolio
24 days ago
IMAGE
Public
deepanshu112/deepanshu-portfolio
24 days ago
IMAGE
Public
1–4 of 4


Not Secure
54.91.250.234:3000
Get the App
Investor Relations
Add restaurant
Log in
Sign up
zomato
Discover the best food & drinks in Patna
Chennai
Search for restaurant, cuisine or a dish

Order Online
Stay home and order to your doorstep
Nightlife and Clubs
Explore the city's top nightlife outlets
Dinning
Views the city's favourite venues

```
root@ip-172-31-21-101:/home/ubuntu# docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED         STATUS         PORTS                                                   NAMES
a0a1ef767110   deepanshu112/zomato:latest "docker-entrypoint.s…"   25 minutes ago  Up 24 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp             zomato
40b28e9fc5d1   sonarqube:lts-community    "/opt/sonarqube/dock…"   8 hours ago     Up 8 hours     0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp             sonar
root@ip-172-31-21-101:/home/ubuntu#
```

```
root@ip-172-31-21-101:/home/ubuntu# docker images

IMAGE                        ID             DISK USAGE   CONTENT SIZE   EXTRA
deepanshu112/zomato:latest   149da4e940b4       1.68GB          250MB    U
sonarqube:lts-community      f709975ab31d       1.02GB          398MB    U
zomato:latest                149da4e940b4       1.68GB          250MB    U
root@ip-172-31-21-101:/home/ubuntu#
```

Phase 5: Monitoring Setup

A separate monitoring server was configured.

Prometheus

- Collected metrics from:
  - Jenkins
  - Node Exporter
  - Kubernetes nodes

Node Exporter

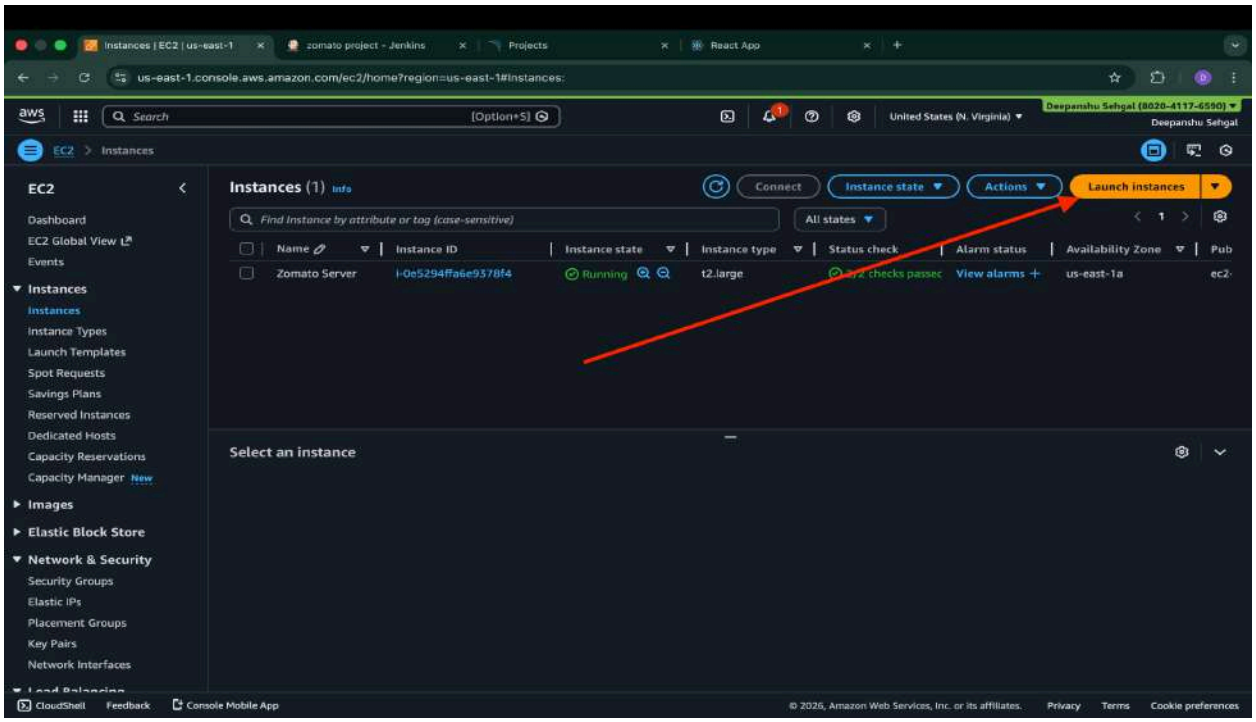
- Provided system-level metrics:
  - CPU usage
  - Memory usage
  - Disk utilization

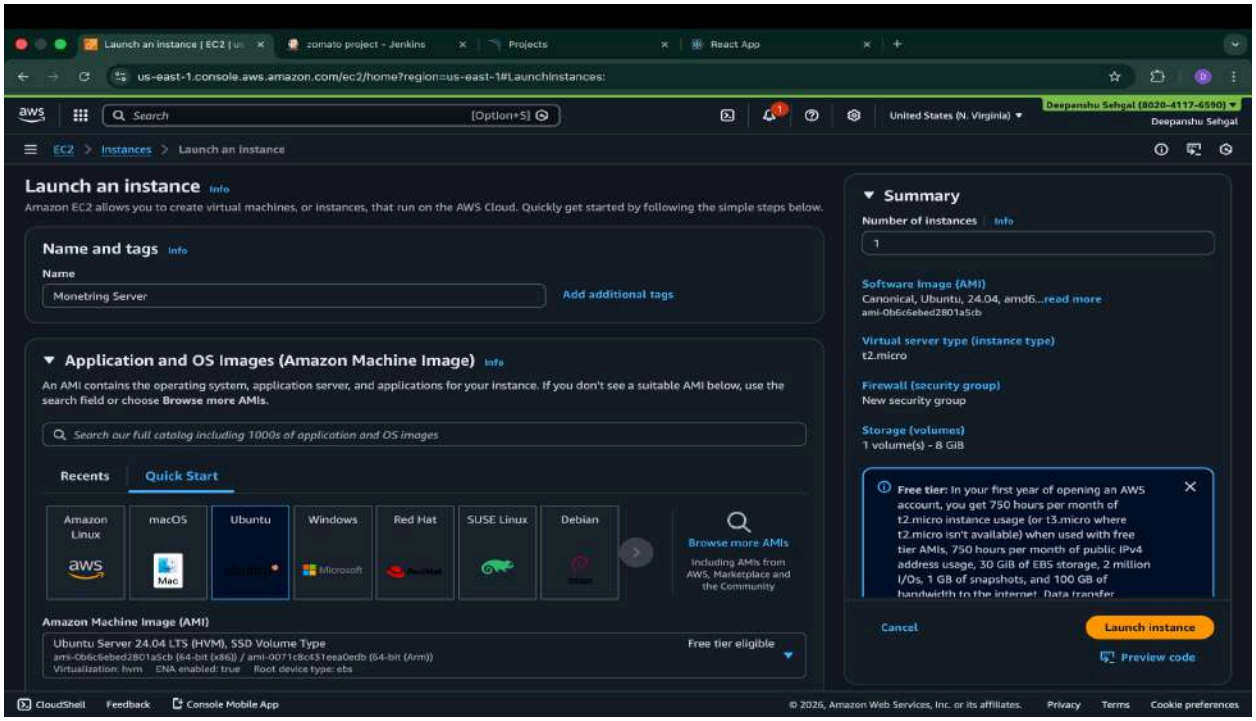
Grafana

- Connected to Prometheus
- Created dashboards
- Visualized metrics in real-time

This provided full observability.

Launch New EC-2 Ubuntu Machine with 29gb of storage





Launch an instance | EC2 | us-east-1 | zomato project - Jenkins | Projects | React App

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws | Search [Option+S] | United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2 > Instances > Launch an instance

Instance type Info | Get advice

Instance type

t2.large
Family: t2 2 vCPU 8 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.1208 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0963 USD per Hour
On-Demand RHEL base pricing: 0.1216 USD per Hour
On-Demand SUSE base pricing: 0.1928 USD per Hour On-Demand Linux base pricing: 0.0928 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

jenkins machine

Create new key pair

Network settings Info

Edit

Network | Info

vpc-0907d014797195056

Subnet | Info

No preference (Default subnet in any availability zone)

Auto-assign public IP | Info

Enable

Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb

Virtual server type (instance type)
t2.large

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer

Cancel | Launch instance | Preview code

CloudShell | Feedback | Console Mobile App

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---

Launch an instance | EC2 | us-east-1 | zomato project - Jenkins | Projects | React App

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws | Search [Option+S] | United States (N. Virginia) | Deepanshu Sehgal (8020-4117-6590) | Deepanshu Sehgal

EC2 > Instances > Launch an instance

Network settings Info

Edit

Network | Info

vpc-0907d014797195056

Subnet | Info

No preference (Default subnet in any availability zone)

Auto-assign public IP | Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Common security groups | Info

Select security groups

launch-wizard-1 sg-0b3a59aa6a111d9b6
VPC: vpc-0907d014797195056

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Configure storage Info

Advanced

1x 29 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb

Virtual server type (instance type)
t2.large

Firewall (security group)
launch-wizard-1

Storage (volumes)
1 volume(s) - 29 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer

Cancel | Launch instance | Preview code

CloudShell | Feedback | Console Mobile App

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```
Last login: Mon Feb  9 11:43:54 on ttys003
deepanshusehgal~$cd Downloads
deepanshusehgal~$ssh -i "jenkins machine.pem" ubuntu@ec2-13-220-59-188.compute-1.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Feb  9 06:14:17 UTC 2026

  System load:  0.02              Processes:             128
  Usage of /:   7.4% of 27.05GB   Users logged in:       0
  Memory usage: 3%                IPv4 address for enX0: 172.31.23.200
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

78 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon Feb  9 06:10:56 2026 from 223.178.212.80
ubuntu@ip-172-31-23-200:~$
```

Installing Prometheus

First, create a dedicated Linux user for Prometheus and download Prometheus

```
root@ip-172-31-23-200:/home/ubuntu# sudo useradd --system --no-create-home --shell /bin/false prometheus
root@ip-172-31-23-200:/home/ubuntu# wget https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-2
.47.1.linux-amd64.tar.gz
```

Extract Prometheus files, move them, and create directories:

```
root@ip-172-31-23-200:/home/ubuntu# tar -xvf prometheus-2.47.1.linux-amd64.tar.gz
prometheus-2.47.1.linux-amd64/
prometheus-2.47.1.linux-amd64/LICENSE
prometheus-2.47.1.linux-amd64/NOTICE
prometheus-2.47.1.linux-amd64/prometheus.yml
prometheus-2.47.1.linux-amd64/consoles/
prometheus-2.47.1.linux-amd64/consoles/prometheus.html
prometheus-2.47.1.linux-amd64/consoles/prometheus-overview.html
prometheus-2.47.1.linux-amd64/consoles/node-cpu.html
prometheus-2.47.1.linux-amd64/consoles/index.html.example
prometheus-2.47.1.linux-amd64/consoles/node.html
prometheus-2.47.1.linux-amd64/consoles/node-disk.html
prometheus-2.47.1.linux-amd64/consoles/node-overview.html
prometheus-2.47.1.linux-amd64/promtool
prometheus-2.47.1.linux-amd64/console_libraries/
prometheus-2.47.1.linux-amd64/console_libraries/prom.lib
prometheus-2.47.1.linux-amd64/console_libraries/menu.lib
prometheus-2.47.1.linux-amd64/prometheus
root@ip-172-31-23-200:/home/ubuntu# cd prometheus-2.47.1.linux-amd64/
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo mkdir -p /data /etc/prometheus
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo mv prometheus promtool /usr/local/bin/
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo mv consoles/ console_libraries/ /etc/prometheus/
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo mv prometheus.yml /etc/prometheus/prometheus.yml
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64#
```

Set ownership for directories:

Create a systemd unit configuration file for Prometheus:

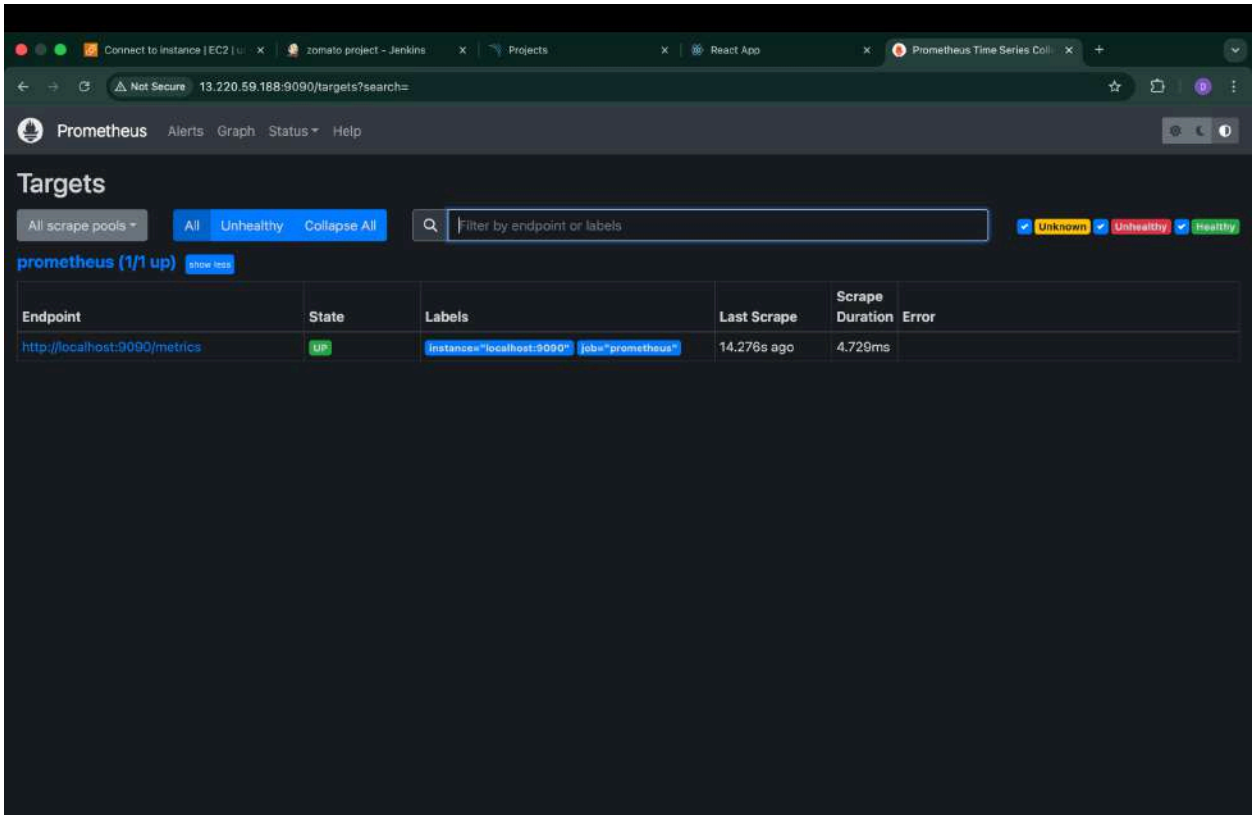
```
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo chown -R prometheus:prometheus /etc/prometheus/ /data/
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo vi /etc/systemd/system/prometheus.service
```

Add the following content to the prometheus.service file:

```
[Unit]
Description=Prometheus
Wants=network-online.target
After=network-online.target

StartLimitIntervalSec=500
StartLimitBurst=5

[Service]
User=prometheus
Group=prometheus
Type=simple
Restart=on-failure
RestartSec=5s
ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/data \
  --web.console.templates=/etc/prometheus/consoles \
  --web.console.libraries=/etc/prometheus/console_libraries \
  --web.listen-address=0.0.0.0:9090 \
  --web.enable-lifecycle

[Install]
WantedBy=multi-user.target
~
~
~
```

Enable and start Prometheus:

```
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo systemctl start prometheus
root@ip-172-31-23-200:/home/ubuntu/prometheus-2.47.1.linux-amd64# sudo systemctl status prometheus
● prometheus.service - Prometheus
     Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-02-09 06:19:52 UTC; 5s ago
   Main PID: 1768 (prometheus)
      Tasks: 7 (limit: 9498)
     Memory: 16.3M (peak: 16.6M)
        CPU: 59ms
     CGroup: /system.slice/prometheus.service
             └─1768 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/data --web.console.templates=/et>

Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.416Z caller=head.go:681 level=info component=tsdb msg="On-disk memory >
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.416Z caller=head.go:689 level=info component=tsdb msg="Replaying WAL, >
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.416Z caller=head.go:760 level=info component=tsdb msg="WAL segment loa>
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.416Z caller=head.go:797 level=info component=tsdb msg="WAL replay comp>
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.420Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.421Z caller=main.go:1048 level=info msg="TSDB started"
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.421Z caller=main.go:1229 level=info msg="Loading configuration file" f>
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.425Z caller=main.go:1266 level=info msg="Completed loading of configur>
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.425Z caller=main.go:1009 level=info msg="Server is ready to receive we>
Feb 09 06:19:52 ip-172-31-23-200 prometheus[1768]: ts=2026-02-09T06:19:52.425Z caller=manager.go:1009 level=info component="rule manager" msg="S>
lines 1-20/20 (END)
```



Installing Node Exporter

Create a system user for Node Exporter and download Node Exporter:

```
root@ip-172-31-23-200:~# sudo useradd --system --no-create-home --shell /bin/false node_exporter
root@ip-172-31-23-200:~# wget https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
--2026-02-09 06:22:06--  https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.114.3
Connecting to github.com (github.com)|140.82.114.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/5509b569-5c34-471e-8598-c05c0733bb7f?sp=r&sv=2018-
11-09&sr=b&spr=https&se=2026-02-09T06%3A55%3A53Z&rscd=attachment%3B+filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&rsct=application%2Foctet-st
ream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-09T05%3A55%3A35Z&ske=2026-02-09T06%3A55%3A
53Z&sks=b&skv=2018-11-09&sig=7KFn7d8gkgZ9RivG7TObovssjX%2BlvW8IN5dAL8NcWuE%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIi
wiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MDYxODQyNiwibmJmIjoxNzcwNjE4MTI2LCJwYXRoIjoicmVsZWFzZWF
zc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.Cjl3KIMPu0v0PT8rTuy60x_FyRJgyEKVW9ut4sXXOOk&response-content-disposition=attachment%3B%20file
name%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-02-09 06:22:06--  https://release-assets.githubusercontent.com/github-production-release-asset/9524057/5509b569-5c34-471e-8598-c05c0733bb7
f?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-09T06%3A55%3A53Z&rscd=attachment%3B+filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&rsct=applica
tion%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-09T05%3A55%3A35Z&ske=2026-0
2-09T06%3A55%3A53Z&sks=b&skv=2018-11-09&sig=7KFn7d8gkgZ9RivG7TObovssjX%2BlvW8IN5dAL8NcWuE%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJ
naXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MDYxODQyNiwibmJmIjoxNzcwNjE4MTI2LCJwYXRo
IjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.Cjl3KIMPu0v0PT8rTuy60x_FyRJgyEKVW9ut4sXXOOk&response-content-disposition=attac
hment%3B%20filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10368103 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.6.1.linux-amd64.tar.gz'

node_exporter-1.6.1.linux-amd64.tar. 100%[===================================================================>]   9.89M  --.-KB/s    in 0.09s

2026-02-09 06:22:06 (111 MB/s) - 'node_exporter-1.6.1.linux-amd64.tar.gz' saved [10368103/10368103]

root@ip-172-31-23-200:~#
```

Extract Node Exporter files, move the binary, and clean up:

```
root@ip-172-31-23-200:~# ls
node_exporter-1.6.1.linux-amd64.tar.gz  snap
root@ip-172-31-23-200:~# tar -xvf node_exporter-1.6.1.linux-amd64.tar.gz
node_exporter-1.6.1.linux-amd64/
node_exporter-1.6.1.linux-amd64/NOTICE
node_exporter-1.6.1.linux-amd64/node_exporter
node_exporter-1.6.1.linux-amd64/LICENSE
root@ip-172-31-23-200:~# sudo mv node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/bin/
root@ip-172-31-23-200:~# rm -rf node_exporter*
root@ip-172-31-23-200:~#
```

Create a systemd unit configuration file for Node Exporter:

sudo vi /etc/systemd/system/node_exporter.service

```
[Unit]
Description=Node Exporter
Wants=network-online.target
After=network-online.target

StartLimitIntervalSec=500
StartLimitBurst=5

[Service]
User=node_exporter
Group=node_exporter
Type=simple
Restart=on-failure
RestartSec=5s
ExecStart=/usr/local/bin/node_exporter --collector.logind

[Install]
WantedBy=multi-user.target
~
~
~
```

Enable and start Node Exporter:

```
root@ip-172-31-23-200:~# sudo vi /etc/systemd/system/node_exporter.service
root@ip-172-31-23-200:~# sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.
root@ip-172-31-23-200:~# sudo systemctl start node_exporter
root@ip-172-31-23-200:~# sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
     Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-02-09 06:24:28 UTC; 7s ago
   Main PID: 1860 (node_exporter)
      Tasks: 4 (limit: 9498)
     Memory: 2.5M (peak: 2.7M)
        CPU: 7ms
     CGroup: /system.slice/node_exporter.service
             └─1860 /usr/local/bin/node_exporter --collector.logind

Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=thermal_zone
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=time
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=timex
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=udp_queues
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=uname
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=vmstat
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=xfs
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=node_exporter.go:117 level=info collector=zfs
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=tls_config.go:274 level=info msg="Listening on" address>
Feb 09 06:24:28 ip-172-31-23-200 node_exporter[1860]: ts=2026-02-09T06:24:28.793Z caller=tls_config.go:277 level=info msg="TLS is disabled." htt>
lines 1-20/20 (END)
```

Configure Prometheus Plugin Integration

```
root@ip-172-31-23-200:~# cd /etc/prometheus/
root@ip-172-31-23-200:/etc/prometheus# ls
console_libraries  consoles  prometheus.yml
root@ip-172-31-23-200:/etc/prometheus# sudo vi prometheus.yml
```

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: 'node_exporter'
    static_configs:
      - targets: ['13.220.59.188:9100']

  - job_name: 'jenkins'
    metrics_path: '/prometheus'
    static_configs:
      - targets: ['54.91.250.234:8080']
~
~
~
~
~
-- INSERT --                                                    37,20         All
```

Check the validity of the configuration file:

```
root@ip-172-31-23-200:/etc/prometheus# promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
  SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax

root@ip-172-31-23-200:/etc/prometheus#
```

Reload the Prometheus configuration without restarting:

```
root@ip-172-31-23-200:/etc/prometheus# curl -X POST http://localhost:9090/-/reload
root@ip-172-31-23-200:/etc/prometheus#
```

Connect to instance | EC2 | u | zomato project - Jenkins | Projects | React App | Prometheus Time Series Coll

Not Secure 13.220.59.188:9090/targets?search=#pool-jenkins

Prometheus Alerts Graph Status Help

Targets

All scrape pools | All | Unhealthy | Collapse All | Filter by endpoint or labels | Unknown | Unhealthy | Healthy

jenkins (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://54.91.250.234:8080/prometheus	UP	instance="54.91.250.234:8080" job="jenkins"	19.999s ago	10.962ms	

node_exporter (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://13.220.59.188:9100/metrics	UP	instance="13.220.59.188:9100" job="node_exporter"	20.514s ago	12.962ms	

prometheus (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	16.321s ago	3.822ms	

Install Grafana

Install Grafana on Monitoring Server;

Step 1: Install Dependencies:
First, ensure that all necessary dependencies are installed:

```
root@ip-172-31-23-200:/etc/prometheus# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
root@ip-172-31-23-200:/etc/prometheus# sudo apt-get install -y apt-transport-https software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.49.3).
software-properties-common set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 78 not upgraded.
Need to get 3970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3970 B]
Fetched 3970 B in 0s (281 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 71752 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Setting up apt-transport-https (2.8.3) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-23-200:/etc/prometheus#
```

Step 2: Add the GPG Key:
Add the GPG key for Grafana:

```
root@ip-172-31-23-200:/etc/prometheus# cd
root@ip-172-31-23-200:~# wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
root@ip-172-31-23-200:~#
```

Step 3: Add Grafana Repository:
Add the repository for Grafana stable releases:

```
root@ip-172-31-23-200:~# echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a /etc/apt/sources.list.d/grafana.list
deb https://packages.grafana.com/oss/deb stable main
root@ip-172-31-23-200:~#
```

Step 4: Update and Install Grafana:

Update the package list and install Grafana:

```
root@ip-172-31-23-200:/etc/prometheus# cd
root@ip-172-31-23-200:~# wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
root@ip-172-31-23-200:~# echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a /etc/apt/sources.list.d/grafana.list
deb https://packages.grafana.com/oss/deb stable main
root@ip-172-31-23-200:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://packages.grafana.com/oss/deb stable InRelease [7660 B]
Get:6 https://packages.grafana.com/oss/deb stable/main amd64 Packages [462 kB]
Fetched 469 kB in 1s (765 kB/s)
Reading package lists... Done
W: https://packages.grafana.com/oss/deb/dists/stable/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRE
CATION section in apt-key(8) for details.
root@ip-172-31-23-200:~# sudo apt-get -y install grafana
```

Step 5: Enable and Start Grafana Service:

To automatically start Grafana after a reboot, enable the service:

```
root@ip-172-31-23-200:~# sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
root@ip-172-31-23-200:~#
```

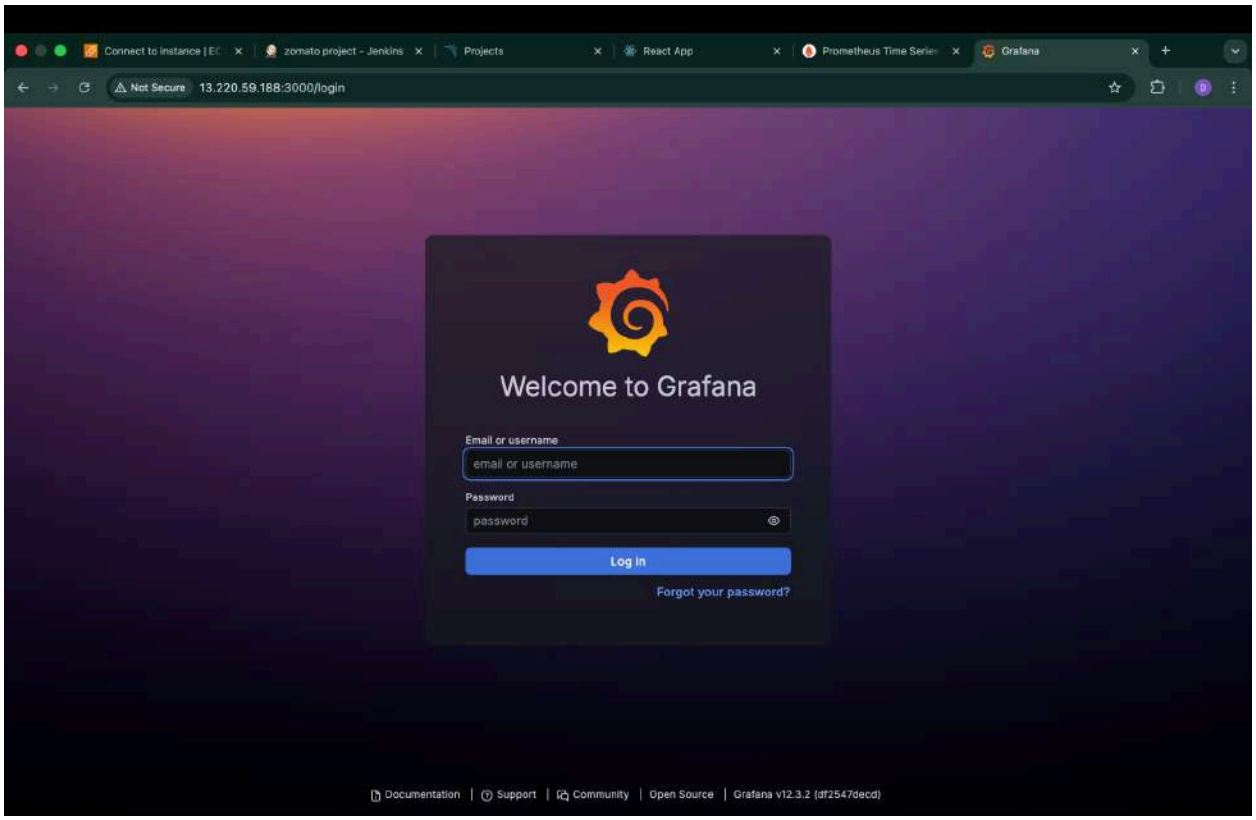
Start Grafana:

Step 6: Check Grafana Status:

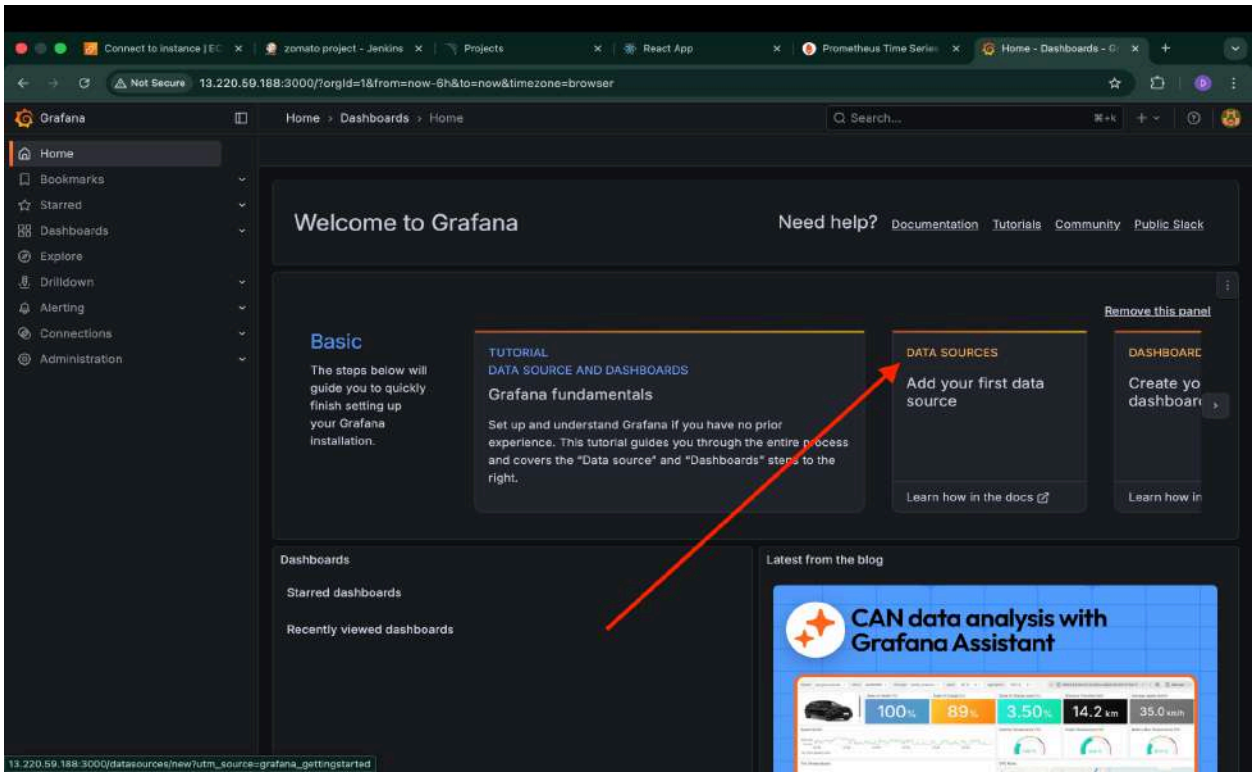
Verify the status of the Grafana service to ensure it's running correctly:

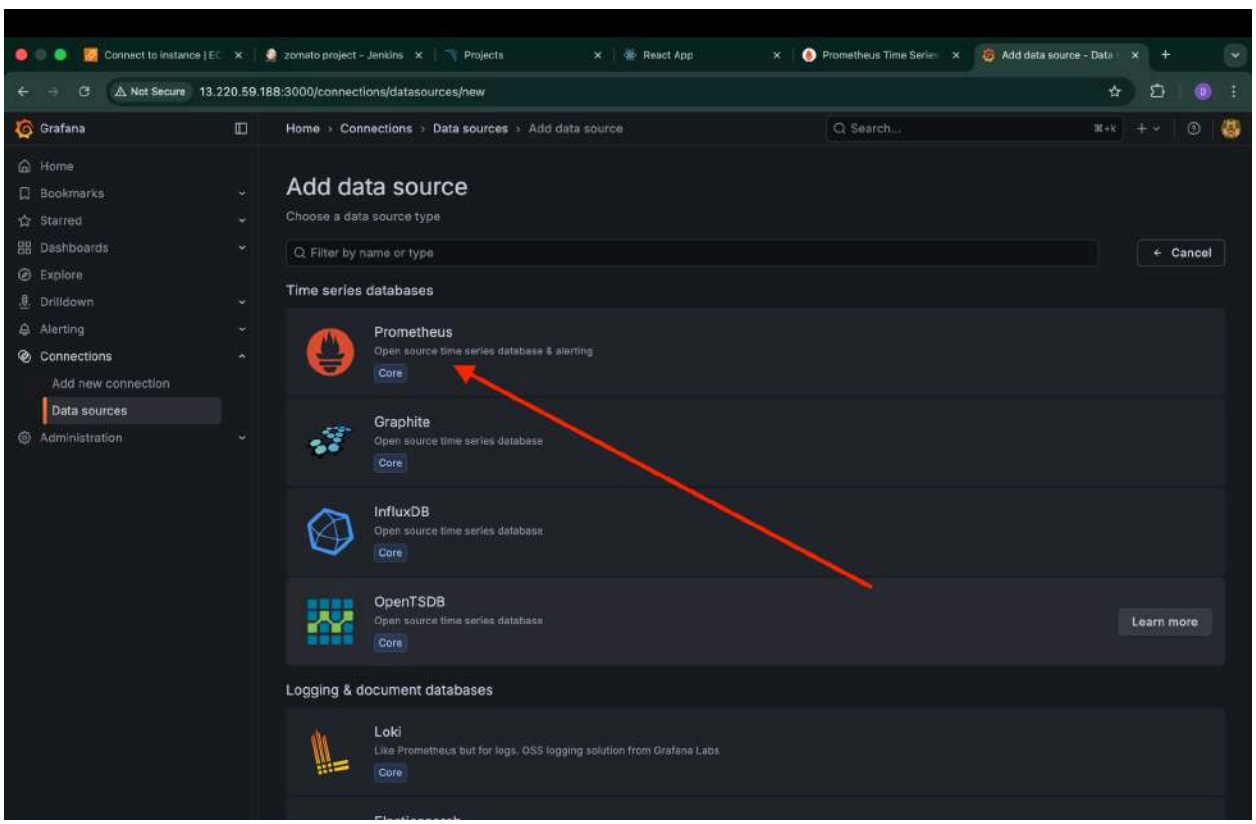
```
root@ip-172-31-23-200:~# sudo systemctl start grafana-server
root@ip-172-31-23-200:~# sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
     Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-02-09 06:41:45 UTC; 24s ago
       Docs: http://docs.grafana.org
   Main PID: 3148 (grafana)
      Tasks: 11 (limit: 9498)
     Memory: 149.4M (peak: 149.8M)
        CPU: 3.932s
     CGroup: /system.slice/grafana-server.service
             └─3148 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packagin>

Feb 09 06:41:53 ip-172-31-23-200 grafana[3148]: logger=plugin.backgroundinstaller t=2026-02-09T06:41:53.791779433Z level=info msg="Installing pl>
Feb 09 06:41:53 ip-172-31-23-200 grafana[3148]: logger=plugin.installer t=2026-02-09T06:41:53.933034007Z level=info msg="Installing plugin" plug>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=installer.fs t=2026-02-09T06:41:54.022055483Z level=info msg="Downloaded and extracted gr>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugins.registration t=2026-02-09T06:41:54.079258114Z level=info msg="Plugin registered" >
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugin.backgroundinstaller t=2026-02-09T06:41:54.079305617Z level=info msg="Plugin succes>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugin.backgroundinstaller t=2026-02-09T06:41:54.079332274Z level=info msg="Installing pl>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugin.installer t=2026-02-09T06:41:54.204193753Z level=info msg="Installing plugin" plug>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=installer.fs t=2026-02-09T06:41:54.290360846Z level=info msg="Downloaded and extracted gr>
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugins.registration t=2026-02-09T06:41:54.352477918Z level=info msg="Plugin registered" >
Feb 09 06:41:54 ip-172-31-23-200 grafana[3148]: logger=plugin.backgroundinstaller t=2026-02-09T06:41:54.352534252Z level=info msg="Plugin succes>
lines 1-21/21 (END)
```

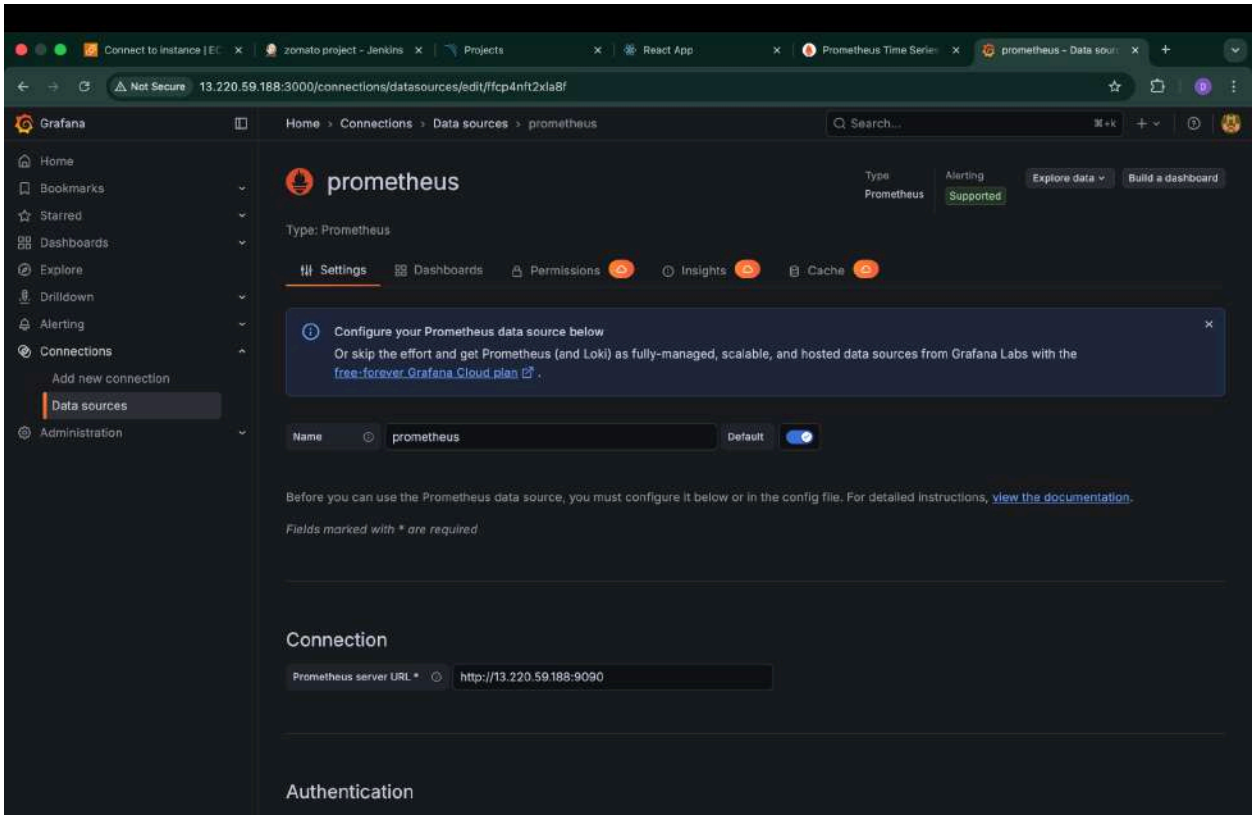
Step 7: Access Grafana Web Interface:
The default port for Grafana is 3000

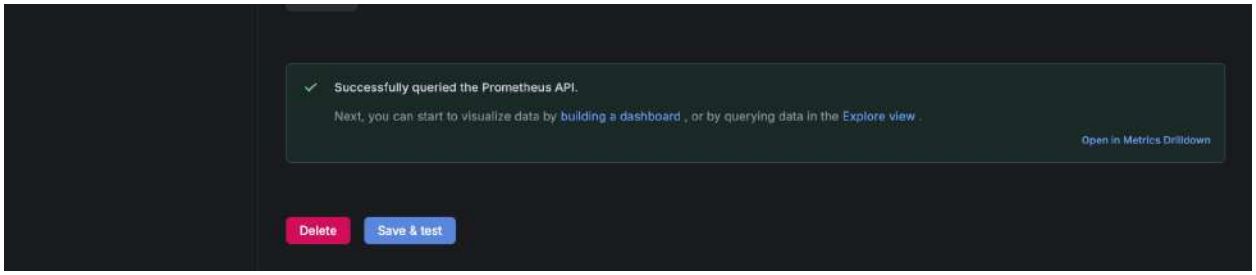


Now we can add the data source;

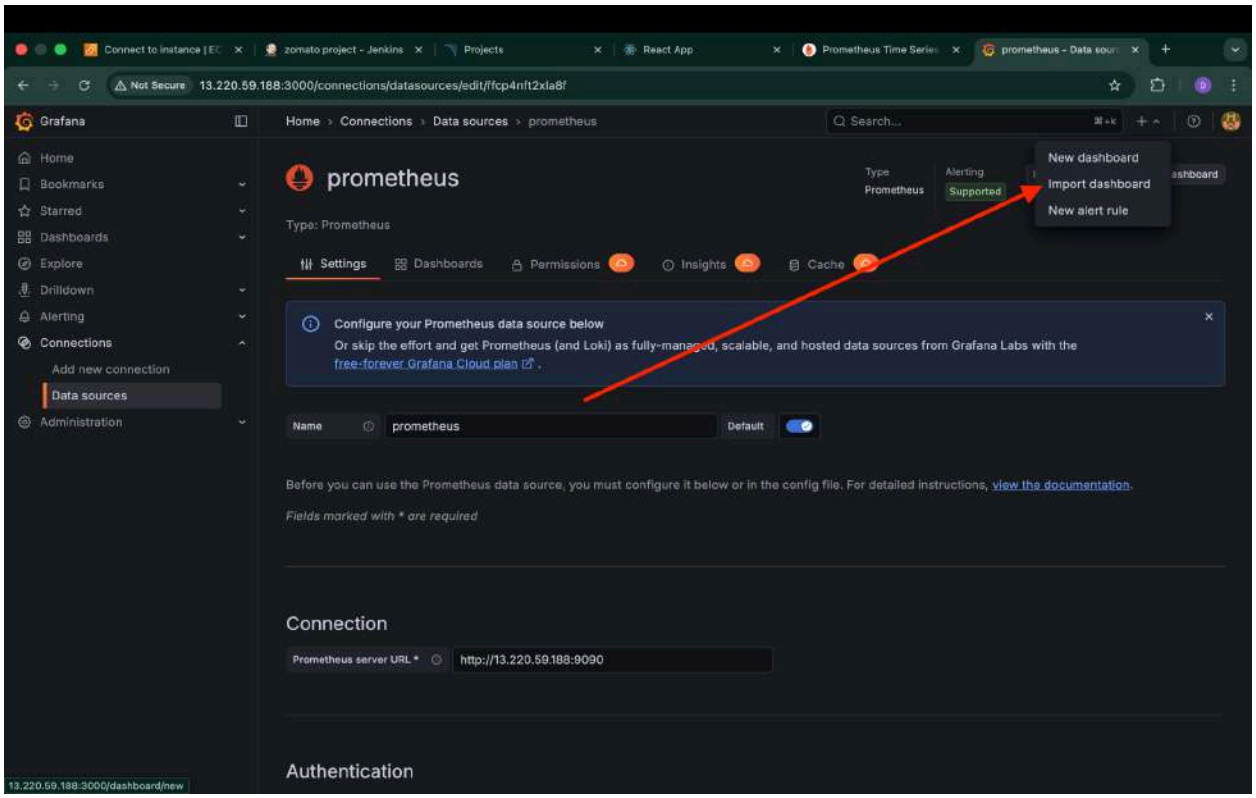


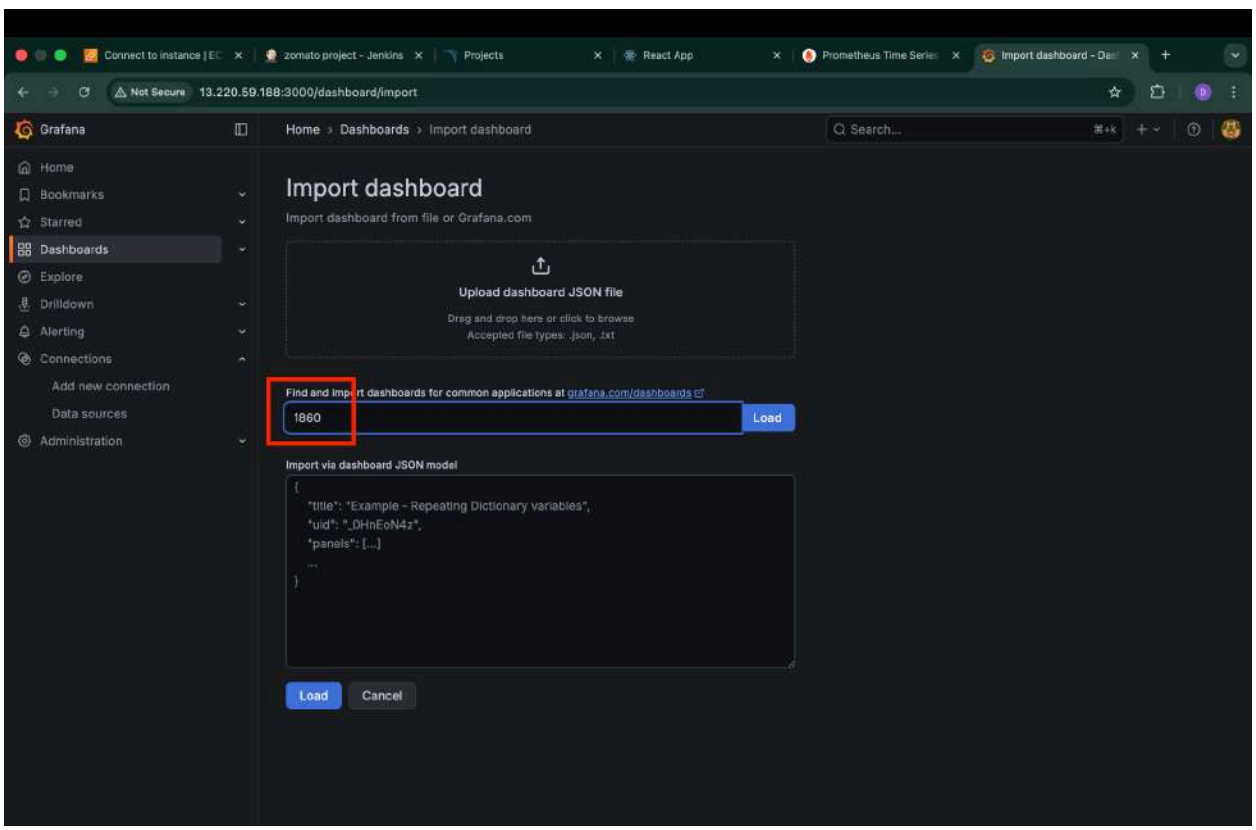


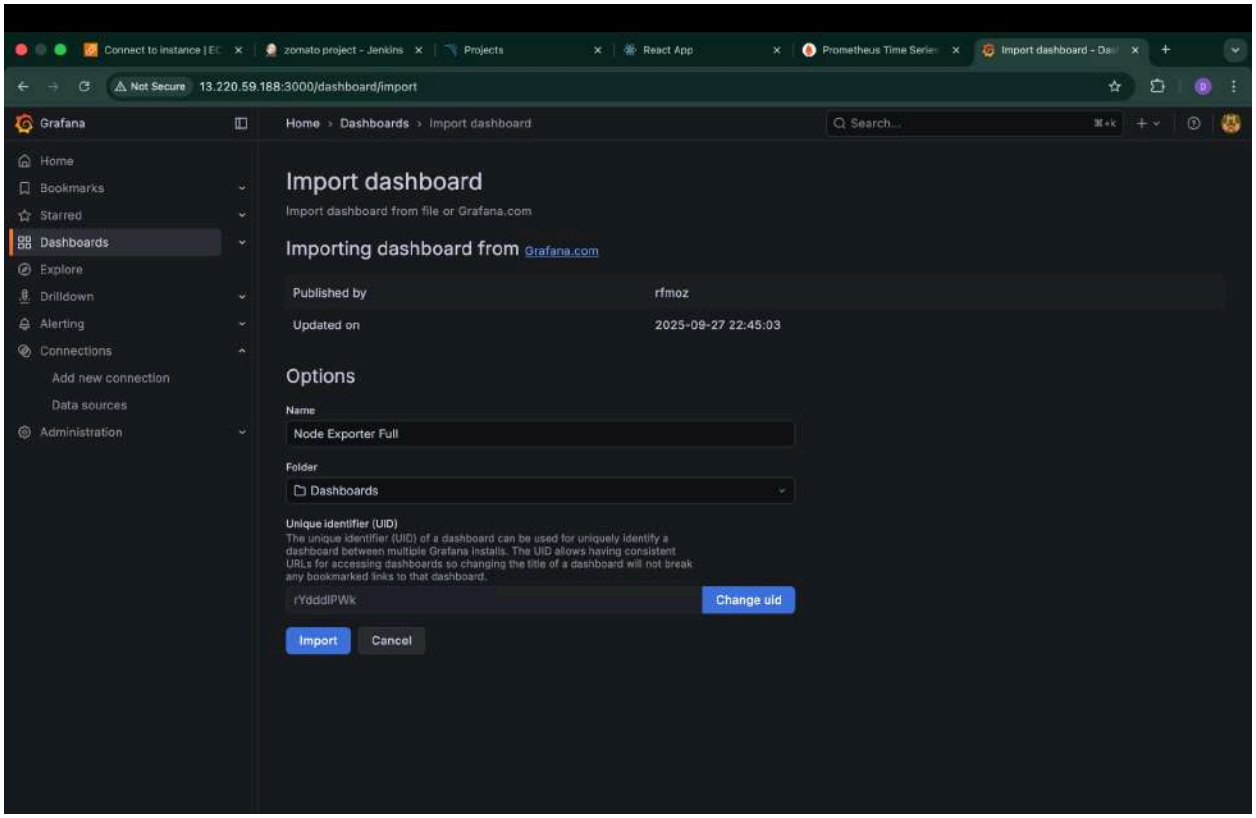

Connect to instance | EC
zomato project - Jenkins
Projects
React App
Prometheus Time Series
prometheus - Data sour
Not Secure
13.220.59.188:3000/connections/datasources/edit/ffcp4nft2xla8f
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Home > Connections > Data sources > prometheus
Search...
prometheus
Type: Prometheus
Type
Prometheus
Alerting
Supported
Explore data
Build a dashboard
Settings
Dashboards
Permissions
Insights
Cache
Configure your Prometheus data source below
Or skip the effort and get Prometheus (and Loki) as fully-managed, scalable, and hosted data sources from Grafana Labs with the free-forever Grafana Cloud plan .
Name
prometheus
Default
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://13.220.59.188:9090
Authentication

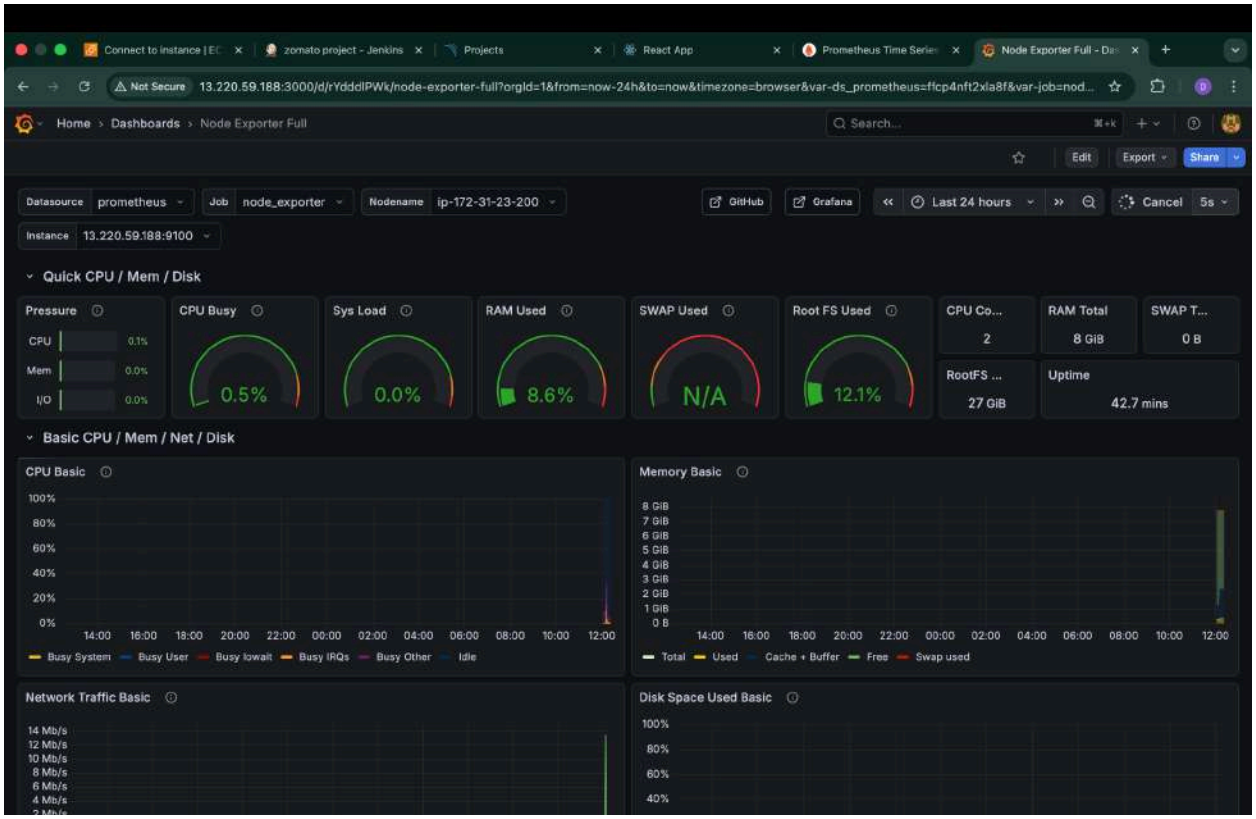

Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard , or by querying data in the Explore view .
Open in Metrics Drilldown
Delete
Save & test

Now add dashboards

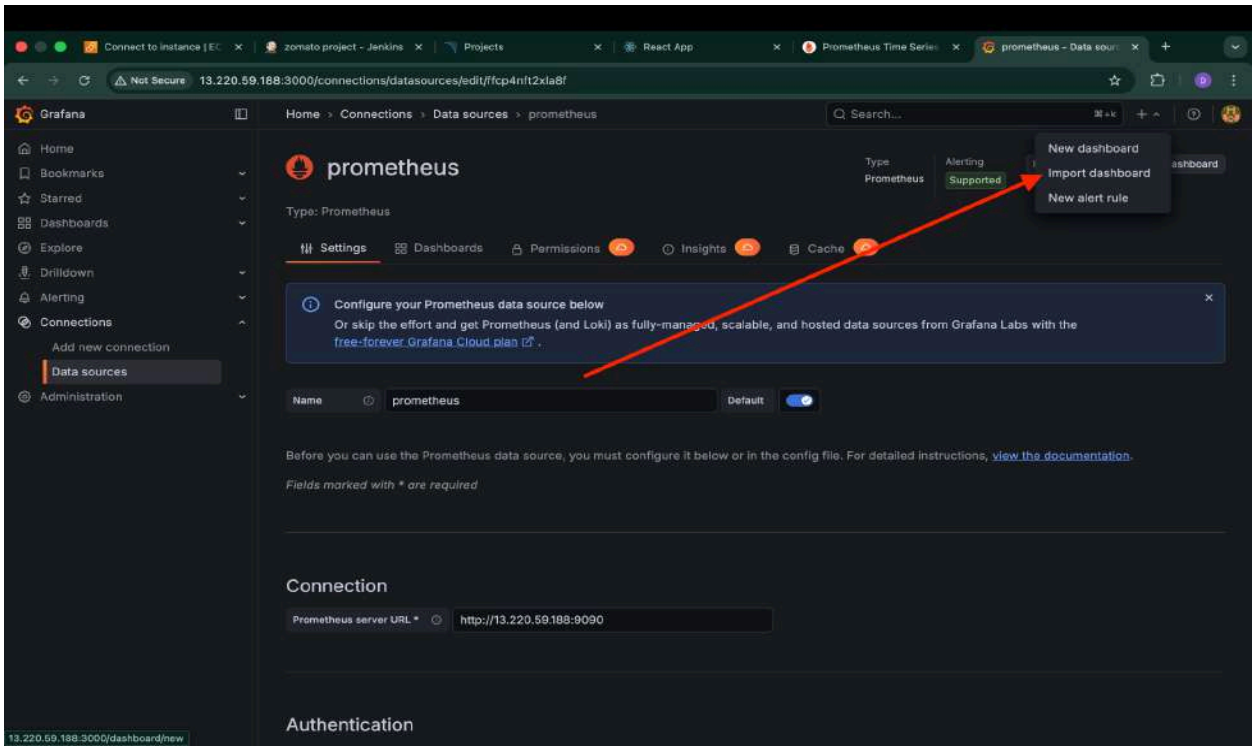


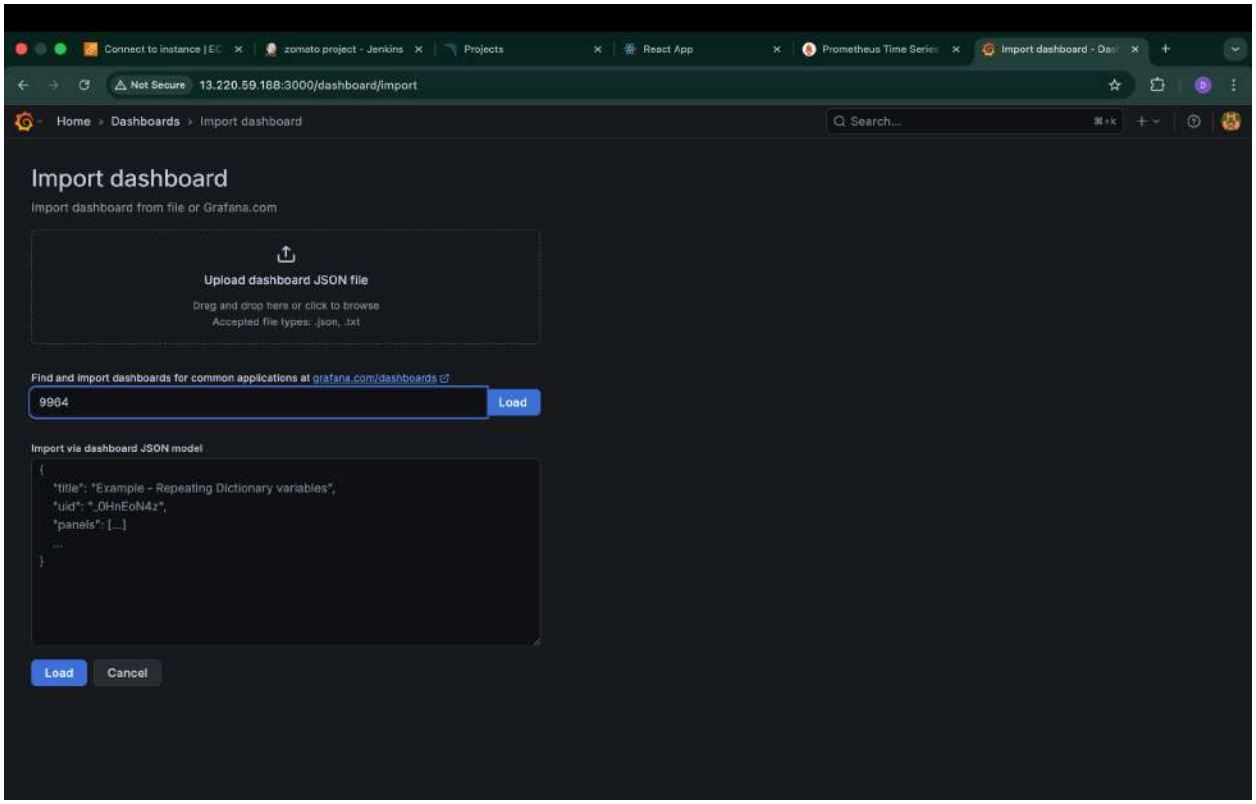


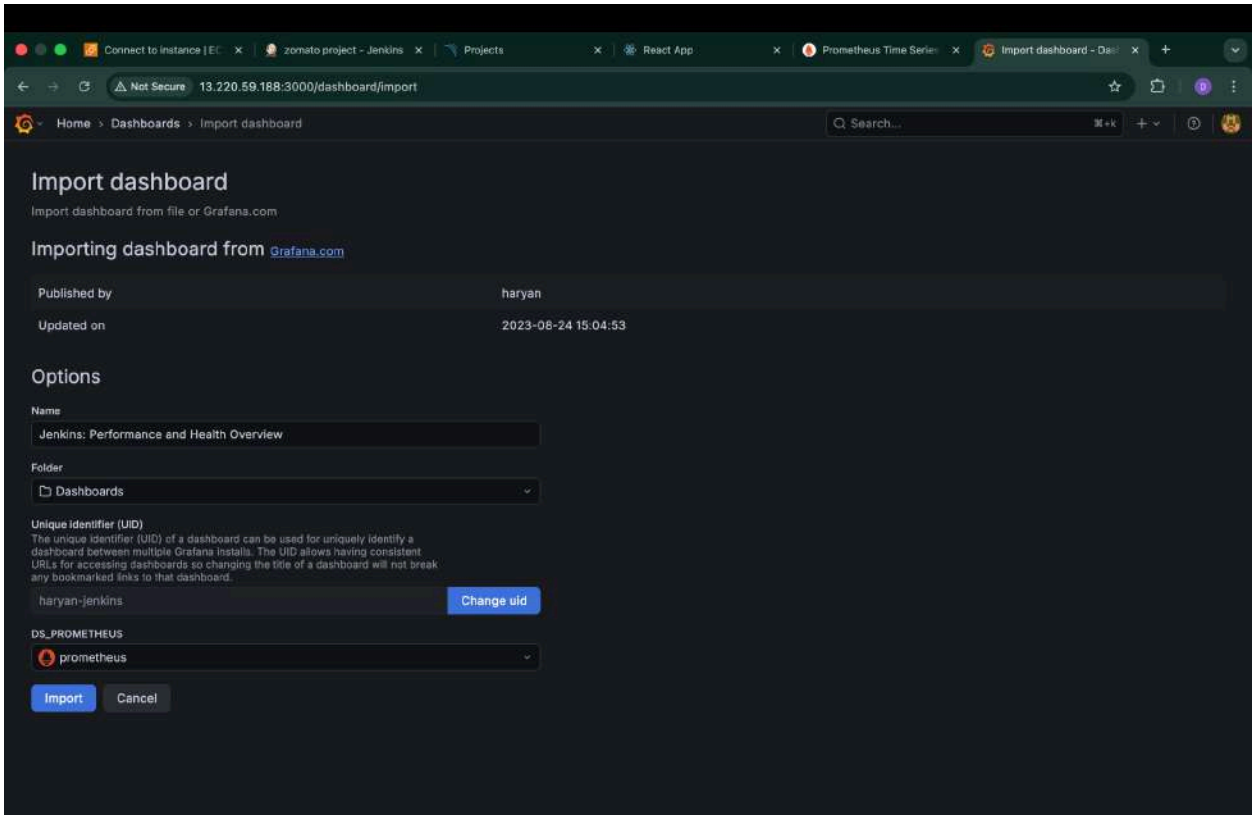

Connect to instance | EC
zomato project - Jenkins
Projects
React App
Prometheus Time Series
Import dashboard - Das
Not Secure 13.220.59.188:3000/dashboard/import
Grafana
Home > Dashboards > Import dashboard
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Import dashboard
Import dashboard from file or Grafana.com
Importing dashboard from Grafana.com
Published by
rfmoz
Updated on
2025-09-27 22:45:03
Options
Name
Node Exporter Full
Folder
Dashboards
Unique identifier (UID)
The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.
rYddclPWk
Change uid
Import
Cancel

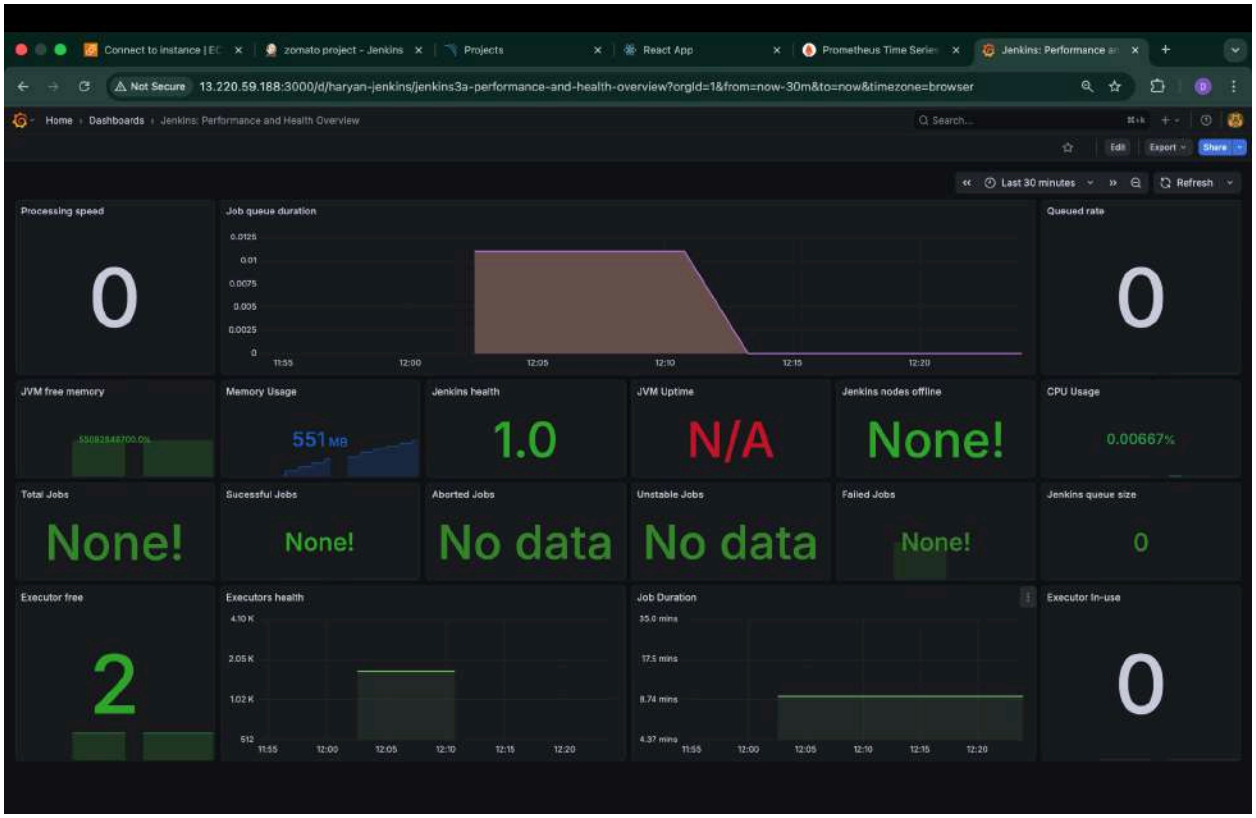

Connect to instance | EC
zomato project - Jenkins
Projects
React App
Prometheus Time Series
Node Exporter Full - Das
Not Secure 13.220.59.188:3000/d/rYddclPWk/node-exporter-full?orgId=1&from=now-24h&to=now&timezone=browser&var-ds_prometheus=ffcp4nft2xla8f&var-job=nod...
Home > Dashboards > Node Exporter Full
Search...
Edit
Export
Share
Datasource prometheus
Job node_exporter
Nodename ip-172-31-23-200
Instance 13.220.59.188:9100
GitHub
Grafana
Last 24 hours
Cancel
5s
Quick CPU / Mem / Disk
Pressure
CPU 0.1%
Mem 0.0%
I/O 0.0%
CPU Busy
0.5%
Sys Load
0.0%
RAM Used
8.6%
SWAP Used
N/A
Root FS Used
12.1%
CPU Co...
2
RAM Total
8 GiB
SWAP T...
0 B
RootFS ...
27 GiB
Uptime
42.7 mins
Basic CPU / Mem / Net / Disk
CPU Basic
100%
80%
60%
40%
20%
0%
14:00 16:00 18:00 20:00 22:00 00:00 02:00 04:00 06:00 08:00 10:00 12:00
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Memory Basic
8 GiB
7 GiB
6 GiB
5 GiB
4 GiB
3 GiB
2 GiB
1 GiB
0 B
14:00 16:00 18:00 20:00 22:00 00:00 02:00 04:00 06:00 08:00 10:00 12:00
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
14 Mb/s
12 Mb/s
10 Mb/s
8 Mb/s
6 Mb/s
4 Mb/s
Disk Space Used Basic
100%
80%
60%
40%

Now add jenkins dashboard




Connect to instance | EC
zomato project - Jenkins
Projects
React App
Prometheus Time Series
Import dashboard - Das
Not Secure
13.220.59.188:3000/dashboard/import
Home › Dashboards › Import dashboard
Search...
Import dashboard
Import dashboard from file or Grafana.com
Upload dashboard JSON file
Drag and drop here or click to browse
Accepted file types: .json, .txt
Find and import dashboards for common applications at grafana.com/dashboards
9964
Load
Import via dashboard JSON model
{
"title": "Example - Repeating Dictionary variables",
"uid": "_0HnEoN4z",
"panels": [...]
...
}
Load
Cancel


Connect to instance | EC
zomato project - Jenkins
Projects
React App
Prometheus Time Series
Import dashboard - Das
Not Secure
13.220.59.188:3000/dashboard/import
Home › Dashboards › Import dashboard
Search...
Import dashboard
Import dashboard from file or Grafana.com
Importing dashboard from Grafana.com
Published by
haryan
Updated on
2023-08-24 15:04:53
Options
Name
Jenkins: Performance and Health Overview
Folder
Dashboards
Unique identifier (UID)
The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.
haryan-jenkins
Change uid
DS_PROMETHEUS
prometheus
Import
Cancel



Phase 6: Kubernetes Deployment (EKS)

An Amazon EKS cluster was created.

Steps

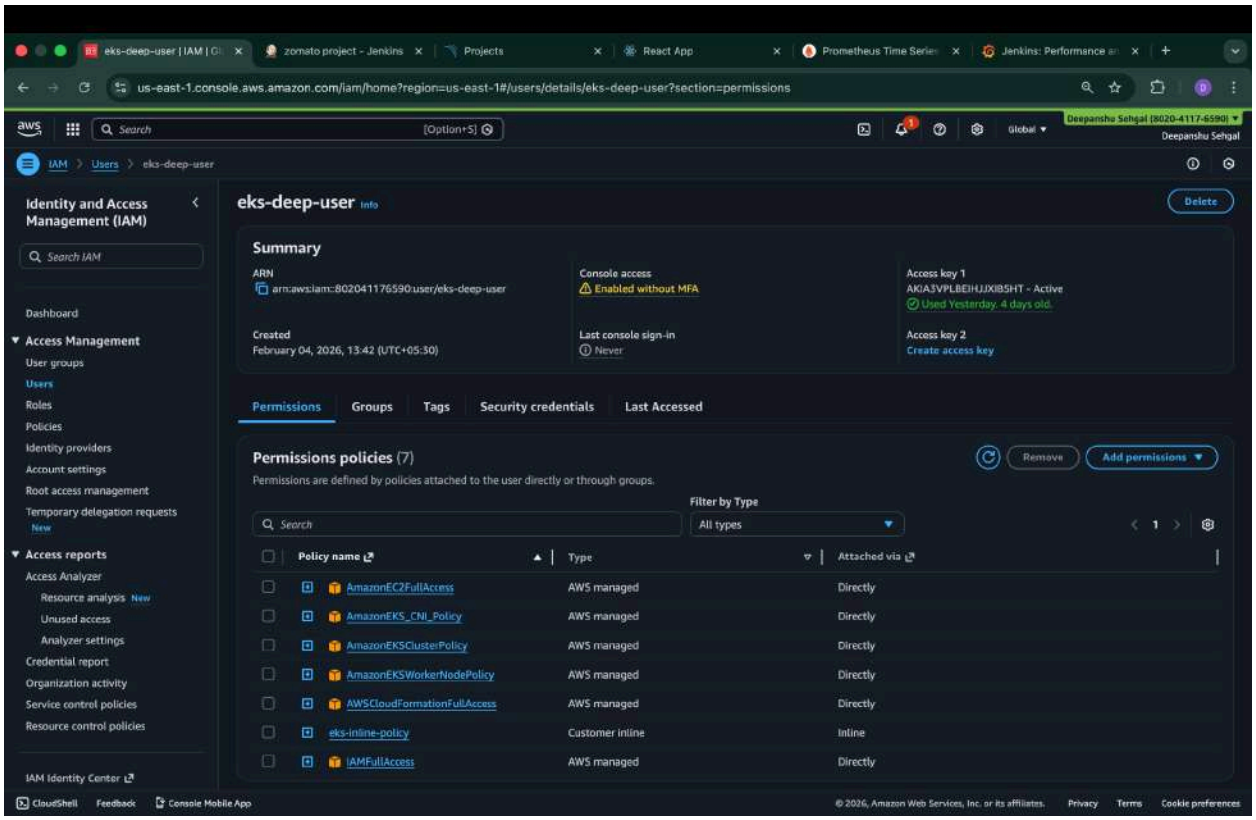
- Created cluster using eksctl
- Associated IAM OIDC provider
- Created managed node groups
- Configured networking and access

Application Deployment

- Kubernetes manifests used
- Pods and services created
- Application exposed via NodePort/LoadBalancer

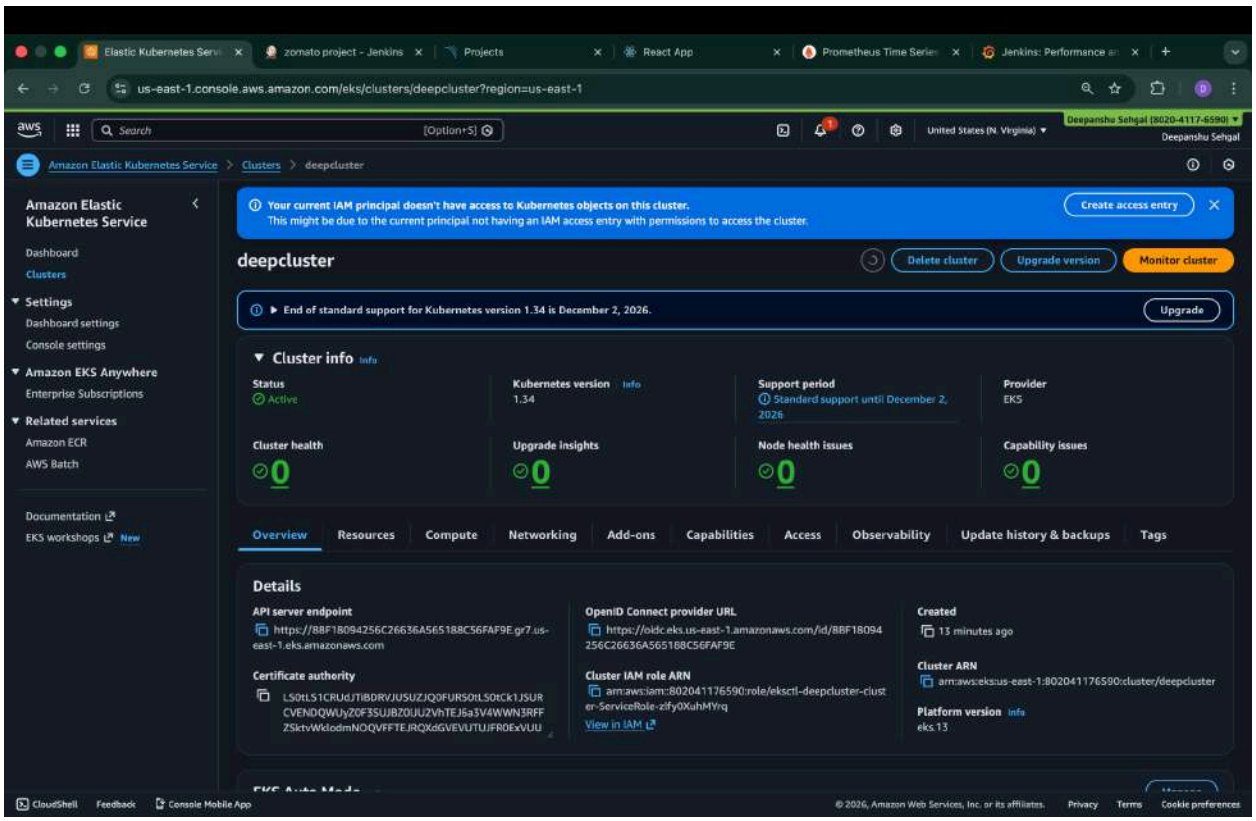
This ensured scalability and orchestration.

Creation of EKS cluster




```
root@ip-172-31-21-101:/home/ubuntu# aws configure
AWS Access Key ID [None]: AKIA3VPLBEIHJJXIB5HT
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-21-101:/home/ubuntu#
```

```
root@ip-172-31-21-101:/home/ubuntu# curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin
kubectl version --short --client
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 57.4M  100 57.4M    0     0  5372k      0  0:00:10  0:00:10 --:--:-- 11.2M
Client Version: v1.19.6-eks-49a6c0
root@ip-172-31-21-101:/home/ubuntu# curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | ta
r xz -C /tmp
sudo mv /tmp/eksctl /usr/local/bin
eksctl version
0.222.0
root@ip-172-31-21-101:/home/ubuntu# eksctl create cluster --name=deepcluster \
                    --region=us-east-1 \
                    --zones=us-east-1a,us-east-1c \
                    --without-nodegroup
2026-02-09 07:24:57 [ℹ]  eksctl version 0.222.0
2026-02-09 07:24:57 [ℹ]  using region us-east-1
2026-02-09 07:24:57 [ℹ]  subnets for us-east-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-02-09 07:24:57 [ℹ]  subnets for us-east-1c - public:192.168.32.0/19 private:192.168.96.0/19
2026-02-09 07:24:57 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons wil
l no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https:
//eksctl.io/usage/auto-mode/
2026-02-09 07:24:57 [ℹ]  using Kubernetes version 1.34
2026-02-09 07:24:57 [ℹ]  creating EKS cluster "deepcluster" in "us-east-1" region with
2026-02-09 07:24:57 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=deepcluster'
2026-02-09 07:24:57 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "deepcluster" in "us-east-1"
2026-02-09 07:24:57 [ℹ]  CloudWatch logging will not be enabled for cluster "deepcluster" in "us-east-1"
2026-02-09 07:24:57 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 -
-cluster=deepcluster'
2026-02-09 07:24:57 [ℹ]  default addons kube-proxy, coredns, metrics-server, vpc-cni were not specified, will install them as EKS addons
2026-02-09 07:24:57 [ℹ]
2 sequential tasks: { create cluster control plane "deepcluster",
    2 sequential sub-tasks: {
        1 task: { create addons },
        wait for control plane to become ready,
    }
}
2026-02-09 07:24:57 [ℹ]  building cluster stack "eksctl-deepcluster-cluster"
2026-02-09 07:24:57 [ℹ]  deploying stack "eksctl-deepcluster-cluster"
```



```
root@ip-172-31-21-101:/home/ubuntu# eksctl get cluster
NAME            REGION          EKSCTL CREATED
deepcluster     us-east-1       True
root@ip-172-31-21-101:/home/ubuntu#
```

Create & Associate IAM OIDC Provider for our EKS Cluster

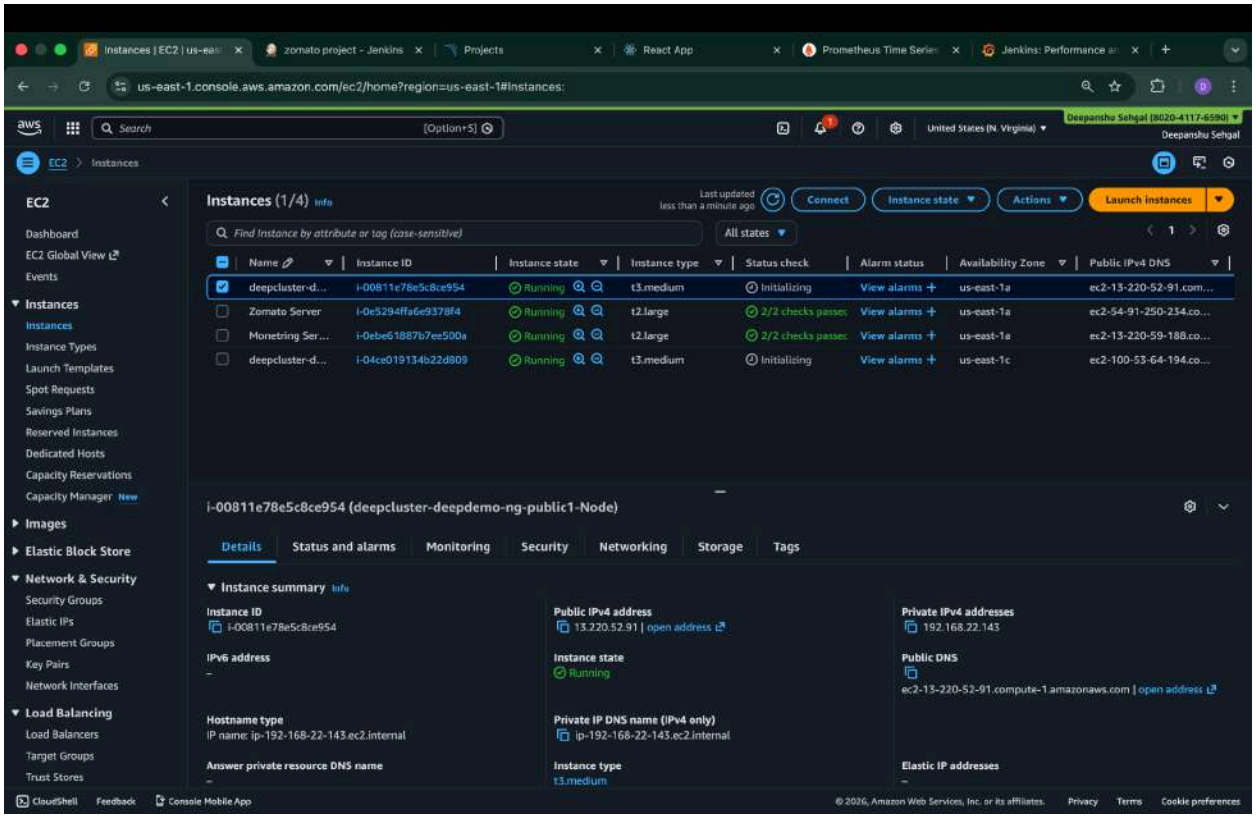
To enable and use AWS IAM roles for Kubernetes service accounts on our EKS cluster, we must create & associate OIDC identity provider.
To do so using eksctl we can use the below commands.

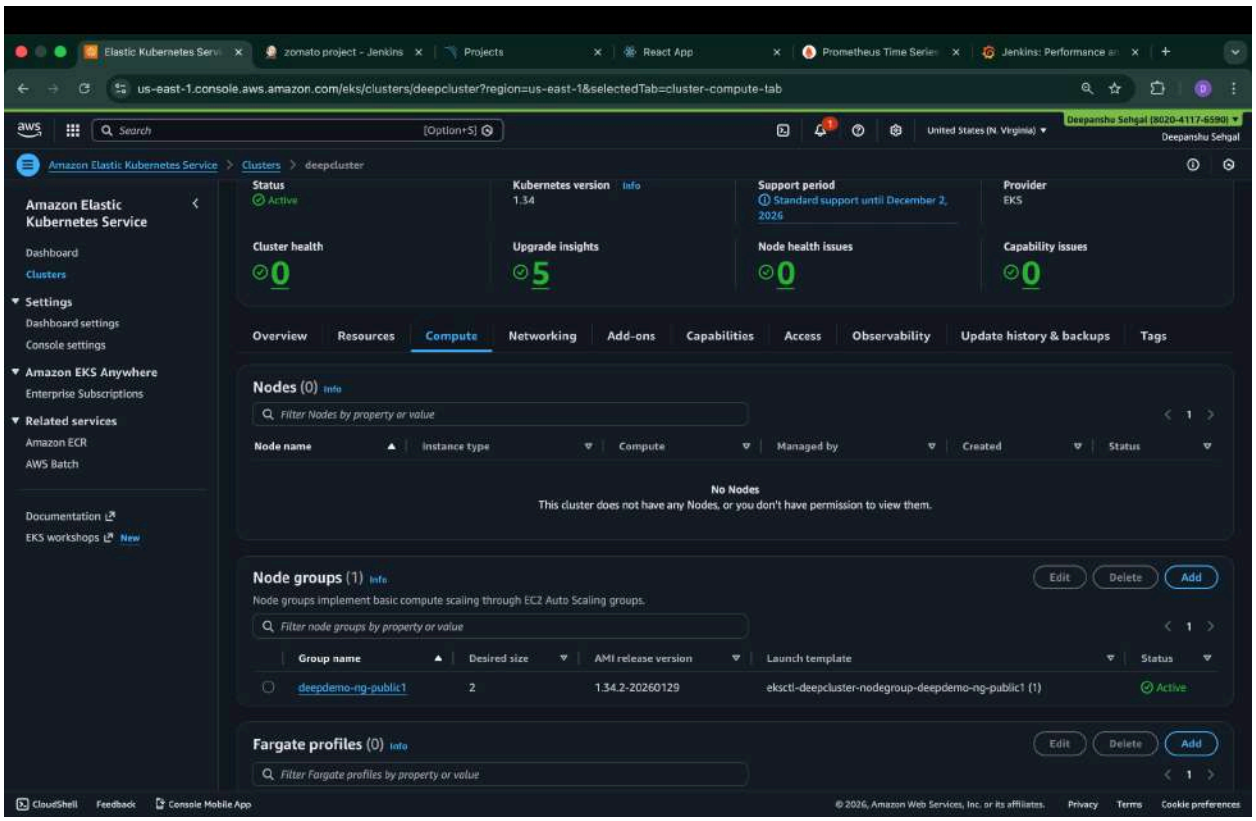
```
root@ip-172-31-21-101:/home/ubuntu# eksctl utils associate-iam-oidc-provider \
    --region us-east-1 \
    --cluster deepcluster \
    --approve
2026-02-09 07:41:39 [ℹ]  will create IAM Open ID Connect provider for cluster "deepcluster" in "us-east-1"
2026-02-09 07:41:39 [✔]  created IAM Open ID Connect provider for cluster "deepcluster" in "us-east-1"
root@ip-172-31-21-101:/home/ubuntu#
```

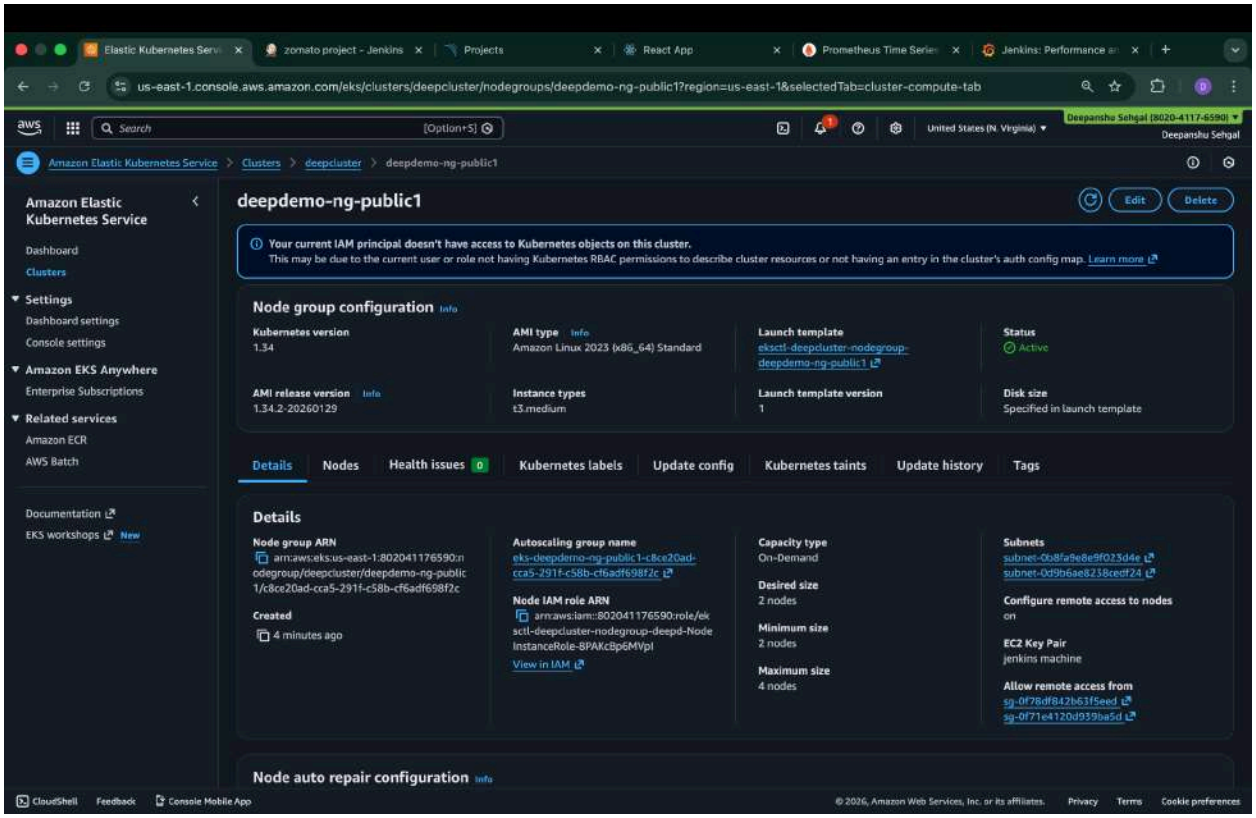
Create a node group with additional Add-Ons in public subnets.

These add-ons will create the respective IAM policies for us automatically within our Node Group role.

```
root@ip-172-31-21-101:/home/ubuntu# eksctl create nodegroup --cluster=deepcluster \
                      --region=us-east-1 \
                      --name=deepdemo-ng-public1 \
                      --node-type=t3.medium \
                      --nodes=2 \
                      --nodes-min=2 \
                      --nodes-max=4 \
                      --node-volume-size=20 \
                      --ssh-access \
                      --ssh-public-key='jenkins machine' \
                      --managed \
                      --asg-access \
                      --external-dns-access \
                      --full-ecr-access \
                      --appmesh-access \
                      --alb-ingress-access
2026-02-09 07:43:44 [ℹ]  will use version 1.34 for new nodegroup(s) based on control plane version
2026-02-09 07:43:45 [ℹ]  nodegroup "deepdemo-ng-public1" will use "" [AmazonLinux2023/1.34]
2026-02-09 07:43:45 [ℹ]  using EC2 key pair "jenkins machine"
2026-02-09 07:43:46 [ℹ]  1 nodegroup (deepdemo-ng-public1) was included (based on the include/exclude rules)
2026-02-09 07:43:46 [ℹ]  will create a CloudFormation stack for each of 1 managed nodegroups in cluster "deepcluster"
2026-02-09 07:43:46 [ℹ]
2 sequential tasks: { fix cluster compatibility, 1 task: { 1 task: { create managed nodegroup "deepdemo-ng-public1" } }
}
2026-02-09 07:43:46 [ℹ]  checking cluster stack for missing resources
2026-02-09 07:43:46 [ℹ]  cluster stack has all required resources
2026-02-09 07:43:46 [ℹ]  building managed nodegroup stack "eksctl-deepcluster-nodegroup-deepdemo-ng-public1"
2026-02-09 07:43:46 [ℹ]  deploying stack "eksctl-deepcluster-nodegroup-deepdemo-ng-public1"
2026-02-09 07:43:46 [ℹ]  waiting for CloudFormation stack "eksctl-deepcluster-nodegroup-deepdemo-ng-public1"
```


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
EC2 > Instances
Instances (1/4) Info
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
deepcluster-d... i-00811e78e5c8ce954 Running t3.medium Initializing View alarms + us-east-1a ec2-13-220-52-91.com...
Zomato Server i-0e5294ffa6e9378f4 Running t2.large 2/2 checks passed View alarms + us-east-1a ec2-54-91-250-234.co...
Monetring Ser... i-0ebe61887b7ee500a Running t2.large 2/2 checks passed View alarms + us-east-1a ec2-13-220-59-188.co...
deepcluster-d... i-04ce019134b22d809 Running t3.medium Initializing View alarms + us-east-1c ec2-100-53-64-194.co...
i-00811e78e5c8ce954 (deepcluster-deepdemo-ng-public1-Node)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-00811e78e5c8ce954
Public IPv4 address
13.220.52.91 | open address
Private IPv4 addresses
192.168.22.143
IPv6 address
Instance state
Running
Public DNS
ec2-13-220-52-91.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-192-168-22-143.ec2.internal
Private IP DNS name (IPv4 only)
ip-192-168-22-143.ec2.internal
Answer private resource DNS name
Instance type
t3.medium
Elastic IP addresses


us-east-1.console.aws.amazon.com/eks/clusters/deepcluster?region=us-east-1&selectedTab=cluster-compute-tab
Amazon Elastic Kubernetes Service > Clusters > deepcluster
Status
Active
Kubernetes version Info
1.34
Support period
Standard support until December 2, 2026
Provider
EKS
Cluster health
0
Upgrade insights
5
Node health issues
0
Capability issues
0
Overview
Resources
Compute
Networking
Add-ons
Capabilities
Access
Observability
Update history & backups
Tags
Nodes (0) Info
No Nodes
This cluster does not have any Nodes, or you don't have permission to view them.
Node groups (1) Info
Node groups implement basic compute scaling through EC2 Auto Scaling groups.
Edit
Delete
Add
Group name
Desired size
AMI release version
Launch template
Status
deepdemo-ng-public1 2 1.34.2-20260129 eksctl-deepcluster-nodegroup-deepdemo-ng-public1 (1) Active
Fargate profiles (0) Info



Phase 7: GitOps with ArgoCD

ArgoCD was installed on the cluster.

Features Used

- Git repository integration
- Continuous synchronization
- Auto-deployment on code changes

Git became the **single source of truth**, enabling GitOps.

Argo CD installation

```
root@ip-172-31-21-101:/home/ubuntu# kubectl create namespace argocd
namespace/argocd created
root@ip-172-31-21-101:/home/ubuntu# kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/v2.4.7/manifests/install.yaml
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io created
```

By default the argo CD server is not publicly exposed, so we need to expose it publicly. To do that, execute the below command;

```
root@ip-172-31-21-101:/home/ubuntu# kubectl patch svc argocd-server -n argocd -p '{"spec": {"type": "LoadBalancer"}}'
service/argocd-server patched
root@ip-172-31-21-101:/home/ubuntu# kubectl get ns
NAME              STATUS   AGE
argocd            Active   92s
default           Active   19m
kube-node-lease   Active   19m
kube-public       Active   19m
kube-system       Active   19m
root@ip-172-31-21-101:/home/ubuntu#
```

Monitor Kubernetes with Prometheus

It used to monitor Kubernetes cluster.
Additionally, you'll install the node exporter using Helm to collect metrics from your cluster nodes.

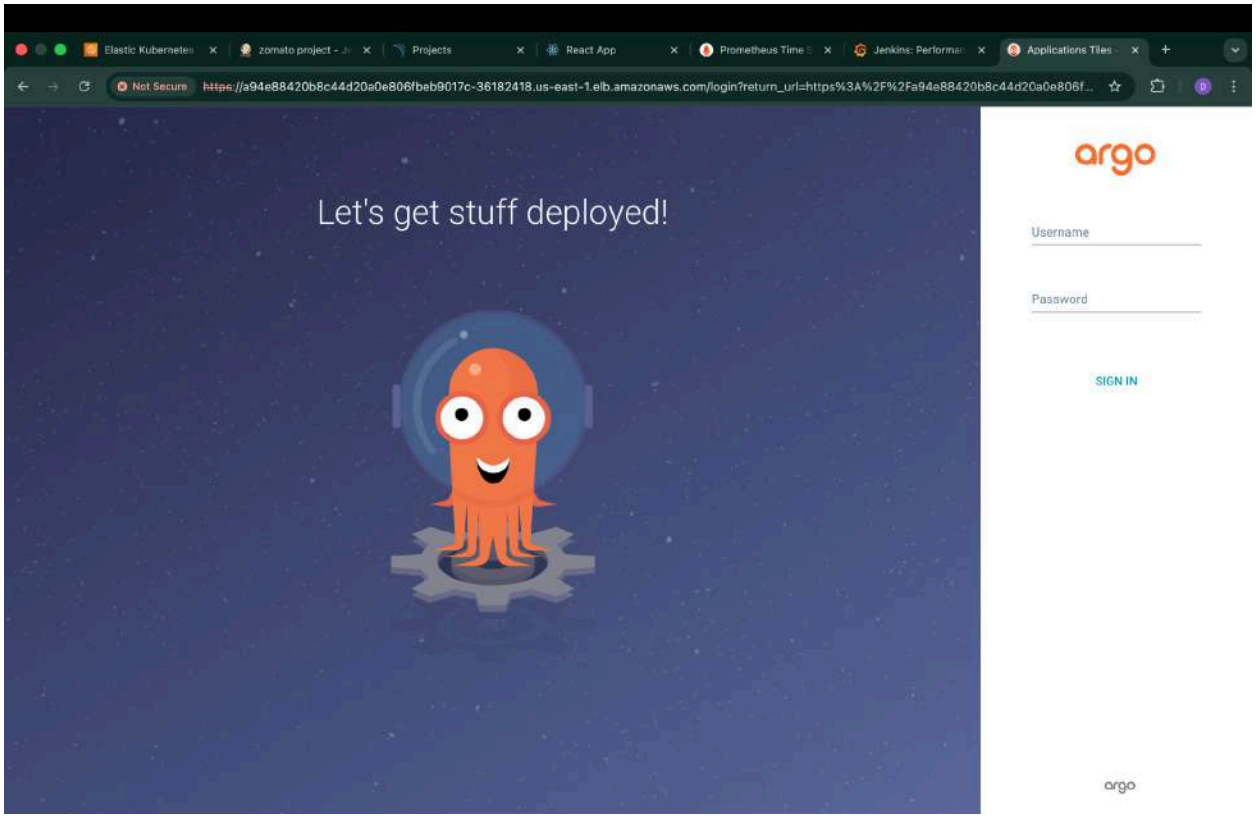
Install Node Exporter using Helm
To begin monitoring your Kubernetes cluster, you'll install the Prometheus Node Exporter. This component allows you to collect system-level metrics from your cluster nodes. Here are the steps to install the Node Exporter using Helm:

Add the Prometheus Community Helm repository:

```
root@ip-172-31-21-101:/home/ubuntu# helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
"prometheus-community" has been added to your repositories
root@ip-172-31-21-101:/home/ubuntu#
```

Create a Kubernetes namespace for the Node Exporter:
Install the Node Exporter using Helm:
Lets continue with load balancer thing of previous step;
To get the loadbalancer url;

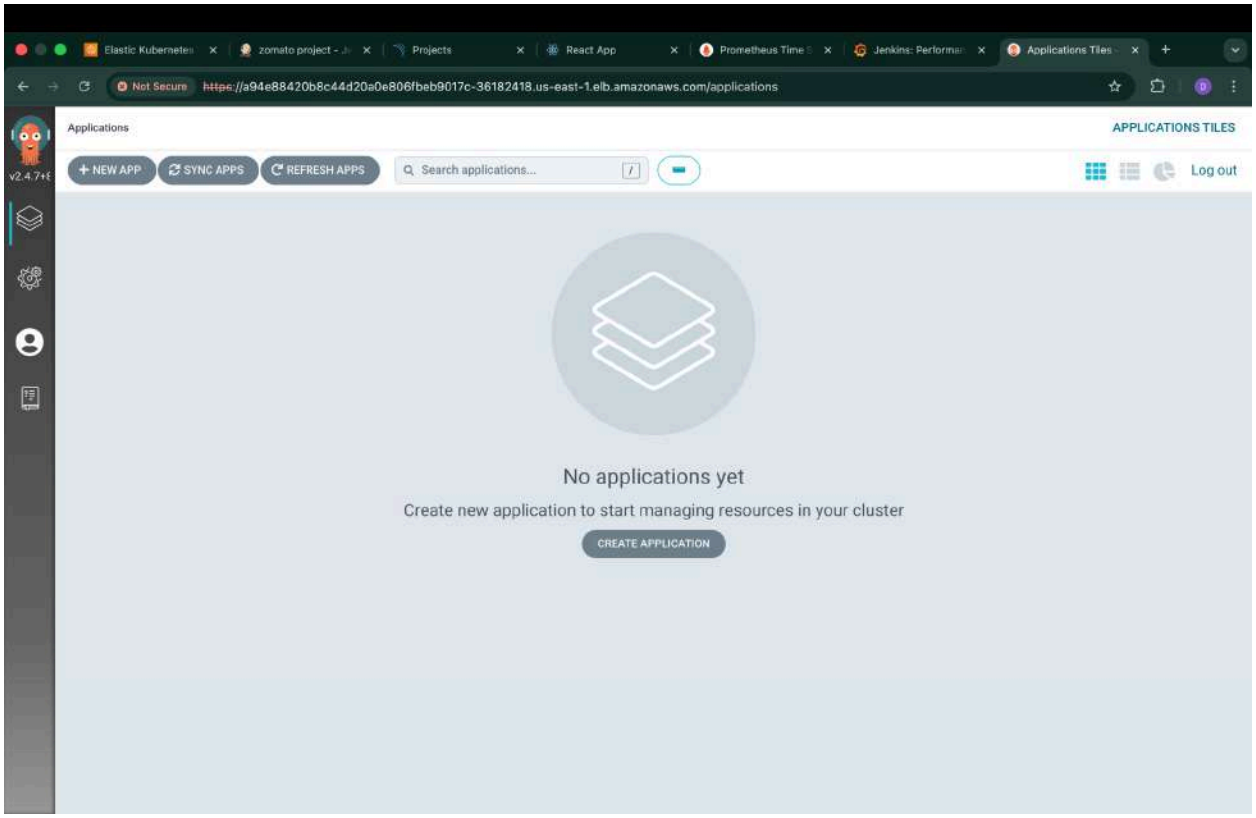
```
root@ip-172-31-21-101:/home/ubuntu# helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
"prometheus-community" has been added to your repositories
root@ip-172-31-21-101:/home/ubuntu# kubectl create namespace prometheus-node-exporter
namespace/prometheus-node-exporter created
root@ip-172-31-21-101:/home/ubuntu# helm install prometheus-node-exporter prometheus-community/prometheus-node-exporter --namespace prometheus-node-exporter
NAME: prometheus-node-exporter
LAST DEPLOYED: Mon Feb  9 07:58:10 2026
NAMESPACE: prometheus-node-exporter
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace prometheus-node-exporter -l "app.kubernetes.io/name=prometheus-node-exporter,app.kubernetes.io/instance=promethe
us-node-exporter" -o jsonpath="{.items[0].metadata.name}")
  echo "Visit http://127.0.0.1:9100 to use your application"
  kubectl port-forward --namespace prometheus-node-exporter $POD_NAME 9100
root@ip-172-31-21-101:/home/ubuntu# export POD_NAME=$(kubectl get pods --namespace prometheus-node-exporter -l "app.kubernetes.io/name=prometheus-node-exporter,a
pp.kubernetes.io/instance=prometheus-node-exporter" -o jsonpath="{.items[0].metadata.name}")
root@ip-172-31-21-101:/home/ubuntu# echo "Visit http://127.0.0.1:9100 to use your application"
  kubectl port-forward --namespace prometheus-node-exporter $POD_NAME 9100
Visit http://127.0.0.1:9100 to use your application
Forwarding from 127.0.0.1:9100 -> 9100
Forwarding from [::1]:9100 -> 9100
root@ip-172-31-21-101:/home/ubuntu# export ARGOCD_SERVER=`kubectl get svc argocd-server -n argocd -o json | jq --raw-output '.status.loadBalancer.ingress[0].host
name'`
root@ip-172-31-21-101:/home/ubuntu# echo $ARGOCD_SERVER
a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com
root@ip-172-31-21-101:/home/ubuntu#
```

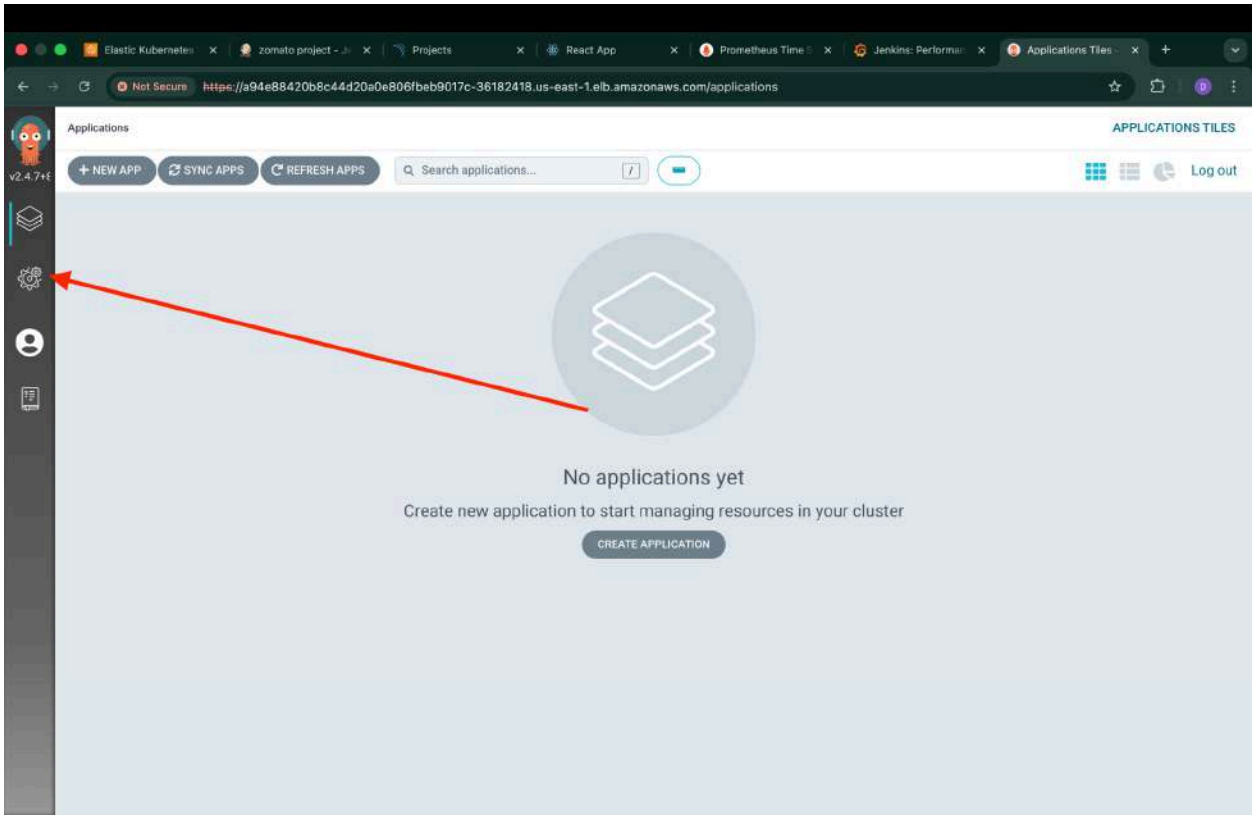


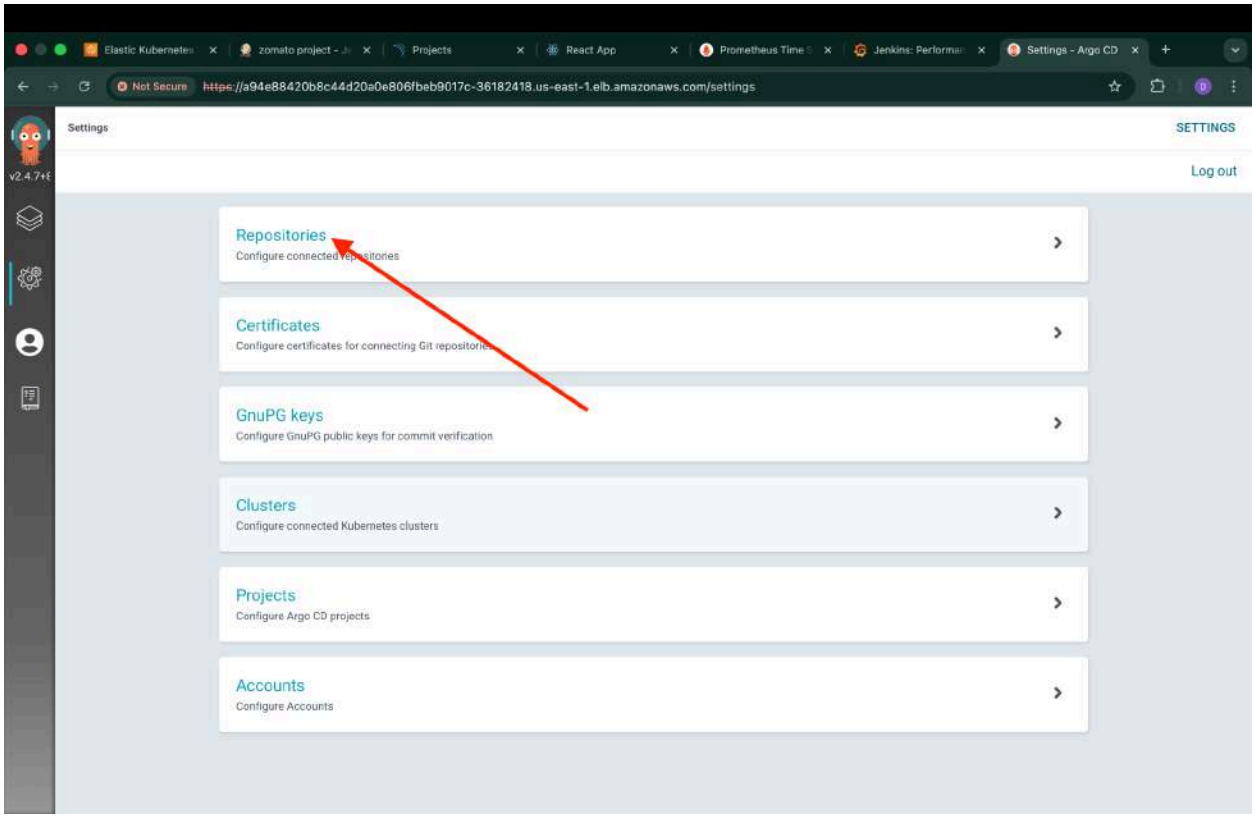
You will see the load balancer url, copy it and paste in browser. You will see the ArgoCD Homepage.
Username is "admin"

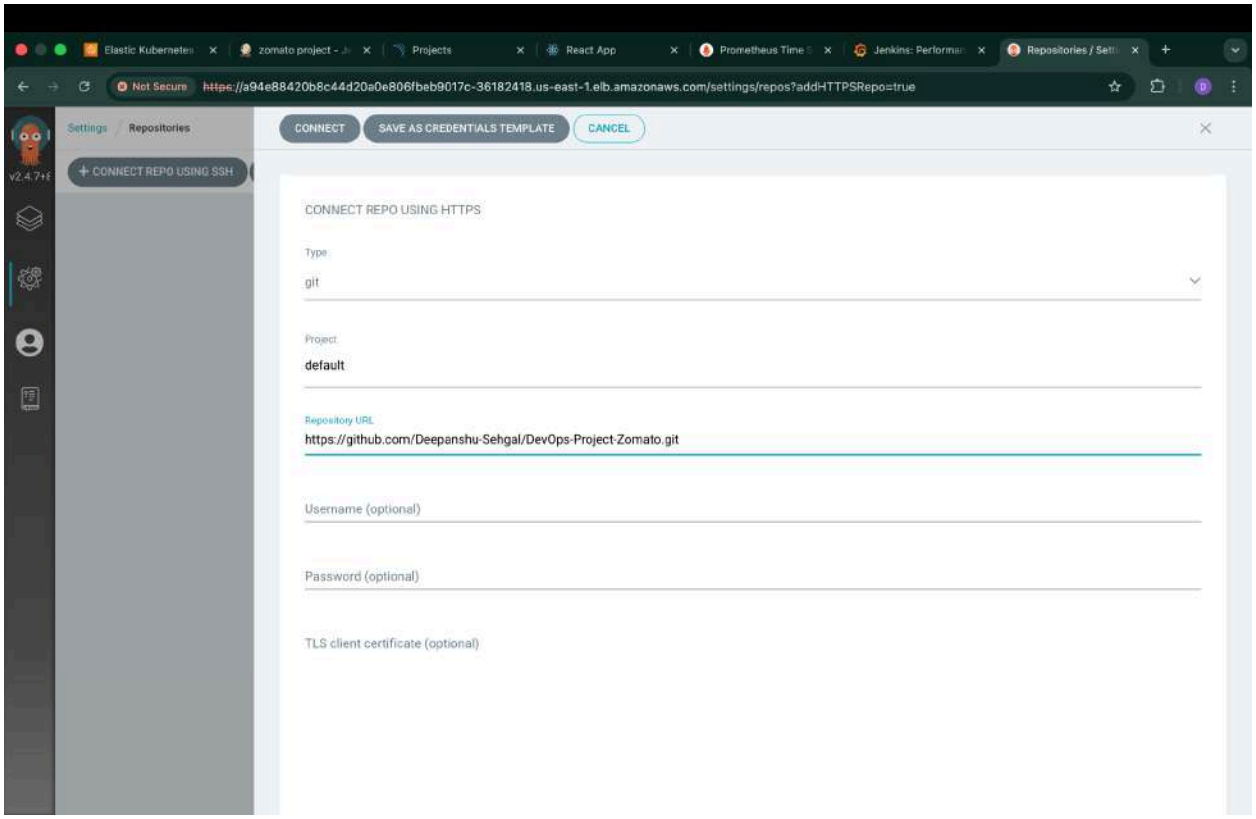
To get the password, execute the below command in vs code editor;

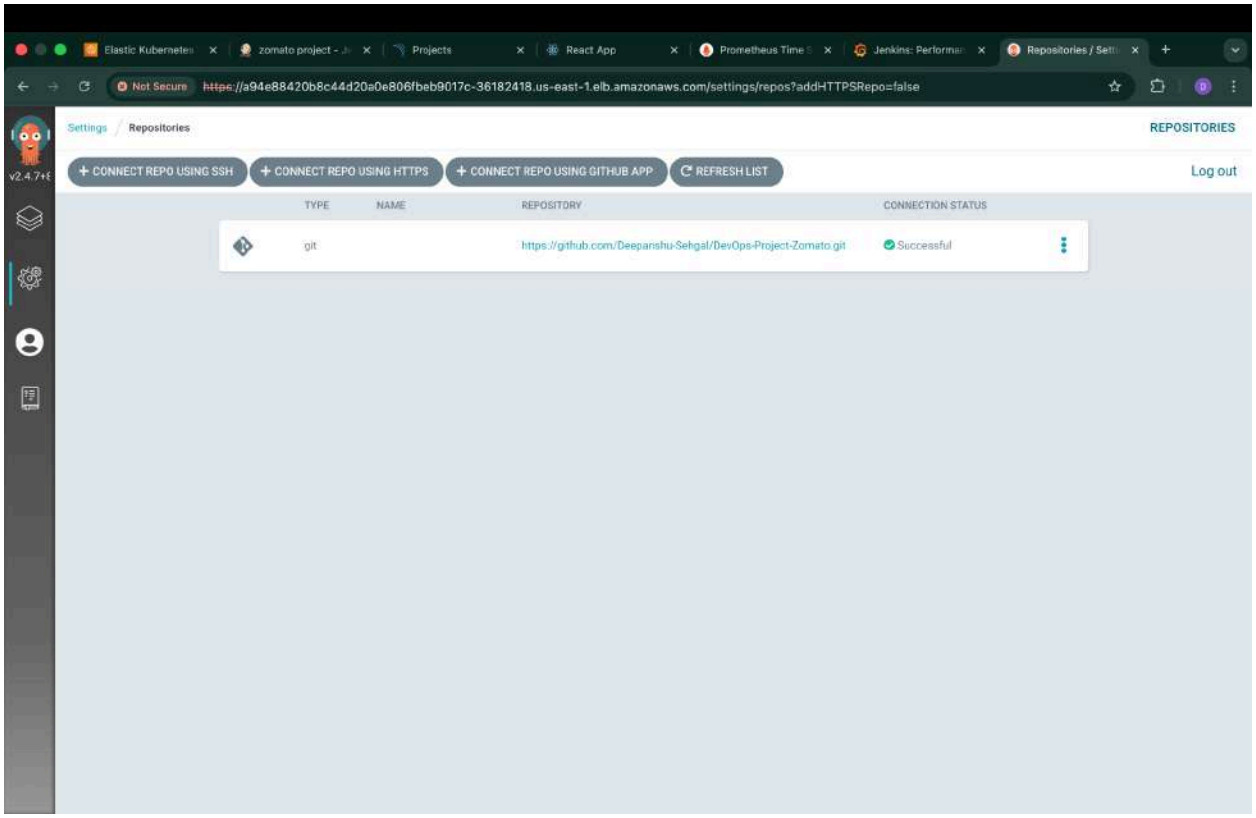
```
root@ip-172-31-21-101:/home/ubuntu# export ARGO_PWD=`kubectl -n argocd get secret argocd-initial-admin-secret -o jsonpath="{.data.password}" | base64 -d`
root@ip-172-31-21-101:/home/ubuntu# echo $ARGO_PWD
a23I0yaW-AcCcDh0
root@ip-172-31-21-101:/home/ubuntu#
```

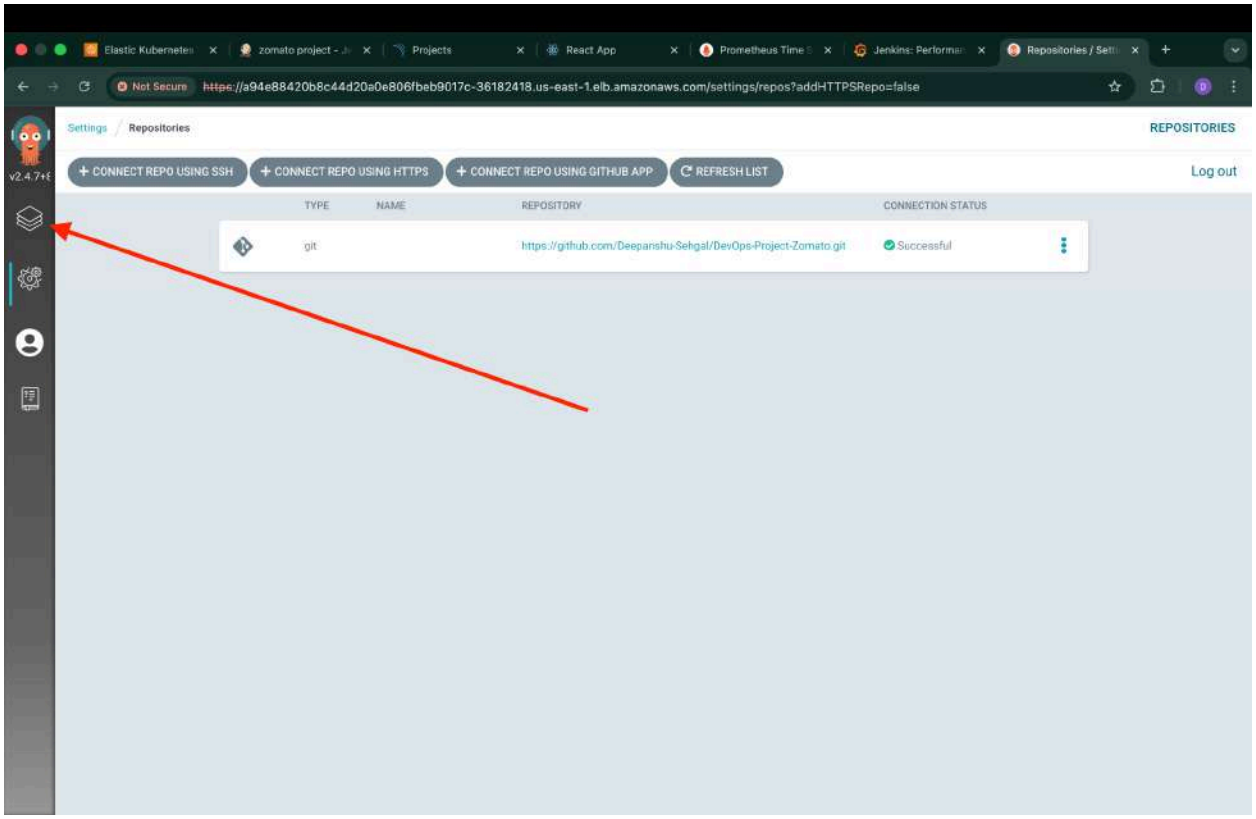

Elastic Kubernetes
zomato project - Jenkins
Projects
React App
Prometheus Time S
Jenkins: Performance
Applications Tiles
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/applications
Applications
APPLICATIONS TILES
v2.4.7+f
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
Log out
No applications yet
Create new application to start managing resources in your cluster
CREATE APPLICATION

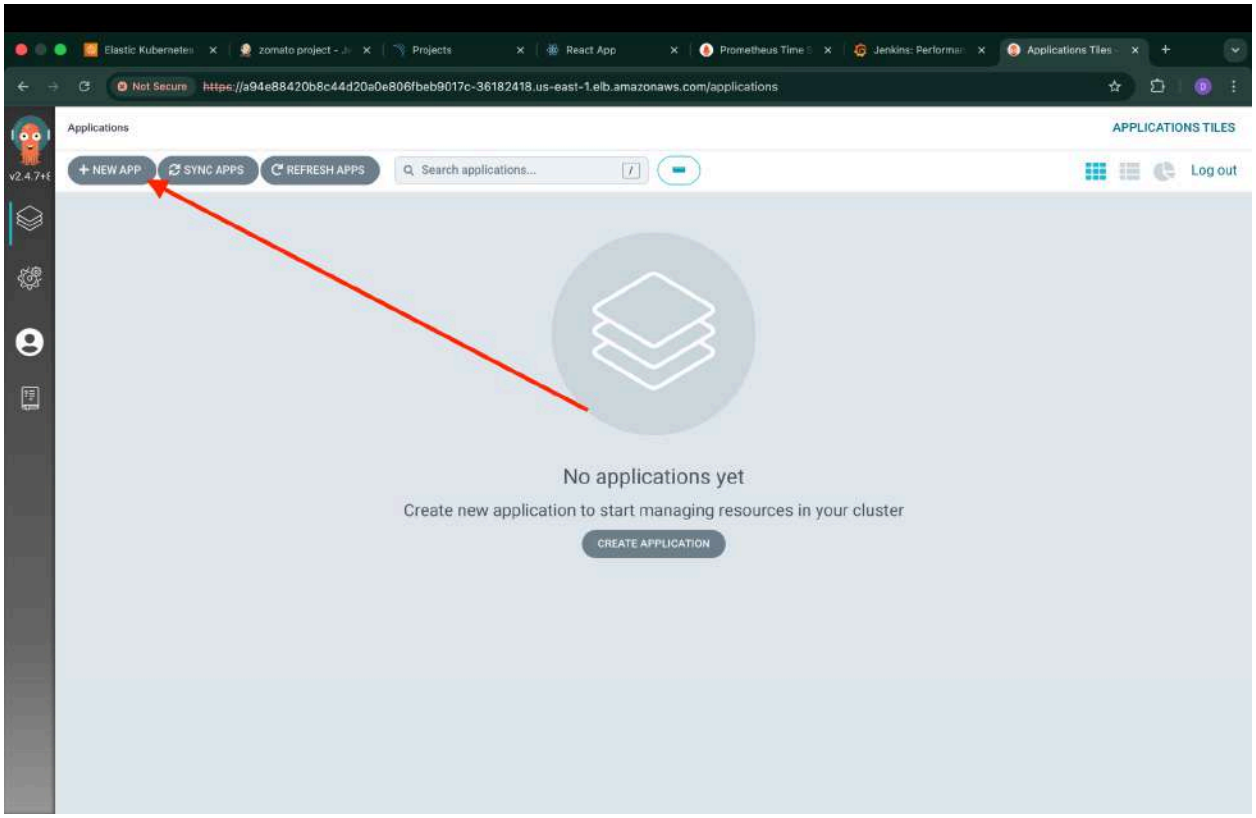

Elastic Kubernetes
zomato project - Jenkins
Projects
React App
Prometheus Time S
Jenkins: Performance
Applications Tiles
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/applications
Applications
APPLICATIONS TILES
v2.4.7+f
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
Log out
No applications yet
Create new application to start managing resources in your cluster
CREATE APPLICATION

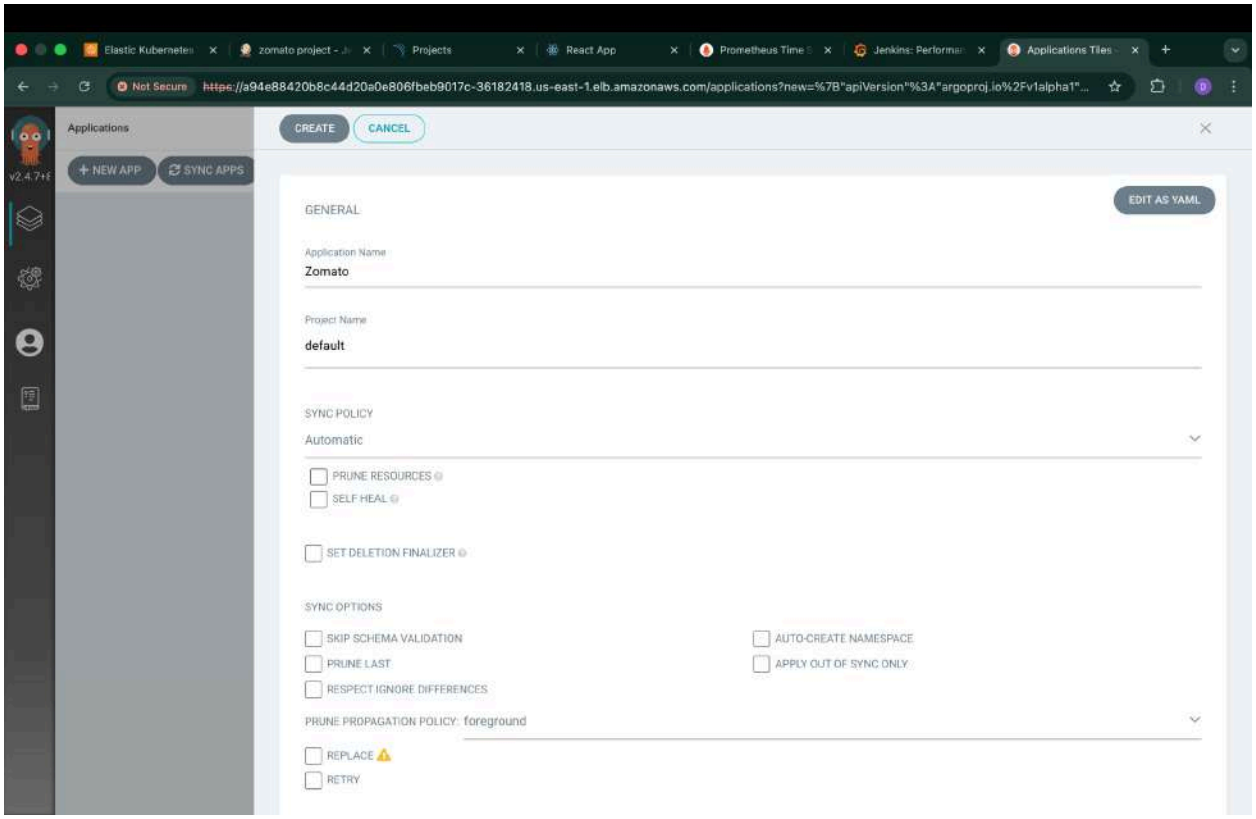

Settings - Argo CD
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/settings
Settings
SETTINGS
Log out
Repositories
Configure connected repositories
Certificates
Configure certificates for connecting Git repositories
GnuPG keys
Configure GnuPG public keys for commit verification
Clusters
Configure connected Kubernetes clusters
Projects
Configure Argo CD projects
Accounts
Configure Accounts

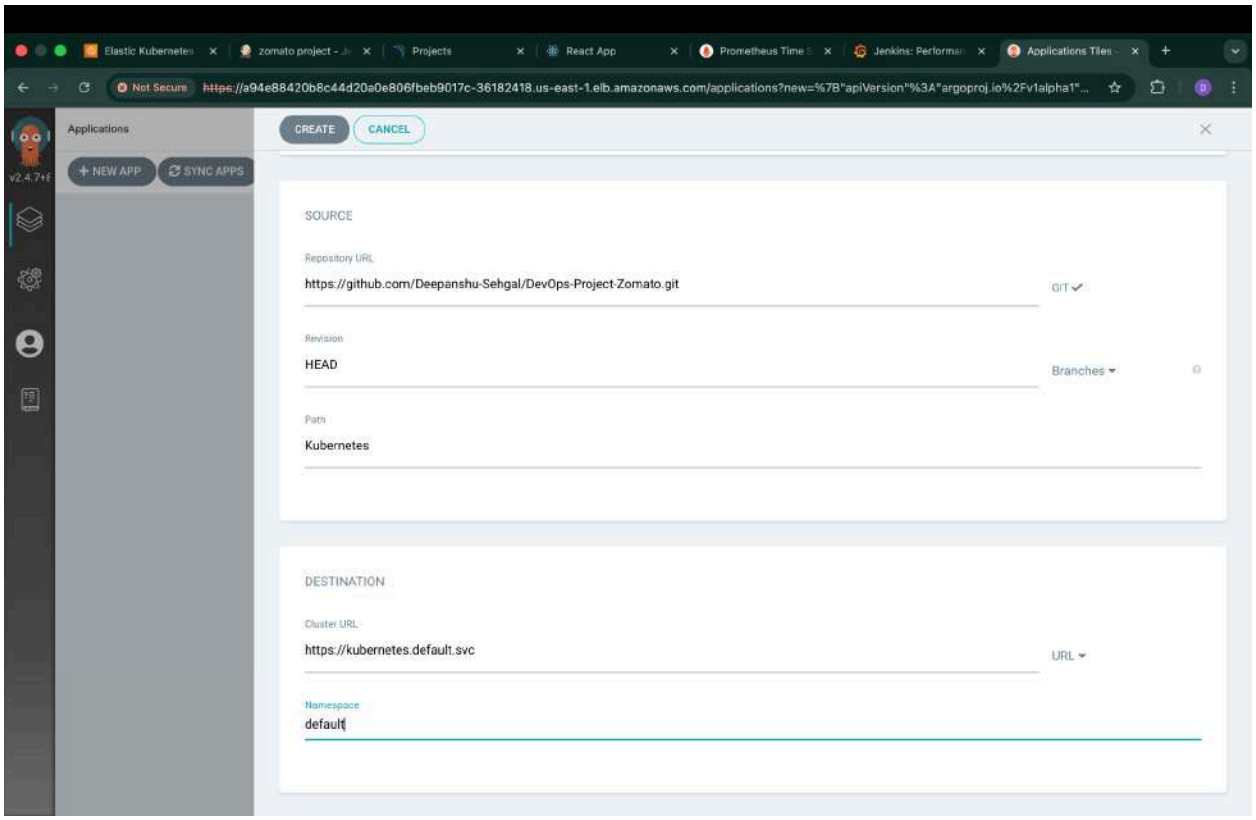

Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/settings/repos?addHTTPSRepo=true
Settings / Repositories
+ CONNECT REPO USING SSH
CONNECT
SAVE AS CREDENTIALS TEMPLATE
CANCEL
CONNECT REPO USING HTTPS
Type
git
Project
default
Repository URL
https://github.com/Deepanshu-Sehgal/DevOps-Project-Zomato.git
Username (optional)
Password (optional)
TLS client certificate (optional)

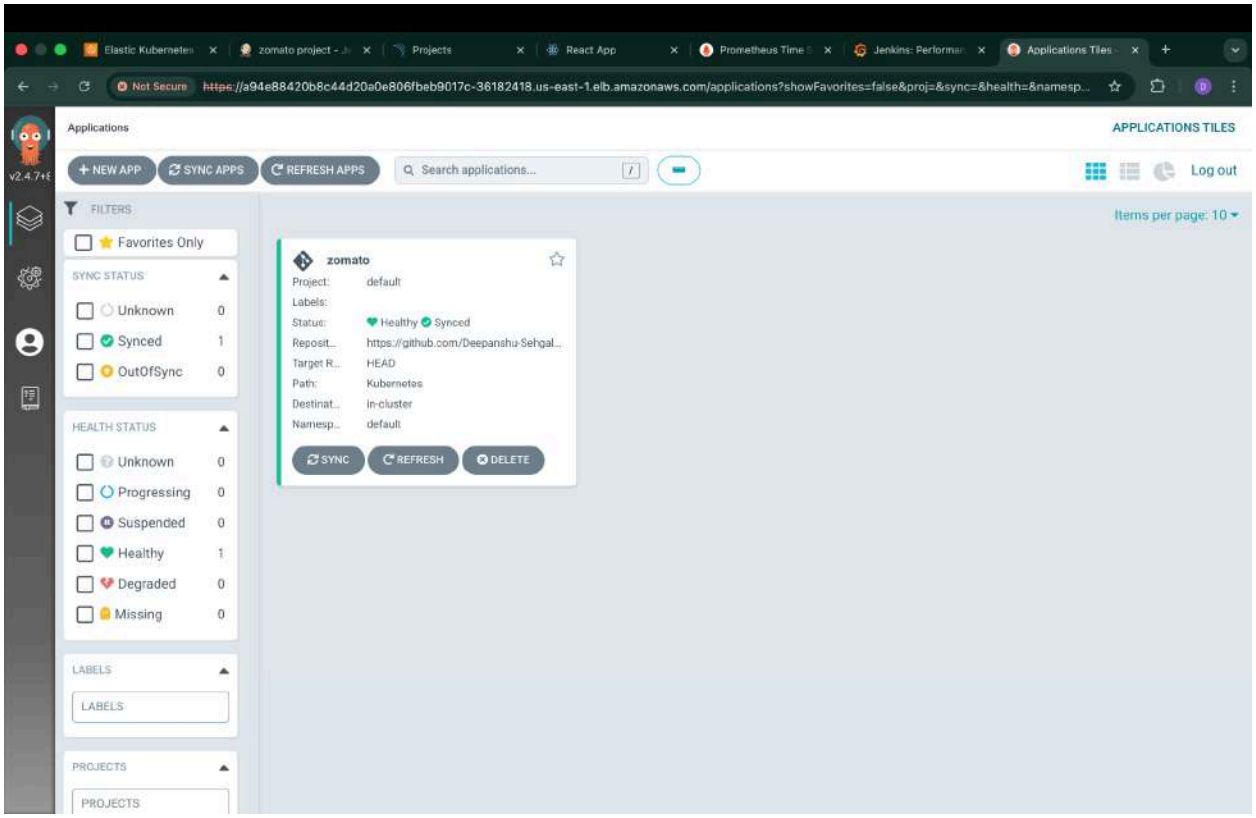

Elastic Kubernetes
zomato project - Jenkins
Projects
React App
Prometheus Time S
Jenkins: Performance
Repositories / Settings
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/settings/repos?addHTTPSRepo=false
v2.4.7+6
Settings / Repositories
REPOSITORIES
+ CONNECT REPO USING SSH
+ CONNECT REPO USING HTTPS
+ CONNECT REPO USING GITHUB APP
REFRESH LIST
Log out
TYPE
NAME
REPOSITORY
CONNECTION STATUS
git
https://github.com/Deepanshu-Sehgal/DevOps-Project-Zomato.git
Successful

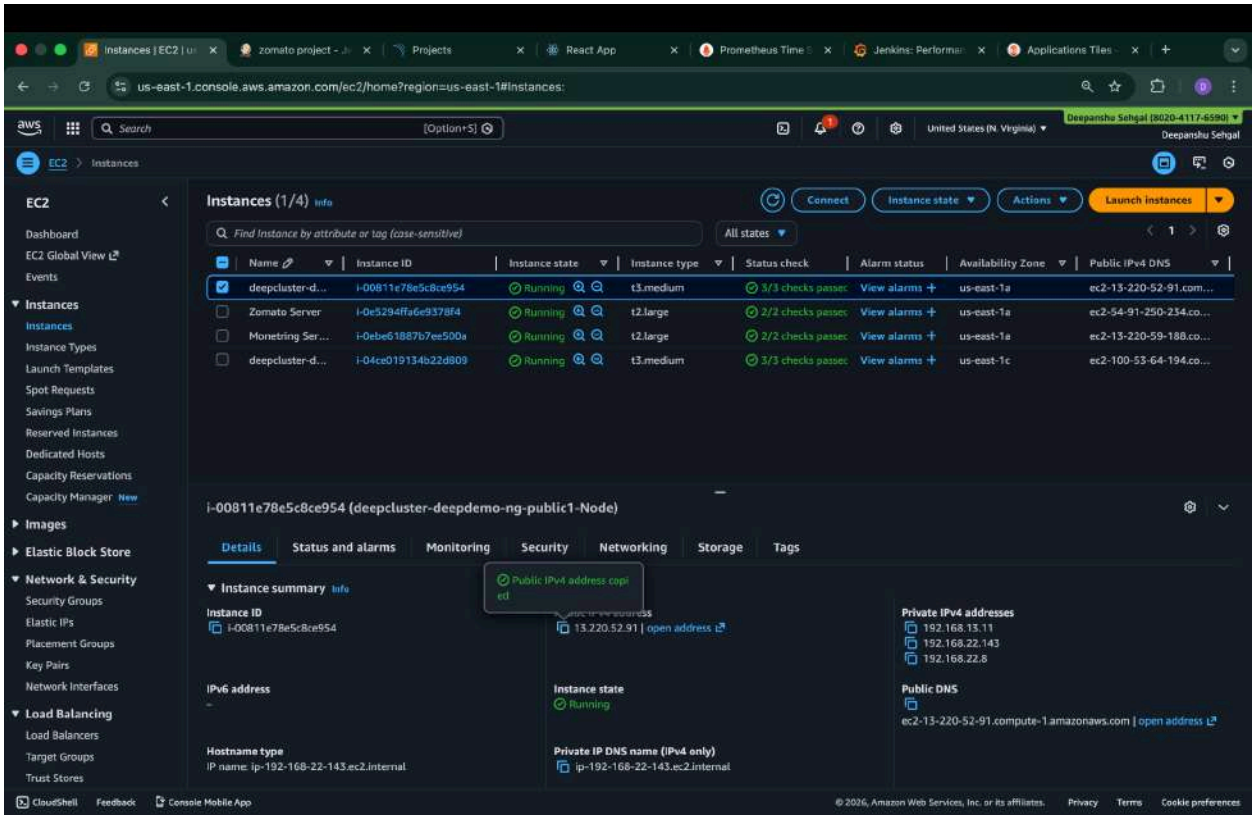

Elastic Kubernetes
zomato project - Jenkins
Projects
React App
Prometheus Time S
Jenkins: Performance
Repositories / Settings
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/settings/repos?addHTTPSRepo=false
v2.4.7+6
Settings / Repositories
REPOSITORIES
+ CONNECT REPO USING SSH
+ CONNECT REPO USING HTTPS
+ CONNECT REPO USING GITHUB APP
REFRESH LIST
Log out
TYPE
NAME
REPOSITORY
CONNECTION STATUS
git
https://github.com/Deepanshu-Sehgal/DevOps-Project-Zomato.git
Successful


Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/applications
Applications
APPLICATIONS TILES
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
Log out
No applications yet
Create new application to start managing resources in your cluster
CREATE APPLICATION


Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/applications?new=%7B"apiVersion"%3A"argoproj.io%2Fv1alpha1"...
Applications
+ NEW APP
SYNC APPS
CREATE
CANCEL
EDIT AS YAML
GENERAL
Application Name
Zomato
Project Name
default
SYNC POLICY
Automatic
PRUNE RESOURCES
SELF HEAL
SET DELETION FINALIZER
SYNC OPTIONS
SKIP SCHEMA VALIDATION
PRUNE LAST
RESPECT IGNORE DIFFERENCES
AUTO-CREATE NAMESPACE
APPLY OUT OF SYNC ONLY
PRUNE PROPAGATION POLICY: foreground
REPLACE
RETRY


Elastic Kubernetes
zomato project - Jenkins
Projects
React App
Prometheus Time Series
Jenkins: Performance
Applications Tiles
Not Secure
https://a94e88420b8c44d20a0e806fbeb9017c-36182418.us-east-1.elb.amazonaws.com/applications?new=%7B"apiVersion"%3A"argoproj.io%2Fv1alpha1"...
v2.4.7+f
Applications
+ NEW APP
SYNC APPS
CREATE
CANCEL
SOURCE
Repository URL
https://github.com/Deepanshu-Sehgal/DevOps-Project-Zomato.git
GIT
Revision
HEAD
Branches
Path
Kubernetes
DESTINATION
Cluster URL
https://kubernetes.default.svc
URL
Namespace
default


Applications
APPLICATIONS TILES
+ NEW APP
SYNC APPS
REFRESH APPS
Search applications...
Log out
FILTERS
Favorites Only
SYNC STATUS
Unknown 0
Synced 1
OutOfSync 0
HEALTH STATUS
Unknown 0
Progressing 0
Suspended 0
Healthy 1
Degraded 0
Missing 0
LABELS
PROJECTS
Items per page: 10
zomato
Project: default
Labels:
Status: Healthy Synced
Reposit... https://github.com/Deepanshu-Sehgal...
Target R... HEAD
Path: Kubernetes
Destinat... in-cluster
Namesp... default
SYNC
REFRESH
DELETE

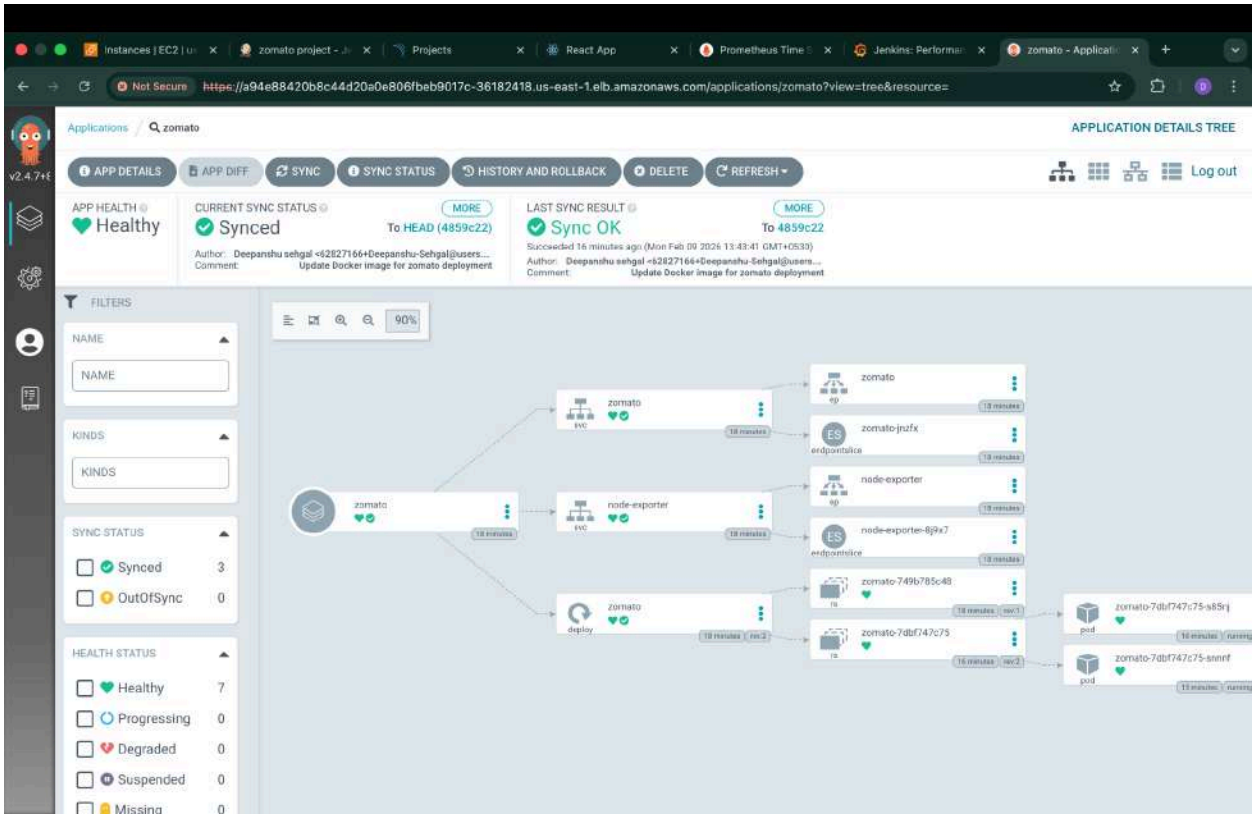

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
EC2 > Instances
Instances (1/4)
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
deepcluster-d... i-00811e78e5c8ce954 Running t3.medium 3/3 checks passed View alarms us-east-1a ec2-13-220-52-91.com...
Zomato Server i-0e5294ffa6e9378f4 Running t2.large 2/2 checks passed View alarms us-east-1a ec2-54-91-250-234.co...
Monetring Ser... i-0ebe61887b7ee500a Running t2.large 2/2 checks passed View alarms us-east-1a ec2-13-220-59-188.co...
deepcluster-d... i-04ce019134b22d809 Running t3.medium 3/3 checks passed View alarms us-east-1c ec2-100-53-64-194.co...
i-00811e78e5c8ce954 (deepcluster-deepdemo-ng-public1-Node)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-00811e78e5c8ce954
13.220.52.91 | open address
Private IPv4 addresses
192.168.13.11
192.168.22.143
192.168.22.8
IPv6 address
Instance state
Running
Public DNS
ec2-13-220-52-91.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-192-168-22-143.ec2.internal
Private IP DNS name (IPv4 only)
ip-192-168-22-143.ec2.internal

sudo vi /etc/prometheus/prometheus.yml

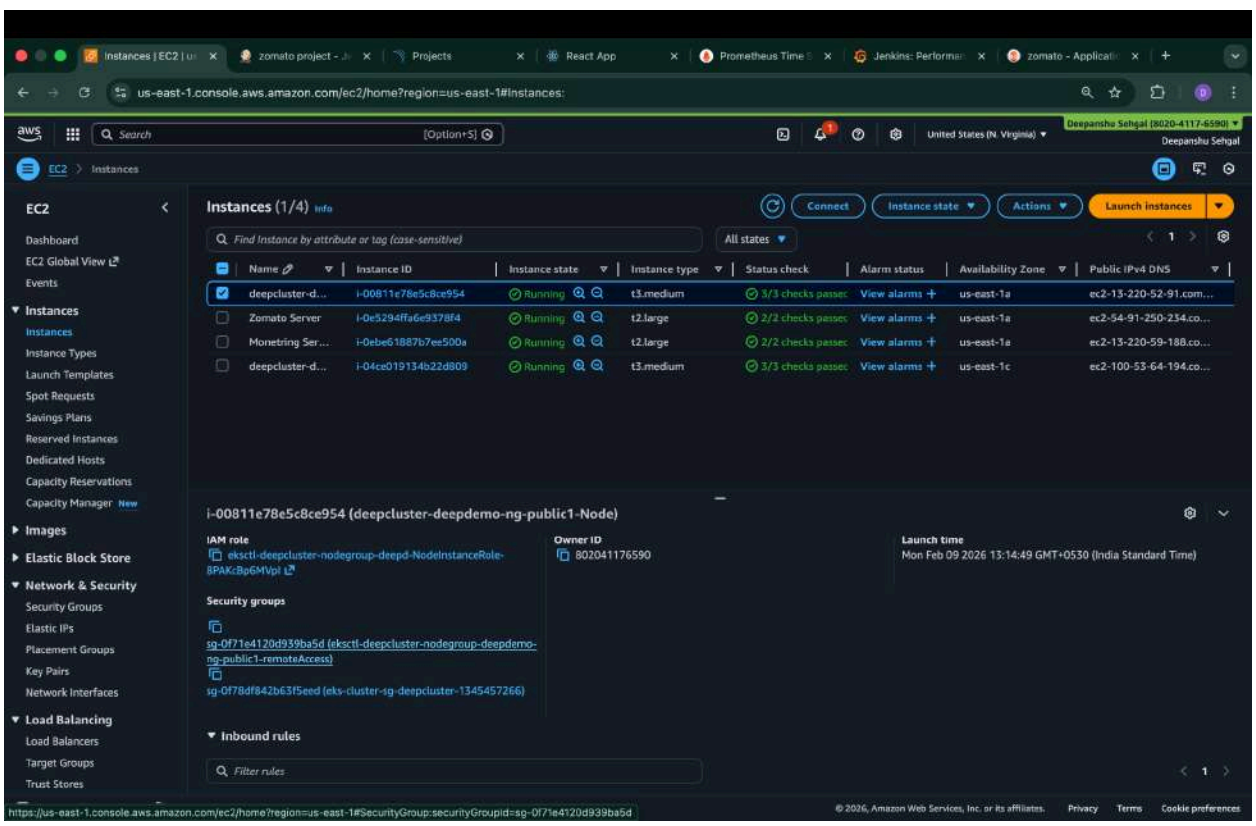
```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: 'node_exporter'
    static_configs:
      - targets: ['13.220.59.188:9100']

  - job_name: 'jenkins'
    metrics_path: '/prometheus'
    static_configs:
      - targets: ['54.91.250.234:8080']

  - job_name: 'k8s'
    metrics_path: '/metrics'
    static_configs:
      - targets: ['13.220.52.9:9100']
                                                                                    43,31          All
```

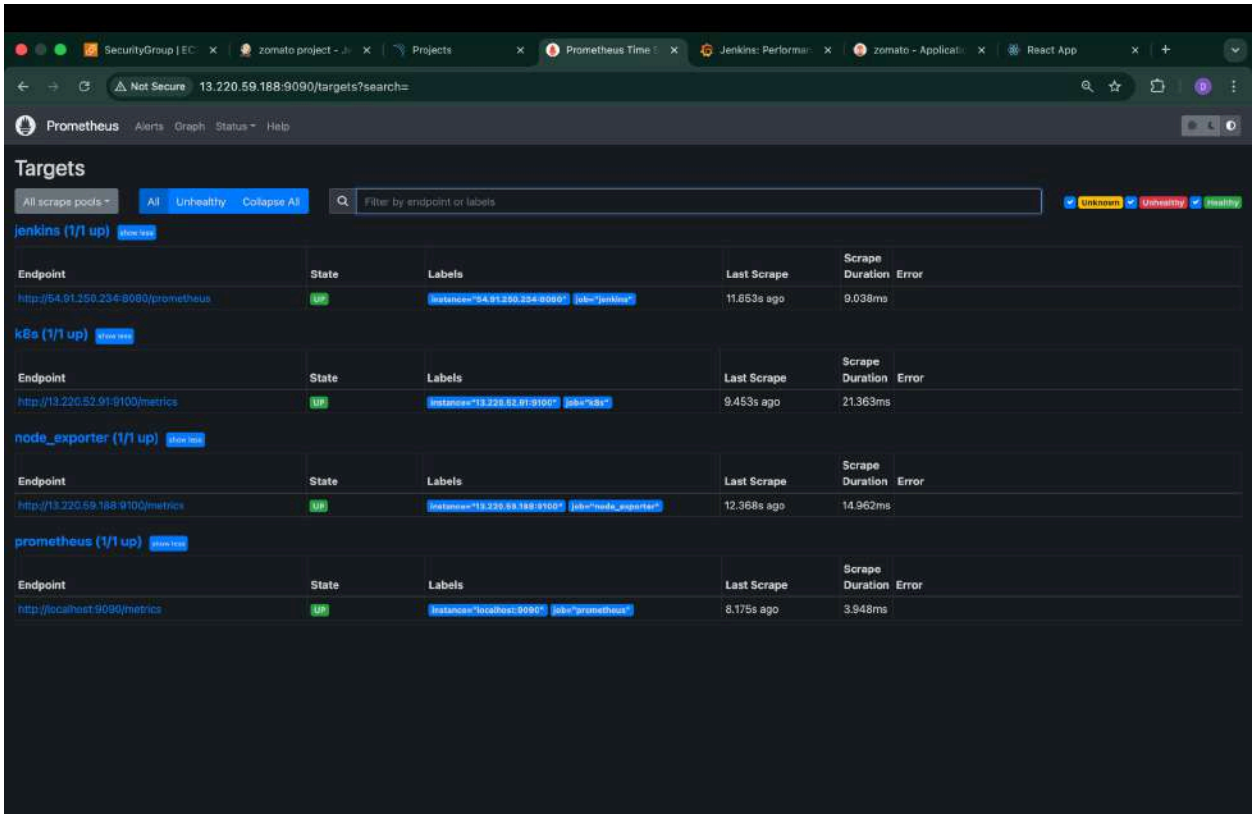
```
root@ip-172-31-23-200:~# promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
 SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax

root@ip-172-31-23-200:~#
```

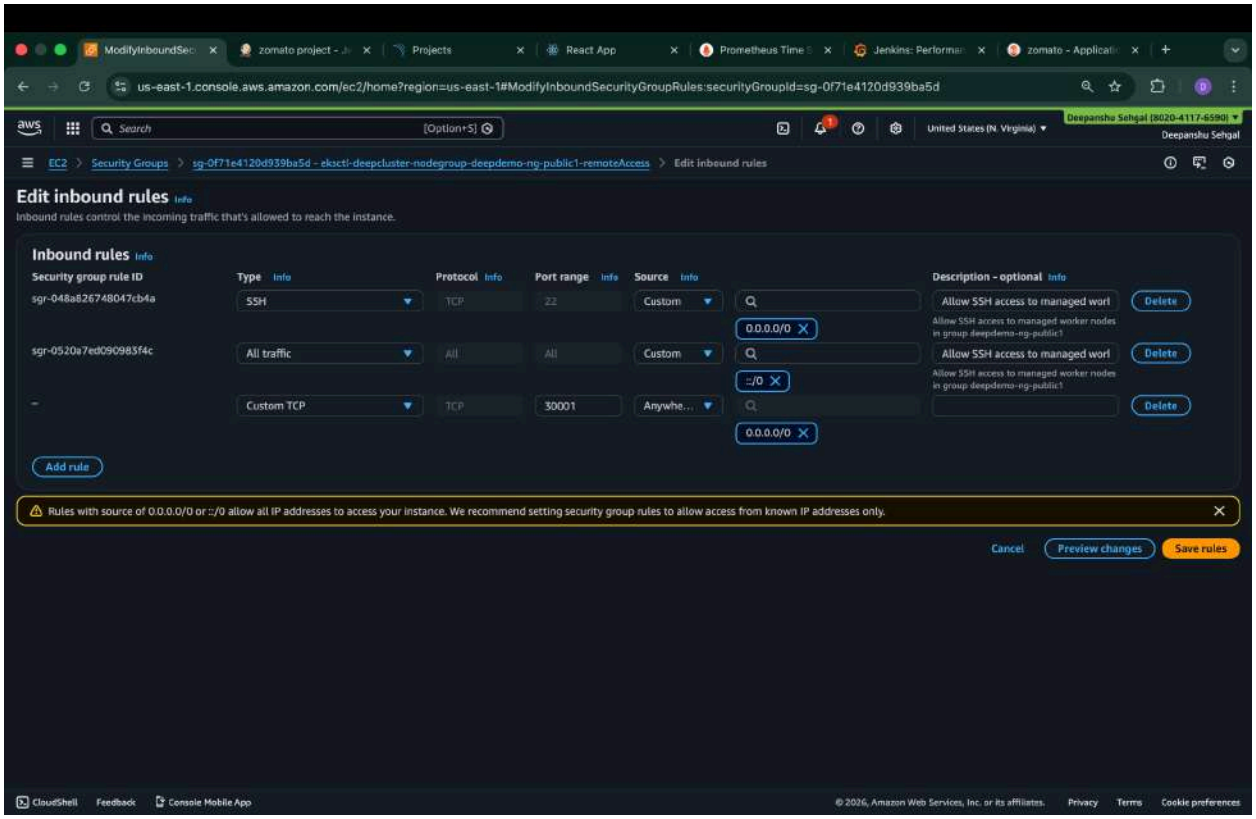
```
root@ip-172-31-23-200:~# curl -X POST http://localhost:9090/-/reload
root@ip-172-31-23-200:~#
```



Open ports to access the application.




Not Secure 13.220.59.188:9090/targets?search=
Prometheus Alerts Graph Status Help
Targets
All scrape pools All Unhealthy Collapse All
Filter by endpoint or labels
Unknown Unhealthy Healthy
jenkins (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://54.91.250.234:8080/prometheus UP instance="54.91.250.234:8080" job="jenkins" 11.853s ago 9.038ms
k8s (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://13.220.52.91:9100/metrics UP instance="13.220.52.91:9100" job="k8s" 9.453s ago 21.363ms
node_exporter (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://13.220.59.188:9100/metrics UP instance="13.220.59.188:9100" job="node_exporter" 12.368s ago 14.962ms
prometheus (1/1 up) show less
Endpoint State Labels Last Scrape Scrape Duration Error
http://localhost:9090/metrics UP instance="localhost:9090" job="prometheus" 8.175s ago 3.948ms


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-0f71e4120d939ba5d
United States (N. Virginia)
Deepanshu Sehgal
EC2 > Security Groups > sg-0f71e4120d939ba5d - eksctl-deepcluster-nodegroup-deepdemo-ng-public1-remoteAccess > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID Type Protocol Port range Source Description - optional
sgr-048a826748047cb4a SSH TCP 22 Custom 0.0.0.0/0 Allow SSH access to managed worl Delete
sgr-0520a7ed090983f4c All traffic All All Custom ::/0 Allow SSH access to managed worl Delete
- Custom TCP TCP 30001 Anywhe... 0.0.0.0/0 Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Cancel Preview changes Save rules
CloudShell Feedback Console Mobile App
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